

# Philosophical Logic

Matthew Tan

August 5, 2026

## Contents

|          |                                                                                                                |           |
|----------|----------------------------------------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Classical Propositional logic, variations, and deviations</b>                                               | <b>2</b>  |
| 1.1      | Classical Propositional Logic . . . . .                                                                        | 2         |
| 1.2      | Alternative Connectives . . . . .                                                                              | 4         |
| 1.3      | Non-classical Logic: Trivalent Semantics . . . . .                                                             | 5         |
| 1.3.1    | Weak Kleene . . . . .                                                                                          | 6         |
| 1.3.2    | (Strong) Kleene . . . . .                                                                                      | 6         |
| 1.3.3    | Łukasiewicz's system . . . . .                                                                                 | 7         |
| 1.4      | Semantic Consequence in the Three-Valued Setting . . . . .                                                     | 8         |
| 1.4.1    | Examples of Semantic Consequence for Kleene . . . . .                                                          | 8         |
| 1.4.2    | Examples of Semantic Consequence for Łukasiewicz . . . . .                                                     | 9         |
| 1.4.3    | Examples of Semantic Consequence for Priest's Logic of Paradox . . . . .                                       | 10        |
| 1.5      | Misc: Determinacy . . . . .                                                                                    | 11        |
| 1.6      | Supervaluationist Semantics and Consequence . . . . .                                                          | 11        |
| <b>2</b> | <b>Modal propositional logic: semantics</b>                                                                    | <b>12</b> |
| 2.1      | Modal Propositional Logic . . . . .                                                                            | 13        |
| 2.1.1    | Semantic Proofs and Counterexamples for K, D, T, B, S4 and S5 . . . . .                                        | 15        |
| 2.2      | Intepretations of MPL: Deontic, Epistemic, Temporal Modalities . . . . .                                       | 20        |
| 2.2.1    | Deontic Logic . . . . .                                                                                        | 20        |
| 2.2.2    | Epistemic Logic . . . . .                                                                                      | 21        |
| 2.2.3    | Tense Logic . . . . .                                                                                          | 21        |
| <b>3</b> | <b>Modal propositional logic: proof theory</b>                                                                 | <b>25</b> |
| 3.1      | Propositional Logic: Axiomatic Proofs . . . . .                                                                | 25        |
| 3.2      | Modal Propositional Logic: Axiomatic Proofs . . . . .                                                          | 27        |
| 3.3      | Proofs of toolkit rules . . . . .                                                                              | 31        |
| 3.4      | Worked Examples . . . . .                                                                                      | 35        |
| <b>4</b> | <b>Metatheory</b>                                                                                              | <b>37</b> |
| 4.1      | Propositional Logic Metatheory . . . . .                                                                       | 37        |
| 4.2      | MPL metatheory . . . . .                                                                                       | 40        |
| <b>5</b> | <b>Classical predicate logic, extensions, and deviations</b>                                                   | <b>45</b> |
| 5.1      | Classical Predicate Logic . . . . .                                                                            | 45        |
| 5.2      | Extensions of PC: identity, descriptor and lambda operators, SOL, Generalised Quantifiers, Functions . . . . . | 49        |
| 5.2.1    | Identity . . . . .                                                                                             | 49        |
| 5.2.2    | The descriptor operator . . . . .                                                                              | 50        |
| 5.2.3    | The lambda operator . . . . .                                                                                  | 50        |
| 5.2.4    | Second Order Logic . . . . .                                                                                   | 50        |
| 5.2.5    | Function Symbols . . . . .                                                                                     | 51        |

|           |                                                               |           |
|-----------|---------------------------------------------------------------|-----------|
| 5.2.6     | Generalised Quantifiers                                       | 52        |
| 5.2.7     | Worked Examples                                               | 53        |
| 5.3       | Basic Metatheory                                              | 55        |
| <b>6</b>  | <b>Simplified Quantified modal logic: constant domains</b>    | <b>56</b> |
| 6.1       | Syntax and Semantics                                          | 56        |
| 6.2       | Showing Validity and Invalidity                               | 58        |
| <b>7</b>  | <b>PC and SQML: Proof Theory</b>                              | <b>60</b> |
| 7.1       | PC and SQML Axiomatic Proofs                                  | 60        |
| 7.2       | PC and SQML Toolkit and worked examples                       | 62        |
| 7.2.1     | PC: Tautologies and Worked Examples                           | 62        |
| 7.2.2     | SQML: Theorems and Worked Examples                            | 63        |
| 7.3       | Basic Metatheory of SQML                                      | 65        |
| <b>8</b>  | <b>Quantified modal logic: variable domains, 2D semantics</b> | <b>66</b> |
| 8.1       | Variable Domain Quantified Modal Logic                        | 66        |
| 8.2       | 2-dimensional quantified modal logic                          | 71        |
| 8.2.1     | Designated World SQML                                         | 71        |
| 8.2.2     | 2DQML                                                         | 72        |
| <b>9</b>  | <b>Counterfactuals</b>                                        | <b>74</b> |
| 9.1       | Stalnaker's Conditional                                       | 74        |
| 9.2       | Lewis' Conditional                                            | 76        |
| <b>10</b> | <b>Appendix</b>                                               | <b>82</b> |
| 10.1      | Mathematical Tools (from Elements of Deductive Logic)         | 82        |
| 10.2      | Soundness and Completeness of Predicate Logic                 | 83        |
| 10.3      | Set Theory                                                    | 83        |

# 1 Classical Propositional logic, variations, and deviations

## 1.1 Classical Propositional Logic

**Definition 1.1 (Primitive Vocabulary for PL).** Sider has the following primitive vocabulary for PL

- Connectives:  $\{\rightarrow, \sim\}$
- Sentence letters:  $P, Q, R, P_1, \dots$
- Parantheses:  $(, )$

**Remark:** Note that Sider only has  $\rightarrow, \sim$  connectives. He uses them to express  $\{\leftrightarrow, \vee, \wedge\}$ .

Similarly, well-formed formulas (wffs) are built from the primitive vocabulary like so

**Definition 1.2 (PL wff).** PL-wffs are defined in the following way:

- Every sentence letter  $\alpha$  is a PL-wff
- If  $\phi$  and  $\psi$  are PL-wffs, then  $(\phi \rightarrow \psi)$  and  $\sim \phi$  are also PL-wffs
- Recursive step:** Only strings that can be shown to be PL-wffs using (i) and (ii) are PL-wffs

**Definition 1.3 (PL-interpretation).** A PL-interpretation is a function  $\mathcal{I}$  that assigns each sentence letter exactly one of the two truth values,  $\{0, 1\}$

**Remark:** If an interpretation function maps a sentence letter to only two truth values, we say that it is a bivalent interpretation.

An interpretation  $\mathcal{I}$  only interprets sentence letters. We can extend  $\mathcal{I}$  to interpret more complex sentences in line with the meaning of connectives via a valuation function  $V_{\mathcal{I}}$ .

**Definition 1.4 (PL-valuations).** Given a PL-interpretation  $\mathcal{I}$ , the PL-valuation for  $\mathcal{I}$ , denoted  $V_{\mathcal{I}}$ , is the unique function that assigns exactly one truth value, 0 or 1, to each wff, such that:

1.  $V_{\mathcal{I}}(\alpha) = \mathcal{I}(\alpha)$  for each sentence letter  $\alpha$
2.  $V_{\mathcal{I}}(\sim \phi) = 1$  iff  $V_{\mathcal{I}}(\phi) = 0$
3.  $V_{\mathcal{I}}(\phi \rightarrow \psi) = 1$  iff  $V_{\mathcal{I}}(\phi) = 0$  or  $V_{\mathcal{I}}(\psi) = 1$

**Definition 1.5 (PL-validity).** A wff  $\phi$  is said to be PL-valid iff  $V_{\mathcal{I}}(\phi) = 1$  for all PL-interpretation  $\mathcal{I}$ . When  $\phi$  is PL-valid, we write  $\models_{\text{PL}} \phi$ .

**Remark:** What Sider calls *valid* is basically what Halbach calls a *tautology*.

**Definition 1.6 (PL Semantic Consequence).** A wff  $\phi$  is a PL semantic consequence of a set of wffs  $\Gamma$  iff  $V_{\mathcal{I}}(\phi) = 1$  for every PL-interpretation  $\mathcal{I}$  such that  $V_{\mathcal{I}}(\gamma) = 1, \forall \gamma \in \Gamma$ . We write,  $\Gamma \models_{\text{PL}} \phi$

**Remark:** This is equivalent to the condition that there does not exist a PL-interpretation  $\mathcal{I}$  such that  $V_{\mathcal{I}}(\phi) = 0$  and  $V_{\mathcal{I}}(\gamma) = 1, \forall \gamma \in \Gamma$ .

**Expressing the other connectives.** We want to express  $\{\vee, \wedge, \leftrightarrow\}$  from the primitives  $\{\sim, \rightarrow\}$ . This is done accordingly:

- $\phi \vee \psi$  abbreviates  $\sim \phi \rightarrow \psi$ . It follows that

$$V_{\mathcal{I}}(\phi \vee \psi) = 1 \text{ iff } V_{\mathcal{I}}(\phi) = 1 \text{ or } V_{\mathcal{I}}(\psi) = 1$$

- $\phi \wedge \psi$  abbreviates  $\sim (\phi \rightarrow \sim \psi)$ . It follows that

$$V_{\mathcal{I}}(\phi \wedge \psi) = 1 \text{ iff } V_{\mathcal{I}}(\phi) = 1 \text{ and } V_{\mathcal{I}}(\psi) = 1$$

- $\phi \leftrightarrow \psi$  abbreviates  $\sim ((\phi \rightarrow \psi) \rightarrow \sim (\psi \rightarrow \phi))$ . It follows that

$$V_{\mathcal{I}}(\phi \leftrightarrow \psi) = 1 \text{ iff } V_{\mathcal{I}}(\phi) = V_{\mathcal{I}}(\psi)$$

**Remark:** Proofs for the propositions above are found in Q1 of Week 1 Exercises for 127 Philosophical Logic. Sider also gives informal semantic arguments to demonstrate facts about semantic consequence. These primarily take the form of *reductio ad absurdum* arguments.

**Example 1.1.** Claim:  $\models_{\text{PL}} \phi \rightarrow (\psi \rightarrow \phi)$ .

Let  $\mathcal{I}$  be a PL-interpretation.

- i. Assume for reductio that  $V_{\mathcal{I}}(\phi \rightarrow (\psi \rightarrow \phi)) = 0$
- ii. By the semantic clause definition of  $\rightarrow$ ,  $V_{\mathcal{I}}(\phi) = 1$  and  $V_{\mathcal{I}}(\psi \rightarrow \phi) = 0$
- iii. Apply the semantic clause definition of  $\rightarrow$  to (ii) again, it must be that  $V_{\mathcal{I}}(\phi) = 0$ , a contradiction
- iv. So, there is no  $\mathcal{I}$  such that  $V_{\mathcal{I}}(\phi \rightarrow (\psi \rightarrow \phi)) = 0$

Hence,  $\models_{\text{PL}} \phi \rightarrow (\psi \rightarrow \phi)$

## Claims about Propositional Logic

**Proposition 1.1.** Let  $\mathcal{I}^+$  be the interpretation such that  $\mathcal{I}^+(\alpha) = 1$  for each sentence letter  $\alpha$ . Show that for any sentence  $\phi$  with no occurrences of negation  $V_{\mathcal{I}^+}(\phi) = 1$

*Proof.* We prove this on induction of the complexity of  $\phi$ . Let  $C(\phi) = n$  denote that  $\phi$  has  $n$  number of connectives.

Base Case:  $C(\phi) = 0$ , so  $\phi := \alpha$  where  $\alpha$  is some sentence letter. By definition,  $\mathcal{I}^+(\alpha) = 1$ .

IH: for all  $m < n$ , if  $C(\phi) = m$  and  $V_{\mathcal{I}^+}(\phi) = 1$ , then  $V_{\mathcal{I}^+}(\psi) = 1$  for  $C(\psi) = n$ .

Assume that  $C(\phi_1) = n_1$  and  $C(\phi_2) = n_2$  and  $V_{\mathcal{I}^+}(\phi_1) = V_{\mathcal{I}^+}(\phi_2) = 1$ , where  $n_1, n_2 < n$ . Then, we have three cases:

1.  $\psi := \phi_1 \wedge \phi_2$ . By IH,  $V_{\mathcal{I}^+}(\phi_1) = V_{\mathcal{I}^+}(\phi_2) = 1$ . Hence,  $V_{\mathcal{I}^+}(\phi_1 \wedge \phi_2) = 1$
2.  $\psi := \phi_1 \vee \phi_2$ . By IH,  $V_{\mathcal{I}^+}(\phi_1) = V_{\mathcal{I}^+}(\phi_2) = 1$ . Hence,  $V_{\mathcal{I}^+}(\phi_1 \vee \phi_2) = 1$
3.  $\psi := \phi_1 \rightarrow \phi_2$ . By IH,  $V_{\mathcal{I}^+}(\phi_1) = V_{\mathcal{I}^+}(\phi_2) = 1$ . Hence,  $V_{\mathcal{I}^+}(\phi_1 \rightarrow \phi_2) = 1$

IH proven, claim established ■

**Proposition 1.2.** Show that if  $\phi$  contains at most one occurrence of each sentence letter then there is an interpretation  $\mathcal{I}$  such that  $V_{\mathcal{I}}(\phi) = 1$  and an interpretation  $\mathcal{J}$  such that  $V_{\mathcal{J}}(\phi) = 0$ .

*Proof.* We prove this by induction on the complexity of  $\phi$ .

Base Case:  $C(\alpha) = 0$ . Then  $V_{\mathcal{I}}(\alpha) = 1$  and  $V_{\mathcal{J}}(\alpha) = 0$ .

IH: for all  $m < n$ , if  $C(\phi) = n$  and  $\phi$  contains at most one occurrence of a sentence letter, there is a  $\mathcal{I}$  and  $\mathcal{J}$  such that  $V_{\mathcal{I}}(\phi) = 1$  and  $V_{\mathcal{J}}(\phi) = 0$ , then there is a  $\mathcal{I}$  and a  $\mathcal{J}$  such that  $V_{\mathcal{I}}(\psi) = 1$  and  $V_{\mathcal{J}}(\psi) = 0$  for  $C(\psi) = n$ , where  $\psi$  contains at most one occurrence of a sentence letter.

Assume that  $C(\phi_1) = n_1$  and  $C(\phi_2) = n_2$  with  $m_1, m_2 < n$  and that  $\phi_1$  and  $\phi_2$  contain at most one occurrence of a sentence letter. Assume that for  $\psi = \sim \phi_1$  or  $\phi_1 \circ \phi_2$ ,  $\psi$  also contains at most one occurrence of a sentence letter. This means that  $Sen(\phi_1) \cap Sen(\phi_2) = \emptyset$ . So, we have 2 cases:

1.  $\psi := \sim \phi_1$ . Then, by assumption, since there is an  $\mathcal{I}$  and  $\mathcal{J}$  such that  $V_{\mathcal{I}}(\phi_1) = 1$  and  $V_{\mathcal{J}}(\phi_1) = 0$ , then  $V_{\mathcal{I}}(\sim \phi_1) = 0$  and  $V_{\mathcal{J}}(\sim \phi_1) = 1$ .
2.  $\psi := \phi_1 \rightarrow \phi_2$ . Since  $Sen(\phi_1) \cap Sen(\phi_2) = \emptyset$ , for the interpretations  $\mathcal{I}$  and  $\mathcal{J}$  such that  $V_{\mathcal{I}}(\phi_1) = 1$  and  $V_{\mathcal{J}}(\phi_1) = 0$ , and the interpretations  $\mathcal{I}'$  and  $\mathcal{J}'$  such that  $V_{\mathcal{I}'}(\phi_2) = 1$  and  $V_{\mathcal{J}'}(\phi_2) = 0$ , there is an interpretation  $\mathcal{I}''$  that follows the assignments of  $\mathcal{I}$  and  $\mathcal{I}'$  for  $\phi_1, \phi_2$  respectively such that  $V_{\mathcal{I}''}(\phi_1 \rightarrow \phi_2) = 1$  and an interpretation  $\mathcal{J}''$  that follows the assignments of  $\mathcal{J}$  and  $\mathcal{J}'$  such that  $V_{\mathcal{J}''}(\phi_1 \rightarrow \phi_2) = 0$ .

IH proven, claim established ■

**Remark:** Notice that because  $\{\sim, \rightarrow\}$  is expressively adequate, we only have two cases to consider.

**Corollary 1.1.** Show that if  $\phi$  contains at most one occurrence of any sentence letter, then  $\models \phi$

*Proof.* An application of Proposition 1.2 ■

**Moral:** it is much harder to prove Corollary 1.1 directly. Sometimes, it is easier to prove a different (stronger) claim that implies the result that you're trying to prove.

## 1.2 Alternative Connectives

Let's consider mainly Peirce's Arrow  $\downarrow$ .

**Definition 1.7 (Peirce's Arrow).** The valuation of Peirce's Arrow is the following:

$$V_{\mathcal{I}}(\phi \downarrow \psi) = \begin{cases} 1 & \text{iff } V_{\mathcal{I}}(\phi) = 0 \text{ and } V_{\mathcal{I}}(\psi) = 0 \\ 0 & \text{otherwise} \end{cases}$$

How do we consider what connectives to include? We must first consider what a truth-function is.

**Definition 1.8 (Truth-Function).** A truth-function is a function from  $\{0,1\}^n$  to  $\{0,1\}$ . Formally, a truth-function  $f$  is

$$f : \{0,1\} \times \cdots \times \{0,1\} \longrightarrow \{0,1\}$$

**Definition 1.9 (Expression).**  $\phi$  expresses the truth function  $f$  just when

$$V_{\mathcal{I}}(\phi(\alpha_1, \dots, \alpha_n)) = f(\mathcal{I}(\alpha_1), \dots, \mathcal{I}(\alpha_n)), \forall \mathcal{I}$$

**Definition 1.10 (Expressive Adequacy).** A set  $C$  of connectives is expressively adequate just when, for every truth-function  $f$ , there is a  $\phi \in \text{Sen}(\mathcal{L}_1)$  with  $\text{Conn}(\phi) \subseteq C$  that expresses  $f$

We will state the next few theorems without proof.

**Theorem 1.2 (Expressive Adequacy for PL).** The set  $\{\sim, \wedge, \vee\}$  is expressively adequate.

**Corollary 1.3.** The sets  $\{\wedge, \sim\}$ ,  $\{\vee, \sim\}$  and  $\{\rightarrow, \sim\}$  are all expressively adequate.

**Corollary 1.4.** Peirce's Arrow,  $\{\downarrow\}$  is expressively adequate.

*Proof.* This is just a sketch. We show that  $\sim \phi$  is equivalent to  $\phi \downarrow \phi$  and  $\phi \rightarrow \psi$  is equivalent to  $((\phi \downarrow \phi) \downarrow \psi) \downarrow ((\phi \downarrow \phi) \downarrow \psi)$ . Then, from Corollary 1.2, we proved Corollary 1.3. ■

### 1.3 Non-classical Logic: Trivalent Semantics

**Motivation:** One property of PL is bivalence. But natural language contains numerous *prima facie* violations of bivalence:

1. **Presupposition failure.** Suppose Hilary doesn't smoke, never has. Is the following claim that 'Hilary has stopped smoking' true or false? If it is true, then Hilary has once smoked. If it is false, then Hilary still smokes. Bivalence implies Hilary smoked at some time or other, but never did
2. **Vagueness.** Henry is borderline tall, borderline non-tall. Is the claim 'Henry is tall' true or false? If the claim is true, then Henry is tall; if the claim is false, then Henry is not tall. But both options seem wrong as he is neither definitely tall nor definitely not tall
3. **Future contingents.** I will flip a coin at future time  $t$ . Is the claim 'The coin lands heads at  $t$ ' true or false? If the claim is true (pre-flip), the coin lands heads at  $t$ ; if the claim is false, then coin does not land heads at  $t$ . But this implies that the outcome is already determined.

One way to cope with these is to deny that the cases are genuine counterexamples to bivalence. Williamson does so specifically on vagueness by taking the position known as epistemicism. Roughly, it is the position that for any vague property such as balding, there is a cut-off sometimes unknowable to us such that the property  $\phi$  is either satisfied or not. In short, there is no real vagueness.

**Alternative response: Trivalent Semantics #.**

**Definition 1.11 (Primitive Vocabulary for three-valued systems).** Our primitive vocabulary must be expanded since the PL-definitions of  $\wedge$  and  $\vee$  might not work in non-classical settings

- Connectives:  $\{\sim, \wedge, \vee, \rightarrow\}$
- Sentence letters:  $P, Q, R, P_1, Q_1, R_1, \dots$
- Parentheses:  $(, )$

**Definition 1.12 (Wff for three-valued systems).** Again, we define wffs recursively

- i. Every sentence letter  $\alpha$  is a wff

- ii. If  $\phi$  and  $\psi$  are wffs, then  $(\phi \rightarrow \psi), (\phi \wedge \psi), (\phi \vee \psi)$  and  $\sim \phi$  are also wffs
- iii. (iii) Only strings that can be shown to be wffs using (i) and (ii) are wffs

**Definition 1.13 (Trivalent Interpretation).** A trivalent interpretation is a function  $\mathcal{I}$  that assigns each sentence letter exactly one of 1, 0 and #

### Evaluating Complex Formulas in the three valued setting

Now, we consider different logical systems that tells us what truth values to assign to a complex formula given a set of truth values.

#### 1.3.1 Weak Kleene

**Intuition:** Weak Kleene basically agrees with the classical tables on all classical rows but whenever any input is #, the output is simply #. This is motivated from the interpretation of # as ‘meaningless’. The truth table is as follows

| $\phi$ | $\psi$ | $\phi \wedge \psi$ | $\phi \vee \psi$ | $\phi \rightarrow \psi$ |
|--------|--------|--------------------|------------------|-------------------------|
| 1      | 1      | 1                  | 1                | 1                       |
| 1      | 0      | 0                  | 1                | 0                       |
| 1      | #      | #                  | #                | #                       |
| 0      | 1      | 0                  | 1                | 1                       |
| 0      | 0      | 0                  | 0                | 0                       |
| 0      | #      | #                  | #                | #                       |
| #      | 1      | #                  | #                | #                       |
| #      | 0      | #                  | #                | #                       |
| #      | #      | #                  | #                | #                       |

and for the negation table, it is simply

| $\phi$ | $\sim \phi$ |
|--------|-------------|
| 1      | 0           |
| 0      | 1           |
| #      | #           |

#### 1.3.2 (Strong) Kleene

**Intuition:** In non-classical rows, we consider how to replace # with either 0 or 1. If all such ways result in 1, then we assign 1; if all such ways result in 0, then we assign 0. Otherwise, if some result in 1 and some in 0, we assign #.

| $\phi$ | $\psi$ | $\phi \wedge \psi$ | $\phi \vee \psi$ | $\phi \rightarrow \psi$ |
|--------|--------|--------------------|------------------|-------------------------|
| 1      | 1      | 1                  | 1                | 1                       |
| 1      | 0      | 0                  | 1                | 0                       |
| 1      | #      | #                  | 1                | #                       |
| 0      | 1      | 0                  | 1                | 1                       |
| 0      | 0      | 0                  | 0                | 1                       |
| 0      | #      | 0                  | #                | 1                       |
| #      | 1      | #                  | 1                | 1                       |
| #      | 0      | 0                  | #                | #                       |
| #      | #      | #                  | #                | #                       |

and for the negation table, it is simply

| $\phi$ | $\sim \phi$ |
|--------|-------------|
| 1      | 0           |
| 0      | 1           |
| #      | #           |

This can be equivalently expressed in words.

**Definition 1.14 (Kleene-valuation).** Given a trivalent-interpretation  $\mathcal{I}$ , the Kleene-valuation for  $\mathcal{I}$ , written  $KV_{\mathcal{I}}$ , is the unique function that assigns exactly one of  $\{0, 1, \#\}$  to each wff such that

1. For each sentence letter  $\alpha$

$$\text{KV}_{\mathcal{J}}(\alpha) = \mathcal{J}(\alpha)$$

2. For  $\sim \phi$

$$\text{KV}_{\mathcal{J}}(\sim \phi) = \begin{cases} 1 & \text{iff } \text{KV}_{\mathcal{J}}(\phi) = 0 \\ 0 & \text{iff } \text{KV}_{\mathcal{J}}(\phi) = 1 \\ \# & \text{otherwise} \end{cases}$$

3. For  $\phi \wedge \psi$

$$\text{KV}_{\mathcal{J}}(\phi \wedge \psi) = \begin{cases} 1 & \text{iff } \text{KV}_{\mathcal{J}}(\phi) = 1 \text{ and } \text{KV}_{\mathcal{J}}(\psi) = 1 \\ 0 & \text{iff } \text{KV}_{\mathcal{J}}(\phi) = 0 \text{ or } \text{KV}_{\mathcal{J}}(\psi) = 0 \\ \# & \text{otherwise} \end{cases}$$

4. For  $\phi \vee \psi$

$$\text{KV}_{\mathcal{J}}(\phi \vee \psi) = \begin{cases} 1 & \text{iff } \text{KV}_{\mathcal{J}}(\phi) = 1 \text{ or } \text{KV}_{\mathcal{J}}(\psi) = 1 \\ 0 & \text{iff } \text{KV}_{\mathcal{J}}(\phi) = 0 \text{ and } \text{KV}_{\mathcal{J}}(\psi) = 0 \\ \# & \text{otherwise} \end{cases}$$

5. For  $\phi \rightarrow \psi$

$$\text{KV}_{\mathcal{J}}(\phi \rightarrow \psi) = \begin{cases} 1 & \text{iff } \text{KV}_{\mathcal{J}}(\phi) = 0 \text{ or } \text{KV}_{\mathcal{J}}(\psi) = 1 \\ 0 & \text{iff } \text{KV}_{\mathcal{J}}(\phi) = 1 \text{ and } \text{KV}_{\mathcal{J}}(\psi) = 0 \\ \# & \text{otherwise} \end{cases}$$

### 1.3.3 Łukasiewicz's system

Łukasiewicz's system differs from Kleene only in a single truth value when  $\text{LV}_{\mathcal{J}}(\phi) = \text{LV}_{\mathcal{J}}(\psi) = \#$ , then  $\text{LV}_{\mathcal{J}}(\phi \rightarrow \psi) = 1$ . Accordingly,

| $\phi$ | $\psi$ | $\phi \wedge \psi$ | $\phi \vee \psi$ | $\phi \rightarrow \psi$ |
|--------|--------|--------------------|------------------|-------------------------|
| 1      | 1      | 1                  | 1                | 1                       |
| 1      | 0      | 0                  | 1                | 0                       |
| 1      | #      | #                  | 1                | #                       |
| 0      | 1      | 0                  | 1                | 1                       |
| 0      | 0      | 0                  | 0                | 1                       |
| 0      | #      | 0                  | #                | 1                       |
| #      | 1      | #                  | 1                | 1                       |
| #      | 0      | 0                  | #                | #                       |
| #      | #      | #                  | #                | 1                       |

We replace  $\text{KV}_{\mathcal{J}}$  with  $\text{LV}_{\mathcal{J}}$  by replicating everything from strong Kleene but replacing the final clause with

$$\text{LV}_{\mathcal{J}}(\phi \rightarrow \psi) = \begin{cases} 1 & \text{iff } \text{LV}_{\mathcal{J}}(\phi) = 0 \text{ or } \text{LV}_{\mathcal{J}}(\psi) = 1 \text{ or } \text{LV}_{\mathcal{J}}(\phi) = \text{LV}_{\mathcal{J}}(\psi) = \# \\ 0 & \text{iff } \text{LV}_{\mathcal{J}}(\phi) = 1 \text{ and } \text{LV}_{\mathcal{J}}(\psi) = 0 \\ \# & \text{otherwise} \end{cases}$$

In PSet 1, we showed that if we order  $1 > \# > 0$  by giving  $\# = \frac{1}{2}$ , then we yield the condition

$$\text{LV}_{\mathcal{J}}(\phi \rightarrow \psi) = \begin{cases} 1 & \text{if } \text{LV}_{\mathcal{J}}(\psi) \geq \text{LV}_{\mathcal{J}}(\phi) \\ 1 - (\text{LV}_{\mathcal{J}}(\phi) - \text{LV}_{\mathcal{J}}(\psi)) & \text{if } \text{LV}_{\mathcal{J}}(\psi) < \text{LV}_{\mathcal{J}}(\phi) \end{cases}$$

## 1.4 Semantic Consequence in the Three-Valued Setting

**Definition 1.15 (Designed Values).** Let  $D$  denote the set of designated values, such that  $D \subseteq \{0, 1, \#\}$

**Definition 1.16 (D-Validity for Kleene).** A wff  $\phi$  is D-valid (with respect to Kleene tables) iff  $KV_{\mathcal{J}}(\phi) \in D$  for every trivalent-interpretation  $\mathcal{J}$

**Definition 1.17 (D-Consequence for Kleene).** A wff  $\phi$  is a D-semantic consequence of a set of wffs  $\Gamma$  (with respect to the Kleene tables) iff  $KV_{\mathcal{J}}(\phi) \in D$  for every trivalent-interpretation such that  $KV_{\mathcal{J}}(\gamma) \in D, \forall \gamma \in \Gamma$

Both definitions carry over to Łukasiewicz and Priest's Logic of Paradox and we just replace  $KV_{\mathcal{J}}$  with  $LV_{\mathcal{J}}$  and  $LP_{\mathcal{J}}$  and the choice of designated values in  $D$ .

The consequence relation we obtain from these three-valued approaches depends on two decisions: (i) the choice of truth-tables, (ii) the choice of designated values.

- For Kleene (K),  $D = \{1\}$ , symbolised by  $\models_K$
- For Łukasiewicz (Ł),  $D = \{1\}$ , symbolised by  $\models_L$
- For Logic of Paradox (LP),  $D = \{1, \#\}$ , symbolised by  $\models_{LP}$

**Remark:** While Kleene and Łukasiewicz regard  $\#$  as being a state that is ‘gappy’ (neither true nor false), Priest regards (being a dialectist) the state of  $\#$  as being *both true and false at the same time*. In turn, for LP,  $LP_{\mathcal{J}}(\phi) = 1$  is the state of being true and only true and similarly for falsity. Naturally for Priest, if semantic consequence is about truth preservation, then the designated values must include both 1 and  $\#$

We must note that many of the classical logical rules fail in the three-valued setting and especially for LP. In particular, one must note that under some settings  $\models$  does not always simply mean  $\rightarrow$  as in PL.

### 1.4.1 Examples of Semantic Consequence for Kleene

#### Rules that hold under Kleene

1. Modus Ponens I:  $P, P \rightarrow Q \models_K Q$

*Proof.* AFC that there is an interpretation  $\mathcal{J}$  such that  $V_{\mathcal{J}}(P) = 1, V_{\mathcal{J}}(P \rightarrow Q) = 1$ , and  $V_{\mathcal{J}}(Q) = \#$  or 0. But if  $V_{\mathcal{J}}(P) = 1, V_{\mathcal{J}}(P \rightarrow Q) = 1$ , then  $V_{\mathcal{J}}(Q) = 1$  by definition of  $\rightarrow$ , a contradiction ■

2. Ex Falso Quod Libet I:  $P, \sim P \models_K Q$

*Proof.* AFC that there is an interpretation  $\mathcal{J}$  such that  $V_{\mathcal{J}}(P) = 1$  and  $V_{\mathcal{J}}(\sim P) = 1$  but  $V_{\mathcal{J}}(Q) = 0$ . But this is impossible, since  $V_{\mathcal{J}}(P) = 1$  implies  $V_{\mathcal{J}}(\sim P) = 0$  which contradicts  $V_{\mathcal{J}}(\sim P) = 1$ . ■

#### Rules that do not hold under Kleene

1. Modus Ponens II:  $\not\models_K P \rightarrow ((P \rightarrow Q) \rightarrow Q)$

*Proof.* Our counterexample is

| $P$ | $Q$  | $P$ | $\rightarrow$ | $((P \rightarrow Q) \rightarrow Q)$ |
|-----|------|-----|---------------|-------------------------------------|
| 1   | 0    | 1   | 0             | 1                                   |
| 1   | $\#$ | 1   | $\#$          | $\#$                                |

2. Ex Falso Quod Libet II:  $\not\models_K (P \wedge \sim P) \rightarrow Q$



*Proof.* Our counterexample is

| $P$  | $Q$  | $(P \wedge \sim P) \rightarrow Q$ |      |      |      |      |      |
|------|------|-----------------------------------|------|------|------|------|------|
| $\#$ | $\#$ | $\#$                              | $\#$ | $\#$ | $\#$ | $\#$ | $\#$ |

■

But notice that there are no Kleene-valid wffs.

**Proposition 1.3.** Show that there are no Kleene-valid wffs

*Proof.* We prove a stronger claim that for a  $\mathcal{J}$  that assigns  $\#$  to all sentence letters, then for any  $\phi$ ,  $V_{\mathcal{J}}(\phi) = \#$ .

Base Case: for any  $\alpha$ ,  $C(\alpha) = 0$ . Then by def,  $\mathcal{J}(\alpha) = \#$ .

IH: if for all  $\phi$  s.t.  $C(\phi) = m < n$ ,  $V_{\mathcal{J}}(\phi) = \#$ , where  $\mathcal{J}$  assigns  $\#$  to all sentence letters then for  $\psi$  s.t.  $C(\psi) = n$ , then  $V_{\mathcal{J}}(\psi) = \#$ . Then since  $\{\sim, \rightarrow\}$  remains expressively adequate, we have two cases

1.  $\psi := \sim \phi$ : By assumption,  $V_{\mathcal{J}}(\phi) = \#, V_{\mathcal{J}}(\sim \phi) = \#$
2.  $\psi := \phi_1 \rightarrow \phi_2$ : By assumption,  $V_{\mathcal{J}}(\phi_1) = V_{\mathcal{J}}(\phi_2) = \#$ . Hence,  $V_{\mathcal{J}}(\phi_1 \rightarrow \phi_2) = \#$

Then,  $\not\models_K \phi$  for all  $\phi$  since for any  $\phi$ ,  $V_{\mathcal{J}}(\phi) = \#$

■

**Proposition 1.4.** Say that one trivalent interpretation  $\mathcal{J}$  refines another  $\mathcal{J}$  iff for any sentence letter  $\alpha$ , if  $\mathcal{J}(\alpha) = 1$ , then  $\mathcal{J}(\alpha) = 1$  and if  $\mathcal{J}(\alpha) = 0$ , then  $\mathcal{J}(\alpha) = 0$ . That is  $\mathcal{J}$  preserves all of  $\mathcal{J}$ 's classical truth values (though it may assign some additional classical values, in cases where  $\mathcal{J}$  assigns  $\#$ ). Show that refining a trivalent interpretation preserves classical values for all wffs, given the Kleene tables. That is, if  $\mathcal{J}$  refines  $\mathcal{J}$ , then for every wff  $\phi$ , if  $KV_{\mathcal{J}}(\phi) = 1$  then  $KV_{\mathcal{J}}(\phi) = 1$  and if  $KV_{\mathcal{J}}(\phi) = 0$ , then  $KV_{\mathcal{J}}(\phi) = 0$

*Proof.* We can prove this by induction on the complexity of  $\phi$ .

Base Case: Consider any  $\alpha$  s.t.  $C(\alpha) = 0$ . By definition, if  $\mathcal{J}(\alpha) = 1$ , then  $\mathcal{J}^+(\alpha) = 1$ , and if  $\mathcal{J}(\alpha) = 0$ , then  $\mathcal{J}^+(\alpha) = 0$ .

IH: if  $\mathcal{J}^+$  preserves the classical truth values of  $\phi$  for  $C(\phi) = m < n$ , then it does so for  $\psi$  where  $C(\psi) = n$ . There are two cases:

1.  $\psi := \sim \phi$ : By definition, if  $V_{\mathcal{J}}(\sim \phi) = 0$ , then  $V_{\mathcal{J}}(\phi) = 1$ . Then  $V_{\mathcal{J}^+}(\phi) = 1, V_{\mathcal{J}^+}(\sim \phi) = 0$ .
2.  $\psi : \phi_1 \rightarrow \phi_2$ : If  $V_{\mathcal{J}}(\phi_1 \rightarrow \phi_2) = 1$ , then either  $V_{\mathcal{J}}(\phi_1) = 0$  or  $V_{\mathcal{J}}(\phi_2) = 1$ . By IH,  $V_{\mathcal{J}^+}(\phi_1) = 0$  or  $V_{\mathcal{J}^+}(\phi_2) = 1$ . Hence,  $V_{\mathcal{J}^+}(\phi_1 \rightarrow \phi_2) = 1$  or  $V_{\mathcal{J}^+}(\phi_1 \rightarrow \phi_2) = 1$ , hence  $V_{\mathcal{J}^+}(\phi_1 \rightarrow \phi_2) = 1$ .  
If  $V_{\mathcal{J}}(\phi_1 \rightarrow \phi_2) = 0$ , then  $V_{\mathcal{J}}(\phi_1) = 1$  and  $V_{\mathcal{J}}(\phi_2) = 0$ . By IH,  $V_{\mathcal{J}^+}(\phi_1) = 1$  and  $V_{\mathcal{J}^+}(\phi_2) = 0$ , hence  $V_{\mathcal{J}^+}(\phi_1 \rightarrow \phi_2) = 0$ .

IH proven and claim established that  $\mathcal{J}^+$  preserves the classical truth values of the Kleene table

■

#### 1.4.2 Examples of Semantic Consequence for Łukasiewicz

##### Rules that hold under Łukasiewicz

1. Modus Ponens I:  $P, P \rightarrow Q \models_L Q$

*Proof.* Let  $V_{\mathcal{J}}(P) = 1$  and  $V_{\mathcal{J}}(P \rightarrow Q) = 1$ . Assume for reductio  $V_{\mathcal{J}}(Q) \neq 1$ . But by the definition of  $\rightarrow$ , if  $V_{\mathcal{J}}(Q) = \#$ , then  $V_{\mathcal{J}}(P \rightarrow Q) = \#$ , and if  $V_{\mathcal{J}}(Q) = 0$ , then  $V_{\mathcal{J}}(P \rightarrow Q) = 0$  which are contradictions. So,  $P, P \rightarrow Q \models_L Q$

■

2. Modus Ponens II:  $\models_L P \rightarrow ((P \rightarrow Q) \rightarrow Q)$

*Proof.* Assume for contradiction that  $V_{\mathcal{J}}(P \rightarrow ((P \rightarrow Q) \rightarrow Q)) \neq 1$ . We have two cases:

- (a)  $V_{\mathcal{J}}(P \rightarrow ((P \rightarrow Q) \rightarrow Q)) = 0$ :  $V_{\mathcal{J}}(P) = 1$  but  $V_{\mathcal{J}}((P \rightarrow Q) \rightarrow Q) = 0$  which implies  $V_{\mathcal{J}}(P \rightarrow Q) = 1$  and  $V_{\mathcal{J}}(Q) = 0$ . So  $V_{\mathcal{J}}(P) = 0$  which is a contradiction.
- (b)  $V_{\mathcal{J}}(P \rightarrow ((P \rightarrow Q) \rightarrow Q)) = \#$ . So either,
- $V_{\mathcal{J}}(P) = 1$  and  $V_{\mathcal{J}}((P \rightarrow Q) \rightarrow Q) = \#$ . But if  $V_{\mathcal{J}}(Q) = 1, 0, \#$  then  $V_{\mathcal{J}}((P \rightarrow Q) \rightarrow Q) = 1, 1, 1$  respectively, which is a contradiction.
  - $V_{\mathcal{J}}(P) = \#$  and  $V_{\mathcal{J}}((P \rightarrow Q) \rightarrow Q) = 0$ . Then,  $V_{\mathcal{J}}(P \rightarrow Q) = 1$  and  $V_{\mathcal{J}}(Q) = 0$ , so  $V_{\mathcal{J}}(P) = 1$ , which is a contradiction.

Hence, the reductio claim is established. ■

### 3. Ex Falso Quod Libet I: $P, \sim P \models_{\mathbf{L}} Q$

*Proof.* AFC that  $V_{\mathcal{J}}(P) = 1$  and  $V_{\mathcal{J}}(\sim P) = 1$  but  $V_{\mathcal{J}}(Q) \neq 1$ . But this is impossible since if  $V_{\mathcal{J}}(P) = 1$ , then  $V_{\mathcal{J}}(\sim P) = 0$  ■

### Rules that do not hold under Łukasiewicz

#### 1. Ex Falso Quod Libet II: $\not\models_{\mathbf{L}} (P \wedge \sim P) \rightarrow Q$ .

*Proof.* Our counterexample is

| $P$  | $Q$ | $(P \wedge \sim P) \rightarrow Q$ |      |      |      |      |   |
|------|-----|-----------------------------------|------|------|------|------|---|
| $\#$ | 0   | $\#$                              | $\#$ | $\#$ | $\#$ | $\#$ | 0 |

Hence,  $\not\models_{\mathbf{L}} (P \wedge \sim P) \rightarrow Q$  ■

**Proposition 1.5.** Show that Proposition 1.4 does not hold if we valuate using Łukasiewicz's tables rather than Kleene's

*Proof.* If we use Łukasiewicz's tables instead, consider an  $\mathcal{J}$  s.t.  $V_{\mathcal{J}}(\phi \rightarrow \psi) = 1$  where  $V_{\mathcal{J}}(\phi) = V_{\mathcal{J}}(\psi) \#$ . Suppose that we have a refinement  $\mathcal{J}^+$  s.t.

$$V_{\mathcal{J}^+}(\phi) = \begin{cases} 0 & \text{if } V_{\mathcal{J}}(\phi) = 0 \\ 1 & \text{otherwise} \end{cases}$$

and

$$V_{\mathcal{J}^+}(\psi) = \begin{cases} 1 & \text{if } V_{\mathcal{J}}(\psi) = 0 \\ 0 & \text{otherwise} \end{cases}$$

Then,  $V_{\mathcal{J}^+}(\phi) = 1$  and  $V_{\mathcal{J}^+}(\psi) = 0$ . Therefore,  $V_{\mathcal{J}^+}(\phi \rightarrow \psi) = 0$  ■

### 1.4.3 Examples of Semantic Consequence for Priest's Logic of Paradox

#### Rules that hold under LP:

- Modus Ponens II:  $\models_{\mathbf{LP}} P \rightarrow ((P \rightarrow Q) \rightarrow Q)$

*Proof.* AFC that  $V_{\mathcal{J}}(P \rightarrow ((P \rightarrow Q) \rightarrow Q)) = 0$ . Then,  $V_{\mathcal{J}}(P) = 1$  and  $V_{\mathcal{J}}((P \rightarrow Q) \rightarrow Q) = 0$ . So,  $V_{\mathcal{J}}(Q) = 0$  and  $V_{\mathcal{J}}(P \rightarrow Q) = 1$ . But if  $V_{\mathcal{J}}(Q) = 0$  and  $V_{\mathcal{J}}(P) = 1$ , then  $V_{\mathcal{J}}(P \rightarrow Q) = 0$ , a contradiction. ■

- Ex Falso Quod Libet II:  $\models_{\mathbf{LP}} (P \wedge \sim P) \rightarrow Q$

*Proof.* AFC that  $V_{\mathcal{J}}((P \wedge \sim P) \rightarrow Q) = 0$ . Then,  $V_{\mathcal{J}}(Q) = 0$  and  $V_{\mathcal{J}}(P \wedge \sim P) = 1$ . Hence  $V_{\mathcal{J}}(P) = 0$  and  $V_{\mathcal{J}}(P) = 1$ . Since  $V_{\mathcal{J}}(\cdot)$  is a function, this is a contradiction. ■

### Rules that do not hold under LP:

- Ex Falso Quod Libet I:  $\phi, \sim \phi \not\models_{\text{LP}} \psi$

*Proof.* Assume  $V_{\mathcal{J}}(P) \neq 0, V_{\mathcal{J}}(\sim P) \neq 0$ . And let  $V_{\mathcal{J}}(Q) = 0$ . If  $V_{\mathcal{J}}(P) = V_{\mathcal{J}}(\sim P) = \#$ , then we have found our counterexample. ■

- Modus Ponens I:  $\phi, \phi \rightarrow \psi \not\models_{\text{LP}} \psi$

*Proof.* Assume  $V_{\mathcal{J}}(P) \neq 0, V_{\mathcal{J}}(\sim P) \neq 0$ , and let  $V_{\mathcal{J}}(Q) = 0$ . If  $V_{\mathcal{J}}(P) = V_{\mathcal{J}}(\sim P) = \#$  while  $V_{\mathcal{J}}(Q) = 0$ . We have found our counterexample. ■

**Remark:** Notice that when the rule is expressed as a tautology in LP, the wff comes out as valid. But when it is expressed as a semantic consequence from a set, it fails to come out as valid.

## 1.5 Misc: Determinacy

One of the most promising applications of three-valued logic is with regards to vagueness. Then, we can think of 1 as representing determinate or a non-vague truth, 0 as representing determinate falsehood and # as representing indeterminacy (Middling Mary is rich).

In turn, we might want to represent indeterminacy in logic via a one-place operator  $\Delta$ . We read  $\Delta\phi$  as “definitely,  $\phi$ ”. Accordingly, we can extend the three-valued systems to incorporate  $\Delta$  via the truth table:

| $\phi$ | $\Delta\phi$ |
|--------|--------------|
| 1      | 1            |
| 0      | 0            |
| #      | 0            |

One serious shortcoming of this approach is that it disallows for “second-order vagueness” or plainly, vagueness about vagueness. It is vague, for instance, if the sentence “Millionaires who has donated a lot of money are rich” is vague – this claim is an example of being an indefinite case of definite richness. But  $\sim \Delta\Delta R \wedge \sim \Delta \sim \Delta R$  (reads, not definitely definitely rich and not definitely not definitely rich) still yields 0 regardless of what truth value we assign  $R$ . Furthermore,  $\Delta R \vee \sim \Delta R$  always yields 1 regardless of the truth value of  $R$ , but it is possible for the statement “Donald Trump is definitely rich or not definitely rich” to be neither true nor false.

According to Sider, the root of these problems is that the approach to vagueness thus far allows us to only represent three states for a given letter: determinate truth (1), indeterminacy (#) and determinate falsity (0). This leaves out determinate determinate falsity, indeterminate determinate falsity and so on.

## 1.6 Supervaluationist Semantics and Consequence

**Intuition:** Kit Fine launches an objection against Kleene due to penumbral connections – logical relations between indefinite sentences. Notice that intuitively, even if  $KV_{\mathcal{J}}(H) = \#$  and  $H \wedge H$  and  $H \vee H$  should also be #; but  $H \wedge \sim H$  should be 0 and  $H \vee \sim H$  should be 1. But on the Kleene scheme, when  $\mathcal{J}(H) = \#$ :

$$KV_{\mathcal{J}}(H \wedge \sim H) = KV_{\mathcal{J}}(H \wedge H) = \# = KV_{\mathcal{J}}(H \vee H) = KV_{\mathcal{J}}(H \vee \sim H)$$

Similar problems arise for any truth-functional three-valued scheme which evaluates  $\sim H$  as #, when  $H$  is assigned #.

Given a gappy, trivalent interpretation  $\mathcal{J}$  the truth-value of a sentence under  $\mathcal{J}$  is determined by considering the ways of removing gaps in  $\mathcal{J}$  to get a bivalent interpretation.

**Definition 1.18 (Refinement).** One trivalent-interpretation  $\mathcal{J}^+$  is said to be a refinement of another  $\mathcal{J}$  if  $\mathcal{J}^+$  preserves the classical truth values of  $\mathcal{J}$ , in that, for each sentence  $\alpha$ :

- If  $\mathcal{J}(\alpha) = 1$ , then  $\mathcal{J}^+(\alpha) = 1$
- If  $\mathcal{J}(\alpha) = 0$ , then  $\mathcal{J}^+(\alpha) = 0$

**Definition 1.19 (Precisification).** If a refinement  $\mathcal{J}^+$  of  $\mathcal{J}$  is bivalent, i.e. assigning no #s,  $\mathcal{J}^+$  is said to be a precisification of  $\mathcal{J}$ .

Truth – called supertruth – under  $\mathcal{J}$  is truth under all precisifications of  $\mathcal{J}$ .

**Definition 1.20 (Supervaluation).** Given a trivalent interpretation  $\mathcal{J}$ , the supervaluation of  $\mathcal{J}$  is the function  $SV_{\mathcal{J}}$  that assigns 0,1 or # to each wffs as follows:

$$SV_{\mathcal{J}}(\phi) = \begin{cases} 1 & \text{if } V_{\mathcal{J}^+}(\phi) = 1 \text{ for every precisification } \mathcal{J}^+ \text{ of } \mathcal{J} \\ 0 & \text{if } V_{\mathcal{J}^+}(\phi) = 0 \text{ for every precisification } \mathcal{J}^+ \text{ of } \mathcal{J} \\ \# & \text{otherwise} \end{cases}$$

**Example 1.2.** Let us assume that  $\mathcal{J}(P) = \mathcal{J}(Q) = \#$ . Then, we consider two precisifications. Then, let  $\mathcal{J}$  that assigns 1 whenever  $\mathcal{J}$  assigns # and let  $\mathcal{J}'$  assigns 0 whenever  $\mathcal{J}$  assigns #. Then, if we consider  $P \wedge Q$ , since under  $SV_{\mathcal{J}}(P \wedge Q) = 1$  and  $SV_{\mathcal{J}'}(P \wedge Q) = 0$ . So,  $SV_{\mathcal{J}}(P \wedge Q) = \#$ .

The main cost of supervaluationist semantics is the rejection of truth-functionality i.e. the determination of a complex wff's truth value solely from its logical connectives and not from the non-logical sentences themselves

**Definition 1.21 (Truth-functionality).** A connective is truth functional if and only if the truth value of the compound sentence cannot be changed by replacing a direct subsentence with another sentence having the same truth-value

**Proposition 1.6.** Supervaluationist semantics is not truth-functional

*Proof.* Consider some wff  $\phi \wedge \psi$ , where  $\mathcal{J}$  assigns # to both  $\phi$  and  $\psi$ . Let  $\phi := P$  and  $\psi := Q$ . Then, for every there is a precisification  $\mathcal{J}^+$  such that  $V_{\mathcal{J}^+}(P \wedge Q) = 1$  i.e. when assigning both the value 1, and a precisification  $\mathcal{J}^+$  such that  $V_{\mathcal{J}^+}(P \wedge Q) = 0$  i.e. when assigning both the value 0. So,  $SV_{\mathcal{J}}(P \wedge Q) = \#$ . Now, let  $\psi := \sim P$ . Notice that  $SV_{\mathcal{J}}(P \wedge \sim P) = 0$ , since under any precisification  $\mathcal{J}^+$ ,  $V_{\mathcal{J}^+}(P) \neq V_{\mathcal{J}^+}(\sim P)$  which implies both cannot take the truth value 1. Hence, the truth of  $\phi \wedge \psi$  depends on what  $\phi$  and  $\psi$  are. ■

## 2 Modal propositional logic: semantics

**Definition 2.1 (Truth-functionality).** A complex wff  $\phi$  is truth-functionl if its truth value is a function of its immediate subformulas

Modal propositional logic extends from propositional logic in capturing non-truth functional statements such as "it is necessary that  $\phi$ ". In other words, PL is a purely extensional language that does not consider the way in which we refer to objects.

**Definition 2.2 (Primitive Vocabulary for MPL).** The primitive vocabulary are the following:

- Connectives:  $\sim, \rightarrow, \Box$
- Sentence Letters:  $P, Q, R, P_1, Q_1, \dots$
- Parantheses:  $(, )$

**Definition 2.3 (Wffs in MPL).** We can construct MPL-wffs in the following way:

- Every sentence letter is a MPL-wff
- If  $\phi$  and  $\psi$  are MPL-wffs, then so are  $(\phi \rightarrow \psi), \sim \phi$  and  $\Box \phi$
- Only strings that can be shown to be MPL-wffs using (i) and (ii) are MPL-wffs

## 2.1 Modal Propositional Logic

We will skip over the discussion of Simplified Modal Propositional Logic. The main point is that we need to impose accessibility relations to get rid of unwarranted validities for various interpretations of modality.

**Definition 2.4 (MPL Truth Condition for necessity).**  $\Box\phi$  is true at  $w$  iff  $\phi$  is true at every world  $u$  accessible from  $w$ .

Informally, we take  $\Diamond\phi$  (it is possible that  $\phi$ ) to abbreviate  $\sim\Box\sim\phi$ .

**Remark:** MPL takes necessary truth to be truth in all accessible worlds. This is contingent on the accessibility relation imposed. Therefore, do not assume that some world is always accessible from itself (requires reflexivity). The SMPL truth-conditions is a special case whereby accessibility is universal on all worlds (i.e. every world is accessible from every world).

Now, we consider **Kripke Semantics** about possible worlds.

**Definition 2.5 (MPL Model/Kripke Semantics).** A MPL-model is a triple  $\langle \mathcal{W}, \mathcal{R}, \mathcal{I} \rangle$  where:

- $\mathcal{W}$  is a non-empty set i.e. the set of possible worlds
- $\mathcal{R}$  is a binary relation over  $\mathcal{W}$  i.e. the accessibility relation
- $\mathcal{I}$  is a two place function that assigns each sentence letter-world pair a truth-value, 1 or 0

**Remark:** We read  $\mathcal{R}wv$  as  $v$  is accessible from  $w$  or informally,  $w$  sees  $v$

**Definition 2.6 (MPL-Valuation).** Given an MPL-model,  $\mathcal{M} = \langle \mathcal{W}, \mathcal{R}, \mathcal{I} \rangle$ , the valuation for  $\mathcal{M}$ ,  $V_{\mathcal{M}}$  is the two place function that assigns 0 or 1 to each MPL-wff and meets the following four conditions, for each  $w \in \mathcal{W}$ :

- $V_{\mathcal{M}}(\alpha, w) = \mathcal{I}(\alpha, w)$  for each sentence letter  $\alpha$
- $V_{\mathcal{M}}(\phi \rightarrow \psi, w) = 1$  iff  $V_{\mathcal{M}}(\phi, w) = 0$  or  $V_{\mathcal{M}}(\psi, w) = 1$
- $V_{\mathcal{M}}(\sim\phi, w) = 1$  iff  $V_{\mathcal{M}}(\phi, w) = 0$
- $V_{\mathcal{M}}(\Box\phi, w) = 1$  iff  $V_{\mathcal{M}}(\phi, v) = 1$  for all  $v \in \mathcal{W}$  such that  $\mathcal{R}wv$

**S-Validity:** Different Modal systems impose different accessibility relations and thus result in different sets of tautologies and consequence relations.

Let  $S$  be one of K, D, T, B, S4 or S5. When an MPL model's accessibility relation  $\mathcal{R}$  meets the associated condition, it is called an  $S$ -model.

**Definition 2.7 (MPL-validity).** Let  $\phi$  be an MPL-wff.

- $\phi$  is valid in an MPL-model  $\mathcal{M} = \langle \mathcal{W}, \mathcal{R}, \mathcal{I} \rangle$  iff  $V_{\mathcal{M}}(\phi, w) = 1$  for every  $w \in \mathcal{W}$
- $\phi$  is valid in  $S$  i.e.  $\models_S \phi$  iff  $\phi$  is valid in every  $S$ -model

**Establishing Validity:** To show  $\models_S \phi$  we can simply assume that  $V_{\mathcal{M}}(\phi, w) = 0$  and show that this results in a contradiction, given the condition on  $\mathcal{R}$  imposed by the model  $S$ .

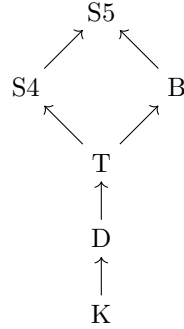
**Establishing Invalidity:** To establish the  $S$ -invalidity of  $\phi$  we need to specify a countermodel – i.e. an  $S$ -model  $\langle \mathcal{W}, \mathcal{R}, \mathcal{I} \rangle$  such that  $V_{\mathcal{M}}(\phi, w) = 0$  for some  $w \in \mathcal{W}$  that satisfies the accessibility relations on  $\mathcal{R}$ .

**Sider's Method for Generating Counterexamples:**

**Relationship between  $S$ -models.** We notice that certain models have relations that are a subset of one another. For instance, the T-model which imposes reflexivity is a stricter condition than a serial relation imposed by D. Generally, the **more accessibility restrictions a model has, the more valid formulas they have** because it is harder to construct a countermodel without violating these restrictions. In general, the hierarchy is as follows:

Table 1: Different Modal Systems and Accessibility Relations

| System | Conditions on $\mathcal{R}$                                                                                                   | Definition                                                                                                                                                                                            |
|--------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| K      | None                                                                                                                          | None                                                                                                                                                                                                  |
| D      | $\mathcal{R}$ is <b>serial</b> on $\mathcal{W}$                                                                               | for each $w \in \mathcal{W}$ , there is some $u$ s.t. $\mathcal{R}wu$<br>i.e. each world has to see another world or a successor relation                                                             |
| T      | $\mathcal{R}$ is <b>reflexive</b> on $\mathcal{W}$                                                                            | for each $w \in \mathcal{W}$ , $\mathcal{R}ww$ i.e. each world sees itself                                                                                                                            |
| B      | $\mathcal{R}$ is <b>reflexive</b> on $\mathcal{W}$<br>$\mathcal{R}$ is <b>symmetric</b>                                       | for each $w \in \mathcal{W}$ , $\mathcal{R}ww$ and for each $w, v$<br>$\mathcal{R}wv$ implies $\mathcal{R}vw$                                                                                         |
| S4     | $\mathcal{R}$ is <b>reflexive</b> on $\mathcal{W}$<br>$\mathcal{R}$ is <b>transitive</b>                                      | for each $w \in \mathcal{W}$ , $\mathcal{R}ww$<br>for each $w, v, u$ , $\mathcal{R}wv$ and $\mathcal{R}vu$ jointly imply $\mathcal{R}wu$                                                              |
| S5     | $\mathcal{R}$ is <b>reflexive</b> on $\mathcal{W}$<br>$\mathcal{R}$ is <b>symmetric</b><br>$\mathcal{R}$ is <b>transitive</b> | for each $w \in \mathcal{W}$ , $\mathcal{R}ww$<br>for each $w, v$ , $\mathcal{R}wv$ implies $\mathcal{R}vw$<br>for each $w, v, u$ , $\mathcal{R}wv$ and $\mathcal{R}vu$ jointly imply $\mathcal{R}wu$ |



whereby the arrow represents the preservation of validity (if a wff is valid in K it is valid in D). For instance, all formulas valid in T are valid in S4 but not all models valid in S4 are valid in T.

**Example 2.1.** Suppose we know that  $\models_T (\Diamond\Box(P \rightarrow Q) \wedge \Box P) \rightarrow \Diamond Q$ . Now, we want to show that  $\models_{S4} (\Diamond\Box(P \rightarrow Q) \wedge \Box P) \rightarrow \Diamond Q$ . We can simply say that since to be S4-valid is to be valid in every S4 model, and since every S4-model is a T-model. So since the formula is valid in all T models, it is valid in all S4 models.

Therefore, another way to put the relationships between models is that the number of models that are permissible (satisfying  $\mathcal{R}$ ) is necessarily less as the restrictions increase. So, since T's accessibility conditions are a subset of S4, the number of models in S4 are a subset of T.

### Some Theorems about the Semantics of modal systems:

**Proposition 2.1.** Let  $O$  be some nonempty string of modalities (e.g.  $\Box\Box\Diamond\Box\dots\phi$ ). We say that  $O_1 \equiv_S O_2$  iff  $\models_S O_1\phi \leftrightarrow O_2\phi$  i.e. that  $O_1$  and  $O_2$  express the same modalities. Then,

1. If  $O_1 \equiv_S O_2$ , then  $OO_1 \equiv_S OO_2$
2. If  $O_1 \equiv_S O_2$ , then  $O_1O \equiv_S O_2O$

Before we prove this, we assume the following lemma

**Lemma 2.1.** Let  $\chi(\phi)$  be some substitution of  $\phi$ . If  $\models_S \phi_1 \leftrightarrow \phi_2$ , then  $\models_S \chi(\phi_1) \leftrightarrow \chi(\phi_2)$

Now, we can prove Proposition 2.1

*Proof.* For the first claim, assume  $O_1 \equiv_S O_2$ , which means  $\models_S O_1\phi \leftrightarrow O_2\phi$ . Then, let  $\chi(\psi)$  be some substitution s.t.  $\chi(\psi) := O\psi$ . Then,  $OO_1 \equiv_S OO_2$ .

For the second claim, assume  $O_1 \equiv_S O_2$  for all  $\phi$  (since it is a tautology), which means  $\models_S O_1\phi \leftrightarrow O_2\phi$  for all  $\phi$ . We want to show  $\models_S O_1O\psi \leftrightarrow O_2O\psi$  for all  $\psi$ . Then fix  $\phi := O\psi$ , which implies  $O_1O \equiv_S O_2O$  ■

**Proposition 2.2.** There are infinitely many modalities in B

*Proof.* A constructive proof is hard. We will try a proof by contradiction ■

**Proposition 2.3.** There are exactly two modalities in S5

*Proof.* We prove this by induction on the length of  $O$  and show that for any string  $O$  longer than length 1 reduces to length 1. See W2 worksheet for the proof. ■

**Proposition 2.4.** There are exactly 6 modalities in S4

*Proof.* We do a sketch of the proof. Basically, it is just a brute force method where we show the following 6 modalities are distinct

1. When  $L(O_i) = 1$ :  $\Box\phi, \Diamond\phi$
2. When  $L(O_i) = 2$ :  $\Box\Diamond\phi, \Diamond\Box\phi$
3. When  $L(O_i) = 3$ :  $\Diamond\Box\Diamond\phi, \Box\Diamond\Box\phi$

Then, we want to show that for any  $O$  such that  $L(O) > 3$ ,  $\models_{S4} O\phi \leftrightarrow \psi$ , where  $\psi$  is defined as one of the 6 modalities above. ■

### 2.1.1 Semantic Proofs and Counterexamples for K, D, T, B, S4 and S5

We will do one semantic proof for validity and one Sider-style counterexample for each system.

#### Semantic Proofs and Counterexamples for K.

**Example 2.2 (Validity for K).** Show that  $\models_K \Box(\phi \rightarrow \psi) \rightarrow (\Box\phi \rightarrow \Box\psi)$

*Proof.* Assume for reductio that there is some K-model  $\mathcal{M}$  such that  $V_{\mathcal{M}}(\Box(\phi \rightarrow \psi) \rightarrow (\Box\phi \rightarrow \Box\psi), w) = 0$ . Then

$$V_{\mathcal{M}}(\Box(\phi \rightarrow \psi), w) = 1 \tag{1}$$

and

$$V_{\mathcal{M}}(\Box\phi \rightarrow \Box\psi, w) = 0 \tag{2}$$

From (1),  $V_{\mathcal{M}}(\phi \rightarrow \psi, w) = 1$  for all  $v \in \mathcal{W}$  s.t.  $\mathcal{R}wv$ . Hence,  $V_{\mathcal{M}}(\phi, v) = 0$  or  $V_{\mathcal{M}}(\psi, v) = 1$  for all  $v \in \mathcal{W}$  s.t.  $\mathcal{R}wv$ .

From (2),  $V_{\mathcal{M}}(\Box\phi, w) = 1$  and  $V_{\mathcal{M}}(\Box\psi, w) = 0$ . Hence,  $V_{\mathcal{M}}(\phi, u) = 1$  for all  $u \in \mathcal{W}$  s.t.  $\mathcal{R}wu$  and  $V_{\mathcal{M}}(\psi, u) = 0$  for some  $u \in \mathcal{W}$  s.t.  $\mathcal{R}wu$ .

Since for all worlds  $v \in \mathcal{W}$  such that  $\mathcal{R}wv$ , it is the case that  $V_{\mathcal{M}}(\phi, v) = 0$  or  $V_{\mathcal{M}}(\psi, v) = 1$ , we first notice that since  $V_{\mathcal{M}}(\phi, u) = 1$  for all  $u \in \mathcal{W}$  s.t.  $\mathcal{R}wu$ , it cannot be the case that  $V_{\mathcal{M}}(\phi, v) = 0$  for any world  $v \in \mathcal{W}$ , since we can just choose  $u = v$ . Then, this means that  $V_{\mathcal{M}}(\psi, v) = 1$  for all  $v \in \mathcal{W}$  s.t.  $\mathcal{R}wv$ . But this cannot hold either, since we can choose a  $v = u$  where  $V_{\mathcal{M}}(\psi, u) = 0$ . Hence, there is a contradiction. ■

**Example 2.3 (Invalidity for K).**  $\not\models_K \Box\phi \rightarrow \Diamond\phi$

*Proof.* We must construct a model  $\mathcal{M}$  for which  $V_{\mathcal{M}}(\Box\phi \rightarrow \Diamond\phi, w) = 0$  for some  $w$ . So,  $V_{\mathcal{M}}(\Box\phi, w) = 1$ , hence  $V_{\mathcal{M}}(\phi, v) = 1$  for all  $v$  s.t.  $\mathcal{R}wv$ .  $V_{\mathcal{M}}(\Diamond\phi, w) = 0$ , hence  $V_{\mathcal{M}}(\phi, u) = 0$  for all  $u$  s.t.  $\mathcal{R}wu$ . Our illustration is

$$r$$

|        |        |               |            |        |
|--------|--------|---------------|------------|--------|
| $\Box$ | $\phi$ | $\rightarrow$ | $\Diamond$ | $\phi$ |
| 1      | 1      | 0             | 0          | 1      |

Our countermodel is

$$\begin{aligned}\mathcal{W} &= \{r\} \\ \mathcal{R} &= \emptyset \\ \mathcal{I}(P, w) &= 1\end{aligned}$$

■

**Remark:** A full counterexample needs to let  $\phi$  be some sentence letter.

### Semantic Proofs and Counterexamples for D.

**Example 2.4 (Validity in D).** Show that  $\models_D \Box\phi \rightarrow \Diamond\phi$

*Proof.* Assume for reductio that there is some  $\mathcal{M}$  s.t.  $V_{\mathcal{M}}(\Box\phi \rightarrow \Diamond\phi, w) = 0$  for some  $w$ . Then,  $V_{\mathcal{M}}(\Box\phi, w) = 1$  and  $V_{\mathcal{M}}(\Diamond\phi, w) = 0$ . This implies

$$V_{\mathcal{M}}(\phi, v) = 1 \text{ for all } v \text{ s.t. } \mathcal{R}wv \quad (1)$$

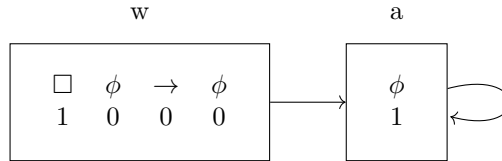
and

$$V_{\mathcal{M}}(\phi, u) = 0 \text{ for all } u \text{ s.t. } \mathcal{R}wu \quad (2)$$

Since  $\mathcal{R}$  is serial on  $\mathcal{W}$ , we know that for any  $w$ , there is at least one world, call this  $u'$ , accessible from  $w$ . From (1), (2) and seriality, there is a  $u'$  s.t.  $\mathcal{R}wu'$  and  $V_{\mathcal{M}}(\phi, u') = 1$  and  $V_{\mathcal{M}}(\phi, u') = 0$ , a contradiction. ■

**Example 2.5 (Invalidity in D).** Show that  $\not\models_D \Box\phi \rightarrow \phi$

*Proof.* Our illustrated countermodel is



The countermodel specification is

$$\begin{aligned}\mathcal{W} &= \{w, a\} \\ \mathcal{R} &= \{\langle w, a \rangle\} \\ V_{\mathcal{M}}(P, r) &= 0 \\ V_{\mathcal{M}}(P, a) &= 1\end{aligned}$$

■

### Semantic Proofs and Counterexamples for T.

**Example 2.6 (Validity in T).** Show that  $\models_T \Box\phi \rightarrow \phi$



*Proof.* Assume for reductio that there is a model  $\mathcal{M}$  s.t.  $V_{\mathcal{M}}(\Box\phi \rightarrow \phi, w) = 0$  for some  $w \in \mathcal{W}$ . Then,

$$V_{\mathcal{M}}(\Box\phi, w) = 1 \text{ for that world } w \quad (1)$$

and

$$V_{\mathcal{M}}(\phi, w) = 0 \text{ for that world } w \quad (2)$$

This means that  $V_{\mathcal{M}}(\phi, v) = 1$  for all  $\mathcal{R}wv$ . Due to reflexivity,  $\mathcal{R}ww$  for all  $w \in \mathcal{W}$ . Hence, from (1), we know that  $V_{\mathcal{M}}(\phi, w) = 1$ . From (2),  $V_{\mathcal{M}}(\phi, w) = 0$ , a contradiction. ■

### Semantic Proofs and Counterexamples for B.

**Example 2.7 (Validity in B).** Show that  $\models_B \phi \rightarrow \Box\Diamond\phi$

*Proof.* AFC that there is a model  $\mathcal{M}$  s.t.  $V_{\mathcal{M}}(\phi \rightarrow \Box\Diamond\phi, w) = 0$  for some  $w \in \mathcal{W}$ . Then

$$V_{\mathcal{M}}(\phi, w) = 1 \quad (1)$$

and

$$V_{\mathcal{M}}(\Box\Diamond\phi, w) = 0 \quad (2)$$

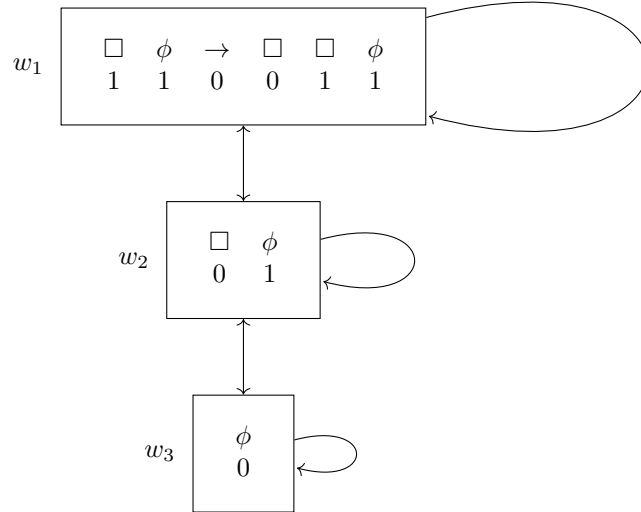
From (2),  $V_{\mathcal{M}}(\Diamond\phi, v) = 0$  for some  $v \in \mathcal{W}$  s.t.  $\mathcal{R}wv$ . Hence,  $V_{\mathcal{M}}(\Diamond\phi, u) = 0$  for all  $u$  s.t.  $\mathcal{R}vu$ . Since  $\mathcal{R}$  is symmetric, we know that if  $\mathcal{R}wv$  then  $\mathcal{R}vw$ . Thus we can conclude that,  $V_{\mathcal{M}}(\Diamond\phi, w) = 0$  by choosing  $u = w$ , contradicting (1) ■

**Example 2.8 (Invalidity in B I).** Show that  $\not\models_B \Box\phi \rightarrow \Box\Box\phi$

*Proof.* We want to find a model  $\mathcal{M}$  s.t.  $V_{\mathcal{M}}(\Box\phi \rightarrow \Box\Box\phi, w) = 0$  Then,

- $V_{\mathcal{M}}(\Box\phi, w) = 1$  implying  $V_{\mathcal{M}}(\phi, v) = 1$  for all  $v$  s.t.  $\mathcal{R}wv$
- $V_{\mathcal{M}}(\Box\Box\phi, w) = 0$ . Then,  $V_{\mathcal{M}}(\Box\phi, u) = 0$  for some  $u$  st.  $\mathcal{R}wu$ , and thus  $V_{\mathcal{M}}(\phi, u') = 0$  for some  $u'$  s.t.  $\mathcal{R}uu'$ .

Our illustrated countermodel is Hence, we illustrate the countermodel as follows:



Our official countermodel is

$$\begin{aligned} \mathcal{W} &= \{w_1, w_2, w_3\} \\ \mathcal{R} &= \{\langle w_1, w_1 \rangle, \langle w_2, w_2 \rangle, \langle w_3, w_3 \rangle, \langle w_1, w_2 \rangle, \langle w_2, w_1 \rangle, \langle w_2, w_3 \rangle, \langle w_3, w_2 \rangle\} \\ V_{\mathcal{M}}(P, w_1) &= 1, V_{\mathcal{M}}(P, w_2) = 1, V_{\mathcal{M}}(P, w_3) = 0 \end{aligned}$$

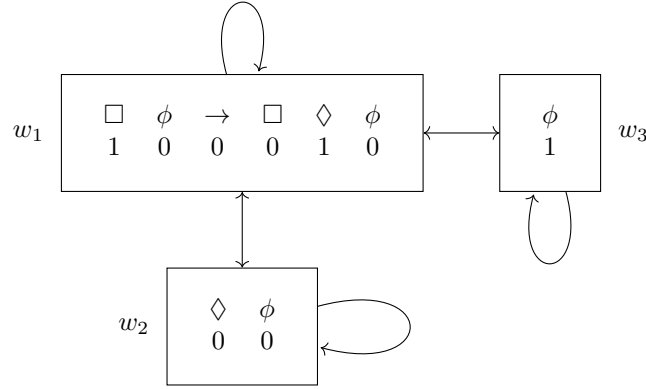
■

**Example 2.9 (Invalidity in B II).** Show that  $\not\models_B \Diamond\phi \rightarrow \Box\Diamond\phi$

*Proof.* Our countermodel is some  $\mathcal{M}$  s.t.  $V_{\mathcal{M}}(\Diamond\phi \rightarrow \Box\Diamond\phi, w) = 0$ . So,

- $V_{\mathcal{M}}(\Diamond\phi, w) = 1$  and  $V_{\mathcal{M}}(\phi, v) = 1$  for some  $v$  s.t.  $\mathcal{R}wv$
- $V_{\mathcal{M}}(\Box\Diamond\phi, w) = 0$ . So,
  - $V_{\mathcal{M}}(\Diamond\phi, u) = 0$  for some  $u$  s.t.  $\mathcal{R}wu$
  - $V_{\mathcal{M}}(\phi, u') = 0$  for all  $u'$  s.t.  $\mathcal{R}uu$

Hence, we illustrate the countermodel as follows:



Our official countermodel is

$$\begin{aligned} \mathcal{W} &= \{w_1, w_2, w_3\} \\ \mathcal{R} &= \{\langle w_1, w_1 \rangle, \langle w_2, w_2 \rangle, \langle w_3, w_3 \rangle, \langle w_1, w_2 \rangle, \langle w_2, w_1 \rangle, \langle w_1, w_3 \rangle, \langle w_3, w_1 \rangle\} \\ V_{\mathcal{M}}(P, w_1) &= 0, V_{\mathcal{M}}(P, w_2) = 0, V_{\mathcal{M}}(P, w_3) = 1 \end{aligned}$$

■

## Semantic Proofs and Counterexamples for S4.

**Example 2.10 (Validity in S4).** Show that  $\models_{S4} \Box\phi \rightarrow \Box\Box\phi$

*Proof.* AFC that  $\mathcal{M}$  is such that  $V_{\mathcal{M}}(\Box\phi \rightarrow \Box\Box\phi, w) = 0$ . Then,

$$V_{\mathcal{M}}(\Box\phi, w) = 1 \tag{1}$$

and

$$V_{\mathcal{M}}(\Box\Box\phi, w) = 0 \tag{2}$$

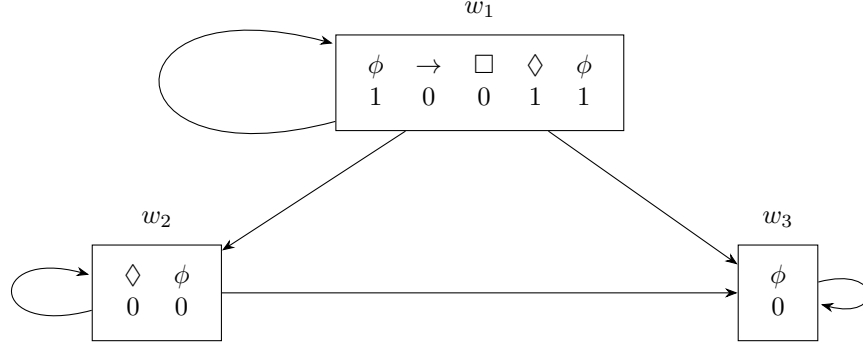
From (1),  $V_{\mathcal{M}}(\phi, v) = 1$  for all  $v \in \mathcal{W}$  s.t.  $\mathcal{R}wv$ .

From (2),  $V_{\mathcal{M}}(\Box\phi, u) = 0$  for some  $u$  s.t.  $\mathcal{R}wu$ . Thus,  $V_{\mathcal{M}}(\phi, u') = 0$  for some  $u'$  s.t.  $\mathcal{R}uu'$ .

Due to transitivity,  $\mathcal{R}wu$  and  $\mathcal{R}uu'$  imply  $\mathcal{R}wu'$ . Then, since  $V_{\mathcal{M}}(\phi, v) = 1$  for all  $v \in \mathcal{W}$  s.t.  $\mathcal{R}wv$ , we can pick  $v = u'$  s.t.  $V_{\mathcal{M}}(\phi, u') = 1$  since  $\mathcal{R}wu'$  for that  $u'$ , contradicting  $V_{\mathcal{M}}(\phi, u') = 0$ . ■

**Example 2.11 (Invalidity in S4 I).** Show that  $\not\models_{S4} \phi \rightarrow \Box\Diamond\phi$

*Proof.* The countermodel diagram is as follows



The official counter model is

$$\begin{aligned}\mathcal{W} &= \{w_1, w_2, w_3\} \\ \mathcal{R} &= \{\langle w_1, w_1 \rangle, \langle w_1, w_2 \rangle, \langle w_2, w_2 \rangle, \langle w_2, w_3 \rangle, \langle w_3, w_3 \rangle\} \\ V_{\mathcal{M}}(P, w_1) &= 1, V_{\mathcal{M}}(P, w_2) = 0, V_{\mathcal{M}}(P, w_3) = 0\end{aligned}$$

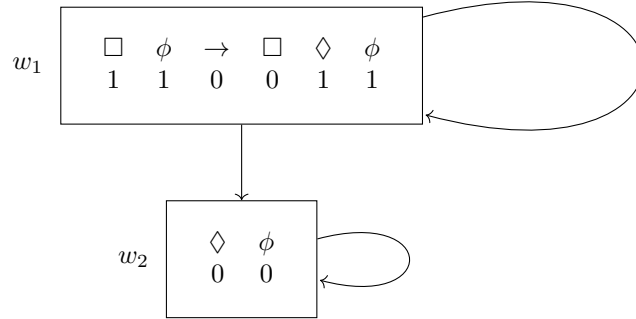
■

**Example 2.12 (Invalidity in S4 II).** Show that  $\not\models_{S4} \Diamond\phi \rightarrow \Box\Diamond\phi$

*Proof.* Our countermodel is some  $\mathcal{M}$  s.t.  $V_{\mathcal{M}}(\Diamond\phi \rightarrow \Box\Diamond\phi, w) = 0$ . So,

- $V_{\mathcal{M}}(\Diamond\phi, w) = 1$  and  $V_{\mathcal{M}}(\phi, v) = 1$  for some  $v$  s.t.  $\mathcal{R}wv$
- $V_{\mathcal{M}}(\Box\Diamond\phi, w) = 0$ . So,
  - $V_{\mathcal{M}}(\Diamond\phi, u) = 0$  for some  $u$  s.t.  $\mathcal{R}wu$
  - $V_{\mathcal{M}}(\phi, u') = 0$  for all  $u'$  s.t.  $\mathcal{R}uu'$

Hence, we illustrate the countermodel as follows:



Our countermodel is officially:

$$\begin{aligned}\mathcal{W} &= \{w_1, w_2\} \\ \mathcal{R} &= \{\langle w_1, w_1 \rangle, \langle w_2, w_2 \rangle, \langle w_1, w_2 \rangle\} \\ V_{\mathcal{M}}(P, w_1) &= 0, V_{\mathcal{M}}(P, w_2) = 0\end{aligned}$$

■

**Semantic Proofs and Counterexamples for S5.**

**Example 2.13 (Validity in S5).** Show that  $\models_{S5} \Diamond\phi \rightarrow \Box\Diamond\phi$

*Proof.* AFC that there is an S5-model  $\mathcal{M}$  s.t.  $V_{\mathcal{M}}(\Diamond\phi \rightarrow \Box\Diamond\phi, w) = 0$  for some  $w$ . Then

$$V_{\mathcal{M}}(\Diamond\phi, w) = 1 \quad (1)$$

and

$$V_{\mathcal{M}}(\Box\Diamond\phi, w) = 0 \quad (2)$$

From (1), there is some  $v$  s.t.  $\mathcal{R}wv$  and  $V_{\mathcal{M}}(\phi, v) = 1$ .

From (2),  $V_{\mathcal{M}}(\Diamond\phi, u) = 0$  for some  $u$  s.t.  $\mathcal{R}wu$ . Hence,  $V_{\mathcal{M}}(\phi, u') = 0$  for all  $u'$  s.t.  $\mathcal{R}uu'$ .

By symmetry,  $\mathcal{R}wu$  implies  $\mathcal{R}uw$ . By transitivity,  $\mathcal{R}wv$  and  $\mathcal{R}uw$  imply  $\mathcal{R}uv$ . Hence, we are permitted to choose  $u' = v$  such that  $V_{\mathcal{M}}(\phi, v) = 0$ , a contradiction to (1)  $\blacksquare$

## 2.2 Interpretations of MPL: Deontic, Epistemic, Temporal Modalities

Metaphysical necessity is not the only way to understand the  $\Box$  operator. We can give a systematic account of a wide range of intensional connectives.

| Modality     | $\Box\phi$ [symbol and gloss]                                                                             | $\Diamond\phi$ [symbol and gloss]                                                      | Member of $\mathcal{W}$ represent                                      |
|--------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Metaphysical | $\Box\phi$ $\phi$ is necessary                                                                            | $\Diamond\phi$ $\phi$ is possible                                                      | metaphysically possible worlds, ways the world could be                |
| Temporal     | $H\phi$ $\phi$ has always been (at all past times)<br>$G\phi$ $\phi$ will always be (at all future times) | $P\phi$ $\phi$ was (at some past time)<br>$F\phi$ $\phi$ will be (at some future time) | times                                                                  |
| Epistemic    | $K\phi$ $S$ knows that $\phi$ , $\phi$ must be (given what $S$ knows)                                     | $\sim K \sim \phi$ $\phi$ could be (for all $S$ knows)                                 | epistemically possible worlds, worlds consistent with what $S$ knows   |
| Deontic      | $O\phi$ $\phi$ is obligatory                                                                              | $P\phi$ $\phi$ is permissible                                                          | permissible worlds, ways the world may be (given the ambient morality) |

### 2.2.1 Deontic Logic

Deontic logic is the study of normative notions. We might interpret  $O\phi$  to mean ‘ $\phi$  is obligatory’ i.e.  $S$  ought to see that  $\phi$ . We then think of  $\mathcal{R}$  as being a relation of “deontic accessibility” and we think of  $v$  as accessible from  $w$  if the goings-on in  $v$  are permitted, given the operative norms in  $w$ . That is,  $\mathcal{R}wv$  iff everything that, in  $w$ , ought to be true is in fact true in  $v$  (thus,  $v$  violates nothing in  $w$  that is mandatory). This is why we can think of  $O$  as representing  $\Box$ .

**Formal Properties on  $\mathcal{R}$ :** One common imposition is seriality since there must always be *some* possibility that morality permits; from every world there is at least one deontically accessible world (perhaps the ideal world). Reflexivity is inappropriate since things that ought to be nevertheless sometimes are not.

If seriality is the sole constraint on  $\mathcal{R}$ , the resulting logic for  $O$  is the logic D.

**Misc Considerations:** Should we impose further constraints on the deontic accessibility relation? For example

- The principle (U) for utopia  $O(O\phi \rightarrow \phi)$  i.e. it ought to be the case that everything that ought to be true is true. If so, then we should find an accessibility relation that makes it valid.

- And can the O operator represent all aspects of deontic reasoning? Suppose, “if you kill the victim, you ought to kill him quickly” can be represented as either  $K \rightarrow OQ$  or  $O(K \rightarrow Q)$ . But neither seems right. Against the first: suppose you do actually kill the victim. Then it follows that one of your obligations is to do the following: kill the victim quickly. But you ought not to kill him at all! Against the second: if it is the right representation of “if you kill the victim, you ought to kill him quickly”, then the right representation of “if you kill the victim, you ought to kill him slowly” should be  $O(K \rightarrow S)$ . But that follows from  $O \sim K$  (given just a K modal logic for O), and “you ought not to kill the victim” certainly does not imply “if you kill the victim, you ought to kill him slowly”

### 2.2.2 Epistemic Logic

In epistemic logic, we read  $\Box$  as meaning “it is known by S to be the case that”. Therefore,  $K\phi$  just means that  $\phi$  is known. Hence, we interpret  $\mathcal{R}$  as epistemic accessibility:  $\mathcal{R}wv$  iff everything known in  $w$  is true in  $v$ .

**Formal Properties on  $\mathcal{R}$ :** We want epistemic accessibility to be reflexive since we think that knowledge implies truth i.e.  $K\phi \rightarrow \phi$  is a valid principle of epistemic logic. S4 and S5 logics are controversial, since S4 would license a controversial principle known as the “KK” principle of positive introspection:  $K\phi \rightarrow KK\phi$ . From the S5-axiom schema, we get negative introspection:  $\sim K\phi \rightarrow K \sim K\phi$ .

### 2.2.3 Tense Logic

Tense logic helps us describe the metaphysics of time. Because time has ‘earlier than’ or ‘later than’ relations, we have two  $\Box$  like operators, G and H. We also use a temporal ordering  $\leq$  to replace  $\mathcal{R}$  (or simply  $<$  if we mean non-inclusive earlier or later than).

**Remark:** We do not yet impose any accessibility constraints on  $\leq$ . This means that we do not assume that  $\leq$  is actually transitive or reflexive.

We have the following operators for tense logic:

- $G\phi$ : it is always going to be the case that  $\phi$
- $H\phi$ : it always has been the case that  $\phi$
- $F\phi$ : it will at some point in the future be the case that  $\phi$
- $P\phi$ : it was at some point in the past be the case that  $\phi$

with the following semantics:

$$\begin{aligned} V_{\mathcal{M}}(G\phi, t) &= 1 \text{ iff for every } t' \text{ such that } t \leq t', V_{\mathcal{M}}(\phi, t') = 1 \\ V_{\mathcal{M}}(H\phi, t) &= 1 \text{ iff for every } t' \text{ such that } t' \leq t, V_{\mathcal{M}}(\phi, t') = 1 \\ V_{\mathcal{M}}(F\phi, t) &= 1 \text{ iff for some } t' \text{ such that } t \leq t', V_{\mathcal{M}}(\phi, t') = 1 \\ V_{\mathcal{M}}(P\phi, t) &= 1 \text{ iff for some } t' \text{ such that } t' \leq t, V_{\mathcal{M}}(\phi, t') = 1 \end{aligned}$$

we call any such model  $\mathcal{M} = \langle \mathcal{T}, \leq, \mathcal{I} \rangle$  a PTL model (Priorean Tense Logic).

**Remark:** The motivation for this logic is from Arthur Prior against Quine’s formalisation of the metaphysics of time via classical predicate logic.

**Formal Properties on  $\leq$ :** We want  $\leq$  to be at least transitive. If we want only a linear conception of time, we want every point in time to be comparable or some sort of ‘connectivity’.

**Definition 2.8 (Connectivity).** Let  $R$  be any binary relation over  $A$ .

- $R$  is strongly connected in  $A$  iff for every  $u, v \in A$ , either  $Ruv$  or  $Rvu$
- $R$  is weakly connected iff for every  $w, v, v'$ , if: either  $Ruv$  and  $Ruv'$ , or  $Rvu$  and  $Rv'u$ , then: either  $Rvv'$  or  $Rv'v$

So, we might require  $\leq$  be strongly connected, or alternatively, weakly connected. We disallow “incomparable pairs of times”. Usually, we want to impose a linear order on  $\leq$ .

**Definition 2.9 (Linear Order).** A linear (or total) order is a binary relation  $\leq$  on some set  $X$ , which satisfies the following for all  $a, b, c \in X$ :

1. Reflexive:  $a \leq a$
2. Anti-symmetry: If  $a \leq b$  and  $b \leq a$ , then  $a = b$
3. Transitivity: If  $a \leq b$  and  $b \leq c$ , then  $a \leq c$
4. Strong connectivity:  $a \leq b$  or  $b \leq a$

**Example 2.14.** Determine whether each of the following is valid in Priorean tense logic (PTL). Provide an informal semantic argument or counterexample in each case:

1.  $\phi \rightarrow \text{HF}\phi$
2.  $\text{PP}\phi \rightarrow \text{P}\phi$
3.  $(\text{F}\phi \wedge \text{F}\psi) \rightarrow (\text{F}(\phi \wedge \text{F}\psi) \vee \text{F}(\psi \wedge \text{F}\phi) \vee \text{F}(\phi \wedge \psi))$

For each of the invalid formula, propose a plausible further constraint on  $\leq$  that renders it valid. Demonstrate this with a semantic argument.

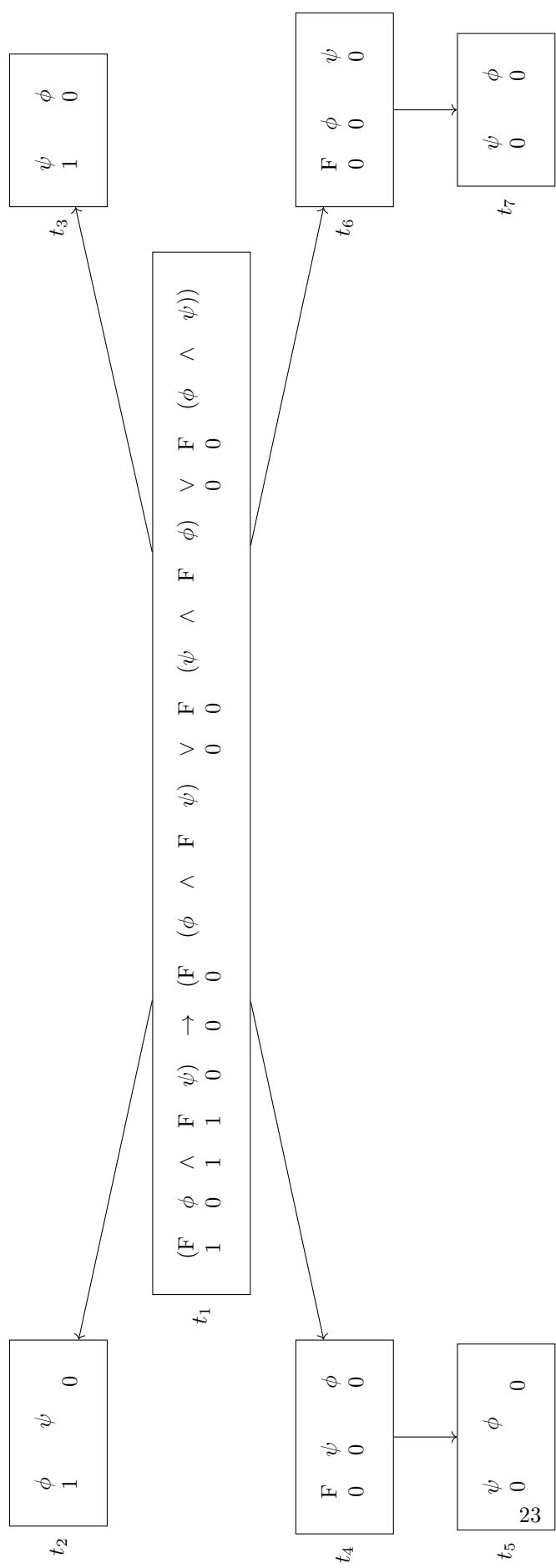
**Solution.** We just do case (3).

First Part: Let there be an  $\mathcal{M}$  s.t.  $V_{\mathcal{M}}((\text{F}\phi \wedge \text{F}\psi) \rightarrow (\text{F}(\phi \wedge \text{F}\psi) \vee \text{F}(\psi \wedge \text{F}\phi) \vee \text{F}(\phi \wedge \psi)), t_1) = 0$ . Then,

1.  $V_{\mathcal{M}}(\text{F}\phi \wedge \text{F}\psi, t_1) = 1$ 
  - (a)  $V_{\mathcal{M}}(\text{F}\phi, t_1) = 1$  and  $V_{\mathcal{M}}(\text{F}\psi, t_1) = 1$
  - (b)  $V_{\mathcal{M}}(\phi, t_2) = 1$  for some  $t_2 \geq t_1$
  - (c)  $V_{\mathcal{M}}(\psi, t_3) = 1$  for some  $t_3 \geq t_1$
2. Then,  $V_{\mathcal{M}}(\text{F}(\phi \wedge \text{F}\psi) \vee \text{F}(\psi \wedge \text{F}\phi) \vee \text{F}(\phi \wedge \psi), t_1) = 0$  implies either one of the three hold
  - (a)  $V_{\mathcal{M}}(\text{F}(\phi \wedge \text{F}\psi), t_1) = 0$
  - (b)  $V_{\mathcal{M}}(\text{F}(\psi \wedge \text{F}\phi), t_1) = 0$
  - (c)  $V_{\mathcal{M}}(\text{F}(\phi \wedge \psi), t_1) = 0$
3. From 2(a),  $V_{\mathcal{M}}(\phi \wedge \text{F}\psi, t_4) = 0$  for all  $t_4 \geq t_1$ . So, for all  $t_4 \geq t_1$ , we have (either  $V_{\mathcal{M}}(\phi, t_4) = 0$  or  $[V_{\mathcal{M}}(\text{F}\psi, t_4) = 0$ , which implies  $V_{\mathcal{M}}(\psi, t_5) = 0$  for all  $t_5 \geq t_4]$ )
4. From 2(b),  $V_{\mathcal{M}}(\psi \wedge \text{F}\phi, t_6) = 0$  for all  $t_6 \geq t_1$ . So for all  $t_6 \geq t_1$ , we have (either  $V_{\mathcal{M}}(\psi, t_6) = 0$  or  $[V_{\mathcal{M}}(\text{F}\phi, t_6) = 0 \implies V_{\mathcal{M}}(\phi, t_7) = 0$  for all  $t_7 \geq t_6]$ )
5. From 2(c),  $V_{\mathcal{M}}(\phi \wedge \psi, t_8) = 0$  for all  $t_8 \geq t_1$ . Then for all  $t_8 \geq t_1$ , either  $(V_{\mathcal{M}}(\phi, t_8) = 0$  or  $V_{\mathcal{M}}(\psi, t_8) = 0)$

TLDR:

- $\phi, \psi$  must be true in one world accessible from  $t_1$  (but not the same world)
- $\text{F}\phi, \text{F}\psi$  must be true in  $t_1$  but false in all worlds accessible from it
- $\phi, \psi$  can never be true in worlds accessible from worlds where  $\text{F}\phi, \text{F}\psi$  are false
- $\phi, \psi$  can never be true in a world accessible from  $t_1$



Our official countermodel is

$$\begin{aligned}\mathcal{W} &= \{t_1, t_2, t_3, t_4, t_5, t_6, t_7\} \\ \mathcal{R} &= \{\langle t_1, t_2 \rangle, \langle t_1, t_3 \rangle, \langle t_1, t_4 \rangle, \langle t_1, t_6 \rangle, \langle t_4, t_5 \rangle, \langle t_6, t_7 \rangle\} \\ V_{\mathcal{M}}(P, t_4) &= V_{\mathcal{M}}(P, t_5) = V_{\mathcal{M}}(P, t_6) = V_{\mathcal{M}}(P, t_7) = 0 \\ V_{\mathcal{M}}(Q, t_4) &= V_{\mathcal{M}}(Q, t_5) = V_{\mathcal{M}}(Q, t_6) = V_{\mathcal{M}}(Q, t_7) = 0\end{aligned}$$

Second part: We impose a constraint of ‘ $\geq$ ’ being a total linear order on  $\mathcal{T}$ . Now we want to assert that  $\models_{\text{PTL}} (F\phi \wedge F\psi) \rightarrow (F(\phi \wedge F\psi) \vee F(\psi \wedge F\phi) \vee F(\phi \wedge \psi))$ .

*Proof.* AFC that  $V_{\mathcal{M}}((F\phi \wedge F\psi) \rightarrow (F(\phi \wedge F\psi) \vee F(\psi \wedge F\phi) \vee F(\phi \wedge \psi)), t_1) = 0$ . Then,

1.  $V_{\mathcal{M}}(F\phi \wedge F\psi, t_1) = 1$ 
  - (a)  $V_{\mathcal{M}}(F\phi, t_1) = 1$  and  $V_{\mathcal{M}}(F\psi, t_1) = 1$
  - (b)  $V_{\mathcal{M}}(\phi, t_2) = 1$  for some  $t_2 \geq t_1$
  - (c)  $V_{\mathcal{M}}(\psi, t_3) = 1$  for some  $t_3 \geq t_1$
2. Then,  $V_{\mathcal{M}}(F(\phi \wedge F\psi) \vee F(\psi \wedge F\phi) \vee F(\phi \wedge \psi), t_1) = 0$  implies either one of the three hold
  - (a)  $V_{\mathcal{M}}(F(\phi \wedge F\psi), t_1) = 0$
  - (b)  $V_{\mathcal{M}}(F(\psi \wedge F\phi), t_1) = 0$
  - (c)  $V_{\mathcal{M}}(F(\phi \wedge \psi), t_1) = 0$
3. From 2(a),  $V_{\mathcal{M}}(\phi \wedge F\psi, t_4) = 0$  for all  $t_4 \geq t_1$ . So, for all  $t_4 \geq t_1$ , we have (either  $V_{\mathcal{M}}(\phi, t_4) = 0$  or  $[V_{\mathcal{M}}(F\psi, t_4) = 0$ , which implies  $V_{\mathcal{M}}(\psi, t_5) = 0$  for all  $t_5 \geq t_4]$ )
4. From 2(b),  $V_{\mathcal{M}}(\psi \wedge F\phi, t_6) = 0$  for all  $t_6 \geq t_1$ . So for all  $t_6 \geq t_1$ , we have (either  $V_{\mathcal{M}}(\psi, t_6) = 0$  or  $[V_{\mathcal{M}}(F\phi, t_6) = 0 \implies V_{\mathcal{M}}(\phi, t_7) = 0$  for all  $t_7 \geq t_6]$ )
5. From 2(c),  $V_{\mathcal{M}}(\phi \wedge \psi, t_8) = 0$  for all  $t_8 \geq t_1$ . Then for all  $t_8 \geq t_1$ , either  $(V_{\mathcal{M}}(\phi, t_8) = 0$  or  $V_{\mathcal{M}}(\psi, t_8) = 0)$
6. Now we proceed by trichotomy:
  - (a) Case 1:  $t_2 = t_3$ : Then it clearly contradicts 2(c) since there is some  $t'$  where  $t' \geq t_1$  s.t.  $V_{\mathcal{M}}(\phi, t') = V_{\mathcal{M}}(\psi, t') = 1$
  - (b) Case 2:  $t_2 < t_3$ , which contradicts 2(a).  
Suppose at  $t_2$ ,  $V_{\mathcal{M}}(\phi, t_2) = 1$ . Then,  $V_{\mathcal{M}}(\psi, t'') = 0$  for all  $t'' \geq t_2$ , but by assumption,  $V_{\mathcal{M}}(\psi, t'') = 0$  for all  $t'' \geq t_2$ .  
By strong connectedness, for any  $t''$ , either  $t'' \geq t_3$  or  $t_3 \geq t''$ . Notice that we can choose  $t'' = t_3$  which satisfies strong connectedness due to reflexivity and antisymmetry of  $\geq$ . Hence,  $V_{\mathcal{M}}(\psi, t'') = 1$  and  $V_{\mathcal{M}}(\psi, t'') = 0$
  - (c) Case 3:  $t_3 < t_2$ , which contradict 2(b).  
Suppose at  $t_3$ ,  $V_{\mathcal{M}}(\psi, t_3) = 1$ . Then,  $V_{\mathcal{M}}(\phi, t''') = 0$  for all  $t''' \geq t_3$ , but by assumption,  $V_{\mathcal{M}}(\phi, t_2) = 1$  for some  $t_2 \geq t_3$ .  
By strong connectedness and anti-symmetry, we can choose a  $t''' = 2$  that satisfies either  $t''' \geq t_2$  or  $t_2 \geq t'''$ . Hence,  $V_{\mathcal{M}}(\phi, t''') = 1$  and  $V_{\mathcal{M}}(\phi, t''') = 0$

Hence, such a model  $\mathcal{M}$  is not possible, and the formula is valid under the constraint. ■



### 3 Modal propositional logic: proof theory

The proof-theoretic approach differs from the semantic approach in the following ways:

- Specify inference rules licensing immediate transitions between formulas, based purely on their syntactic properties (e.g. natural deduction rules)
- Specify how these immediate transitions may be chained together to make a proof (e.g. natural deduction proof tree)
- Define logical consequence in terms of the existence of an appropriate proof

The axiomatic approach differs from the sequent calculus approach in the following ways:

- You cannot introduce and discharge assumptions, so it disallows conditional or reductio ad absurdum
- You can only use inference rules and axiom-schemas

**Definition 3.1 (Axiomatic proof from a set).** An axiomatic proof of a wff  $\phi$  from a set of wffs  $\Gamma$  in system S is a finite sequence of wffs:

$$\begin{array}{c} \varphi_1 \\ \varphi_2 \\ \vdots \\ \varphi_n \end{array}$$

where the last line,  $\varphi_n$ , is  $\phi$  and for each line,  $\varphi_i$  ( $i = 1, \dots, n$ ), either:

- $\varphi_i$  is an S-axiom, or
- $\varphi_i$  is a member of  $\Gamma$ , or
- $\varphi_i$  follows from earlier wffs in the sequence via an S-rule:

$$\frac{\varphi_{j_1} \dots \varphi_{j_n}}{\varphi_i}$$

with  $j_1, \dots, j_n < i$ .

**Definition 3.2 (Axiomatic Proof).** An axiomatic proof of  $\phi$  is an axiomatic proof of  $\phi$  from  $\emptyset$  i.e., a finite sequence of wffs whose last line is  $\phi$ , in which each line either (i) is an axiom, or (ii) follows from earlier wffs in the sequence via a rule.

#### 3.1 Propositional Logic: Axiomatic Proofs

**Axiomatic system for PL.**

- *Rule:* All PL-instances of Modus Ponens (MP) are PL-rules:

$$\frac{\phi \rightarrow \psi \quad \phi}{\psi} \quad (\text{MP})$$

- *Axioms:* All PL-instances of the following axiom schemas are PL-axioms:

$$\phi \rightarrow (\psi \rightarrow \phi) \quad (\text{PL1})$$

$$(\phi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \chi)) \quad (\text{PL2})$$

$$(\sim \psi \rightarrow \sim \phi) \rightarrow ((\sim \psi \rightarrow \phi) \rightarrow \psi) \quad (\text{PL3})$$

**Definition 3.3 (PL-instance).** A PL-instance of a schema is the result of uniformly replacing each schematic letter ( $\phi, \psi, \dots$ ) with a PL-wff

**Example 3.1.** The following are PL-instances of (PL1):

$$P \rightarrow (Q \rightarrow P), \quad P \rightarrow (P \rightarrow P), \quad (P \rightarrow Q) \rightarrow (\sim\sim R \rightarrow (P \rightarrow Q))$$

**Deduction Meta-Rules:** Meta-rules allow us to demonstrate the existence of PL-proofs without writing them out in full – facts about provability that we can establish in the metatheory.

**Theorem 3.1 (Cut1).** If  $\Gamma \vdash_S \delta$  and  $\Sigma, \delta \vdash_S \phi$ , then  $\Gamma, \Sigma \vdash_S \phi$

**Theorem 3.2 (Cut).** If  $\Gamma_1 \vdash_S \delta_1, \dots, \Gamma_n \vdash_S \delta_n$  and  $\Sigma, \delta_1, \dots, \delta_n \vdash_S \phi$ , then  $\Gamma_1, \dots, \Gamma_n, \Sigma \vdash_S \phi$

**Theorem 3.3 (Deduction Theorem).** If  $\Gamma_1, \phi \vdash_{PL} \psi$ , then  $\Gamma \vdash_{PL} \phi \rightarrow \psi$

**Remark:** The converse of the deduction theorem is also true – it is merely an application of modus ponens.

**Theorem 3.4 (Deduction Theorem I).**  $\Gamma_1, \phi \vdash_{PL} \psi$  iff  $\Gamma \vdash_{PL} \phi \rightarrow \psi$

**Remark:** The other direction is not really a theorem.

### Sider's Toolkit

**Summary of the rules to be added into our toolkit:** if permitted, we can use these results directly.

We reserve the proof for after the summary

1. Transitivity of conditional:  $\phi \rightarrow \psi, \psi \rightarrow \chi \vdash \phi \rightarrow \chi$
2. Contraposition I:  $\sim \psi \rightarrow \sim \phi \vdash \phi \rightarrow \psi$
3. Contraposition II:  $\phi \rightarrow \psi \vdash \sim \psi \rightarrow \sim \phi$
4. Ex falso quodlibet:  $\phi, \sim \phi \vdash \psi$
5. Negated Conditional I:  $\sim (\phi \rightarrow \psi) \vdash \phi$
6. Negated Conditional II:  $\sim (\phi \rightarrow \psi) \vdash \sim \psi$
7. Excluded Middle MP:  $\phi \rightarrow \psi, \sim \phi \rightarrow \psi \vdash \psi$
8. Double Negation I:  $\vdash \phi \rightarrow \sim\sim \phi$
9. Double Negation II:  $\vdash \sim\sim \phi \rightarrow \phi$
10. Permutation:  $\vdash [\phi \rightarrow (\psi \rightarrow \chi)] \rightarrow [\psi \rightarrow (\phi \rightarrow \chi)]$

**Remark:** We sometimes call transitivity of the conditional ‘syllogism’.

### 3.2 Modal Propositional Logic: Axiomatic Proofs

#### Axiomatic system for K.

- *Rule:* All MPL-instances of Modus Ponens (MP) are PL-rules:

$$\frac{\phi \rightarrow \psi \quad \phi}{\psi} \text{MP} \quad \frac{\phi}{\Box \phi} \text{NEC} \quad \frac{\phi \rightarrow \psi}{\Box \phi \rightarrow \Box \psi} \text{Becker}$$

**Remark:** the Becker meta rule combines MP, NEC and K

- *Axioms:* All MPL-instances of the following axiom schemas are K-axioms:

$$\phi \rightarrow (\psi \rightarrow \phi) \quad (\text{PL1})$$

$$(\phi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \chi)) \quad (\text{PL2})$$

$$(\sim \psi \rightarrow \sim \phi) \rightarrow ((\sim \psi \rightarrow \phi) \rightarrow \psi) \quad (\text{PL3})$$

- All MPL-instances of K-schema are K-axioms:

$$\Box(\phi \rightarrow \psi) \rightarrow (\Box \phi \rightarrow \Box \psi) \quad (\text{K})$$

Systems D and T each add further axioms to K.

#### Axiomatic System for D

- All K-axioms and -rules are D-axioms and -rules
- All MPL-instances of the D-schema are D-axioms:

$$\Box \phi \rightarrow \Diamond \phi \quad (\text{D})$$

#### Axiomatic System for T

- All K-axioms and -rules are T-axioms and -rules
- All MPL-instances of the T-schema are T-axioms:

$$\Box \phi \rightarrow \phi \quad (\text{T})$$

B, S4 and S5 add further axioms to T:

#### Axiomatic System for B

- All T-axioms and -rules are B-axioms and -rules
- All MPL-instance of the B-schema are B-axioms:

$$\Diamond \Box \phi \rightarrow \phi \quad (\text{B})$$

#### Axiomatic System for S4

- All T-axioms and -rules are S4-axioms and -rules
- All MPL-instances of the S4-schema are S4-axioms:

$$\Box \phi \rightarrow \Box \Box \phi \quad (\text{S4})$$

### Axiomatic System for S5

- All T-axioms and -rules are S5-axioms and -rules
- All MPL-instances of the S5-schema are S5-axioms:

$$\Diamond \Box \phi \rightarrow \Box \phi \quad (S5)$$

Unfortunately, we lose access to Cut and the Deduction Theorem. Furthermore, we must note the following relations between semantic validity and provability.

**Definition 3.4 (Strong Soundness).** If  $\Gamma \vdash \phi$ , then  $\Gamma \models \phi$

**Definition 3.5 (Soundness).** If  $\vdash \phi$ , then  $\models \phi$

Strong soundness fails for K but soundness still holds. The reason is because the NEC axiom does not preserve truth-at- $w$  in  $\mathcal{M}$  but preserves truth-at-all- $w$  in  $\mathcal{M}$  i.e. preserves  $\mathcal{M}$  validity.

### Abbreviating MPL-proofs

We want to make our lives easier – we do this by invoking PL theorems without proof into our MPL axiomatic proof. In other words, we can invoke MPL-tautologies.

**Definition 3.6 (MPL tautology).** An MPL-wff  $\phi$  is an MPL tautology if  $\phi$  is the result of uniformly substituting MPL-wffs for sentence letters in a PL-tautology.

**Remark:** An MPL-tautology is NOT the same as being MPL-valid.

**Example 3.2.** Both  $\Box \sim P \rightarrow \sim \Diamond P$  and  $\sim \Box P \rightarrow \Diamond \sim P$  are MPL-valid, but only the first is an MPL-tautology. This is because  $\Box \sim P \rightarrow \sim \Diamond P$  is  $\Box \sim P \rightarrow \sim \sim \Box \sim P$  which is just  $\phi \rightarrow \sim \sim \phi$ .

So, in an MPL axiomatic proof, any MPL-tautology (i.e. derived rule from PL) may be written down without proof (i.e. an axiom), and we reference it as simply (PL).

Notice that systems K, T and S4 do not give us any way to deal with  $\Diamond$ s, other than converting it to  $\sim \Box \sim$  as per definition. So, we two ways to give us short cuts: (i) Modal negation theorem schemas or (ii) have analog axioms that give us short cuts.

**Modal Negation Schemas.** We have the following theorem schemas:

- $\vdash_K \sim \Box \phi \rightarrow \Diamond \sim \phi$

*Proof.*

- |    |                                                               |                   |
|----|---------------------------------------------------------------|-------------------|
| 1. | $\sim \Box \phi \rightarrow \sim \Box \phi$                   | PL                |
| 2. | $\sim \Box \phi \rightarrow \sim \sim \Diamond \sim \phi$     | def of $\Diamond$ |
| 3. | $\sim \sim \Diamond \sim \phi \rightarrow \Diamond \sim \phi$ | PL                |
| 4. | $\sim \Box \phi \rightarrow \Diamond \sim \phi$               | MP2,3             |

■

- $\vdash_K \sim \Diamond \phi \rightarrow \Box \sim \phi$

*Proof.*

- |    |                                                           |               |
|----|-----------------------------------------------------------|---------------|
| 1. | $\sim \Diamond \phi \rightarrow \sim \Diamond \phi$       | PL            |
| 2. | $\sim \Diamond \phi \rightarrow \sim \sim \Box \sim \phi$ | def of $\Box$ |
| 3. | $\sim \sim \Box \sim \phi \rightarrow \Box \sim \phi$     | PL            |
| 4. | $\sim \Diamond \phi \rightarrow \Box \sim \phi$           | MP2,3         |

■

- $\vdash_K \Diamond \sim \phi \rightarrow \sim \Box \phi$

*Proof.* Analogous to the first ■

- $\vdash_K \Box \sim \phi \rightarrow \sim \Diamond \phi$

*Proof.* Analogous to the second ■

#### Analog for K

- All MPL-instances of the K analog schema are K-axioms:

$$\Box(\phi \rightarrow \psi) \rightarrow (\Diamond \phi \rightarrow \Diamond \psi) \quad (K\Diamond)$$

#### Analog for T

- All MPL-instances of the T analog schema are T-axioms:

$$\phi \rightarrow \Diamond \phi \quad (T\Diamond)$$

#### Analog for B

- All MPL-instances of the B analog schema are B axioms:

$$\phi \rightarrow \Box \Diamond \phi \quad (B\Diamond)$$

#### Analog for S4

- All MPL-instances of the S4 analog schema are S4 axioms:

$$\Diamond \Diamond \phi \rightarrow \Diamond \phi \quad (S4\Diamond)$$

#### Analog for S5

- All MPL-instances of the S5 analog schema are S5 axioms:

$$\Diamond \phi \rightarrow \Box \Diamond \phi \quad (S5\Diamond)$$

### Metatheorems for MPL

**Theorem 3.5 (Substitution of equivalents).** Where S is any of our modal systems, if  $\vdash_S \alpha \leftrightarrow \beta$ , then  $\vdash_S \chi \leftrightarrow \chi'$

Roughly, you can substitute provably equivalent wffs within complex wffs. More carefully, call two wffs “ $\alpha/\beta$ ” iff they differ only in that in zero or more places, wff  $\alpha$  occurs in one where wff  $\beta$  occurs in the other. Thus, you can turn one into the other by changing (zero or more)  $\alpha$ s to  $\beta$ s or  $\beta$ s to  $\alpha$ s. For example,  $P \rightarrow (Q \rightarrow P)$  and  $\sim S \rightarrow (Q \rightarrow \sim S)$  are  $P/\sim S$  variants, as are  $P \rightarrow (Q \rightarrow P)$  and  $\sim S \rightarrow (Q \rightarrow P)$ .

*Proof.* Suppose  $\vdash_S \alpha \leftrightarrow \beta$ . Argue by induction that the following holds for any wff,  $\chi$ :

$$\vdash_S \chi \leftrightarrow \chi', \text{ for any } \alpha/\beta \text{ variant } \chi' \text{ of } \chi$$

Base Case: here  $\chi$  is a sentence letter. Let  $\chi'$  be any  $\alpha/\beta$  variant of  $\chi$ . If  $\chi$  is neither  $\alpha$  nor  $\beta$  then  $\chi'$  is just  $\chi$  itself. If on the other hand  $\chi$  is either  $\alpha$  or  $\beta$  then  $\chi'$  is either  $\alpha$  or  $\beta$ . Either way, we have one of the following cases:  $\chi' = \chi$ , or  $\chi = \alpha$  and  $\chi' = \beta$ , or  $\chi = \beta$  and  $\chi' = \alpha$ . Since  $\vdash_S \alpha \leftrightarrow \beta$  and S includes PL,

$\vdash_S (\chi \leftrightarrow \chi')$  in each case.

Induction case: Now we assume the inductive hypothesis, that wffs  $\chi_1$  and  $\chi_2$  obey the theorem:

$$\begin{aligned} \vdash_S \chi_1 &\leftrightarrow \chi'_1, \text{ for any } \alpha/\beta \text{ variant } \chi'_1 \text{ of } \chi_1 \\ \vdash_S \chi_2 &\leftrightarrow \chi'_2, \text{ for any } \alpha/\beta \text{ variant } \chi'_2 \text{ of } \chi_2 \end{aligned}$$

We must show that the theorem holds for  $\sim \chi_1$ ,  $\chi_1 \rightarrow \chi_2$ , and  $\Box \chi_1$ .

Take the first case. We must show that the theorem holds for  $\sim \chi_1$  – i.e., that  $\vdash_S \sim \chi_1 \leftrightarrow \phi$ , for any  $\alpha/\beta$  variant  $\phi$  of  $\sim \chi_1$ . Suppose first that  $\phi$  has the form  $\sim \chi'_1$ , where  $\chi'_1$  is an  $\alpha/\beta$  variant of  $\chi_1$ . By the inductive hypothesis,  $\vdash_S \chi_1 \leftrightarrow \chi'_1$ ; since S include PL,  $\vdash_S \sim \chi_1 \leftrightarrow \sim \chi'_1$ , i.e.,  $\vdash_S \sim \chi_1 \leftrightarrow \phi$ . If, on the other hand,  $\phi$  does not have the form  $\sim \chi'_1$  for some  $\alpha/\beta$  variant  $\chi'_1$  of  $\chi_1$ , then  $\phi$  must result from changing the whole of  $\sim \chi_1$  from  $\alpha$  to  $\beta$  or from  $\beta$  to  $\alpha$ . Thus, each of  $\sim \chi_1$  and  $\phi$  must be either  $\alpha$  or  $\beta$ . But then, as in the base,  $\vdash_S \sim \chi_1 \leftrightarrow \phi$ .

The second case is  $\chi_1 \rightarrow \chi_2$  i.e. that  $\vdash_S (\chi_1 \rightarrow \chi_2) \leftrightarrow \phi$ , for any  $\alpha/\beta$  variant  $\phi$  of  $\chi_1 \rightarrow \chi_2$ . Suppose first that  $\phi$  has the form  $\chi'_1 \rightarrow \chi'_2$  where  $\chi_1$  is an  $\alpha/\beta$  variant of  $\chi_1$  and  $\chi'_2$  of  $\chi_2$ . By inductive hypothesis,  $\vdash_S \chi_1 \leftrightarrow \chi'_1$  and  $\vdash_S \chi_2 \leftrightarrow \chi'_2$ . Then

1.  $(\chi'_1 \rightarrow \chi_1) \rightarrow ((\chi_1 \rightarrow \chi_2) \rightarrow (\chi'_1 \rightarrow \chi_2))$  PL
2.  $\chi'_1 \rightarrow \chi_1$  Assumption
3.  $(\chi_1 \rightarrow \chi_2) \rightarrow (\chi'_1 \rightarrow \chi_2)$  MP1,2
4.  $(\chi_2 \rightarrow \chi'_2) \rightarrow ((\chi'_1 \rightarrow \chi_2) \rightarrow (\chi'_1 \rightarrow \chi'_2))$  PL
5.  $\chi_2 \rightarrow \chi'_2$
6.  $(\chi'_1 \rightarrow \chi_2) \rightarrow (\chi'_1 \rightarrow \chi'_2)$  MP4,5
7.  $(\chi_1 \rightarrow \chi_2) \rightarrow (\chi'_1 \rightarrow \chi'_2)$  PL3,6

The proof of the other direction  $(\chi'_1 \rightarrow \chi'_2) \rightarrow (\chi_1 \rightarrow \chi_2)$  is exactly the same. So,  $\vdash_S (\chi_1 \rightarrow \chi_2) \leftrightarrow (\chi'_1 \rightarrow \chi'_2)$ . If on the other hand,  $\phi$  does not have the form  $\chi'_1 \rightarrow \chi'_2$ , then  $\phi$  must result from changing the whole of  $\chi_1 \rightarrow \chi_2$  from  $\alpha$  to  $\beta$  or from  $\beta$  to  $\alpha$ . Thus each of  $\chi_1 \rightarrow \chi_2$  and  $\phi$  must be either  $\alpha$  or  $\beta$ . But then as in the base case,  $\vdash_S (\chi_1 \rightarrow \chi_2) \leftrightarrow \phi$ .

The last case is exactly the same as the first. We must show that the theorem holds for  $\Box \chi_1$  – i.e., that  $\vdash_S \Box \chi_1 \leftrightarrow \phi$ , for any  $\alpha/\beta$  variant  $\phi$  of  $\Box \chi_1$ . Suppose first that  $\phi$  has the form  $\Box \chi'_1$ , where  $\chi'_1$  is an  $\alpha/\beta$  variant of  $\chi_1$ . By the inductive hypothesis,  $\vdash_S \chi_1 \leftrightarrow \chi'_1$ ; since S include PL,  $\vdash_S \Box \chi_1 \leftrightarrow \Box \chi'_1$ , i.e.,  $\vdash_S \Box \chi_1 \leftrightarrow \phi$ . If, on the other hand,  $\phi$  does not have the form  $\Box \chi'_1$  for some  $\alpha/\beta$  variant  $\chi'_1$  of  $\chi_1$ , then  $\phi$  must result from changing the whole of  $\Box \chi_1$  from  $\alpha$  to  $\beta$  or from  $\beta$  to  $\alpha$ . Thus, each of  $\Box \chi_1$  and  $\phi$  must be either  $\alpha$  or  $\beta$ . But then, as in the base,  $\vdash_S \Box \chi_1 \leftrightarrow \phi$ . ■

Our next metatheorem applies ONLY to S5.

**Theorem 3.6 (Modal reduction theorem for S5).** Where  $\mathcal{O}_1 \dots \mathcal{O}_n$  are modal operators and  $\phi$  is a wff:

$$\vdash_S \mathcal{O}_1 \dots \mathcal{O}_n \phi \leftrightarrow \mathcal{O}_n \phi$$

Basically, a string of modal operators always boils down to the innermost operator.

*Proof.* The following equivalences are all theorems of S5:

- (a)  $\Diamond \Box \phi \leftrightarrow \Box \phi$
- (b)  $\Box \Box \phi \leftrightarrow \Box \phi$
- (c)  $\Box \Diamond \phi \leftrightarrow \Diamond \phi$
- (d)  $\Diamond \Diamond \phi \leftrightarrow \Diamond \phi$

The left-to-right direction of (a) is just S5; the right-to-left is T $\Diamond$ ; (b) is T and S4; (c) is T and S5 $\Diamond$ ; and (d) is S4 $\Diamond$  and T $\Diamond$ . Now consider  $\mathcal{O}_1 \mathcal{O}_2 \dots \mathcal{O}_n \phi$ . Depending on which two modal operators  $\mathcal{O}_1$  and  $\mathcal{O}_2$  are, one of (a)-(d) tells us that  $\vdash_{S5} \mathcal{O}_1 \mathcal{O}_2 \dots \mathcal{O}_n \phi \leftrightarrow \mathcal{O}_2 \dots \mathcal{O}_n \phi$ . Repeating this process  $n - 1$  times, we have  $\vdash_{S5} \mathcal{O}_1 \dots \mathcal{O}_n \phi \leftrightarrow \mathcal{O}_n \phi$ . (We can convert this argument into a more rigorous inductive proof.) ■

### 3.3 Proofs of toolkit rules

#### PL Toolkit Rules

1. Transitivity of Conditional:  $\phi \rightarrow \psi, \psi \rightarrow \chi \vdash_S \phi \rightarrow \chi$

*Proof.*

- |    |                         |         |
|----|-------------------------|---------|
| 1. | $\phi \rightarrow \psi$ | premise |
| 2. | $\psi \rightarrow \chi$ | premise |
| 3. | $\phi$                  | premise |
| 4. | $\psi$                  | 1,3, MP |
| 5. | $\chi$                  | 2,4, MP |

By the deduction theorem, we can move from  $\phi \rightarrow \psi, \psi \rightarrow \chi, \phi \vdash \chi$  to  $\phi \rightarrow \psi, \psi \rightarrow \chi \vdash \phi \rightarrow \chi$  ■

2. Contraposition I:  $\sim \psi \rightarrow \sim \phi \vdash \phi \rightarrow \psi$

*Proof.*

- |    |                                                 |                   |
|----|-------------------------------------------------|-------------------|
| 1. | $\sim \psi \rightarrow \sim \phi$               | premise           |
| 2. | $\phi$                                          | premise           |
| 3. | $\phi \rightarrow (\sim \psi \rightarrow \phi)$ | PL1               |
| 4. | $\sim \psi \rightarrow \phi$                    | 2,3, MP           |
| 5. | $\psi$                                          | PL3, 1, MP, 3, MP |

By DT, we move from  $\sim \psi \rightarrow \sim \phi, \phi \vdash \psi$  to  $\sim \psi \rightarrow \sim \phi \vdash \phi \rightarrow \psi$ . ■

3. Contraposition II:  $\phi \rightarrow \psi \vdash \sim \psi \rightarrow \sim \phi$

*Proof.*

- |    |                                             |                                       |
|----|---------------------------------------------|---------------------------------------|
| 1. | $\phi \rightarrow \psi$                     | premise                               |
| 2. | $\psi \rightarrow \sim \sim \psi$           | PL (double negation I)                |
| 3. | $\sim \sim \phi \rightarrow \phi$           | PL (double negation II)               |
| 4. | $\sim \sim \phi \rightarrow \sim \sim \psi$ | 3,1,2 transitivity                    |
| 5. | $\phi \rightarrow \sim \sim \phi$           | PL (double negation I)                |
| 6. | $\sim \sim \psi \rightarrow \psi$           | PL (double negation II)               |
| 7. | $\sim \psi \rightarrow \sim \phi$           | 5,4,6, transitivity, contraposition I |

4. Ex Falso Quodlibet:  $\phi, \sim \phi \vdash \psi$

*Proof.*

- |    |                                   |               |
|----|-----------------------------------|---------------|
| 1. | $\phi$                            | premise       |
| 2. | $\sim \phi$                       | premise       |
| 3. | $\sim \psi \rightarrow \phi$      | PL1, 1, MP    |
| 4. | $\sim \psi \rightarrow \sim \phi$ | PL1,2,MP      |
| 5. | $\psi$                            | PL3,4,MP,3,MP |

5. Negated Conditional I:  $\sim(\phi \rightarrow \psi) \vdash \phi$

*Proof.*

- |    |                                                          |                             |
|----|----------------------------------------------------------|-----------------------------|
| 1. | $\sim \phi \rightarrow (\phi \rightarrow \psi)$          | PL (ex falso quodlibet), DT |
| 2. | $\sim(\phi \rightarrow \psi) \rightarrow \sim \sim \phi$ | 1, contraposition II        |
| 3. | $\sim(\phi \rightarrow \psi)$                            | premise                     |
| 4. | $\sim \sim \phi$                                         | 2,3, MP                     |
| 5. | $\phi$                                                   | 4, PL (double negation I)   |

■

6. Negated Conditional II:  $\sim(\phi \rightarrow \psi) \vdash \sim \psi$

*Proof.*

- |    |                                                     |                      |
|----|-----------------------------------------------------|----------------------|
| 1. | $\psi \rightarrow (\phi \rightarrow \psi)$          | PL1                  |
| 2. | $\sim(\phi \rightarrow \psi) \rightarrow \sim \psi$ | 1, contraposition II |
| 3. | $\sim(\phi \rightarrow \psi)$                       | premise              |
| 4. | $\psi$                                              | 2,3,MP               |

■

7. Excluded Middle MP:  $\phi \rightarrow \psi, \sim \phi \rightarrow \psi \vdash \psi$

*Proof.*

- |    |                                        |                                    |
|----|----------------------------------------|------------------------------------|
| 1. | $\phi \rightarrow \psi$                | premise                            |
| 2. | $\sim \phi \rightarrow \psi$           | premise                            |
| 3. | $\sim \psi \rightarrow \sim \phi$      | 1, contraposition II               |
| 4. | $\sim \psi \rightarrow \sim \sim \phi$ | 2, contraposition II               |
| 5. | $\sim \psi \rightarrow \phi$           | 4, double negation I, transitivity |
| 6. | $\psi$                                 | PL3, 3, MP, 5, MP                  |

■

8. Double Negation I:  $\vdash \sim \sim \phi \rightarrow \phi$

*Proof.*

- |     |                                                                                                                                                                                                           |          |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1.  | $\sim \sim \phi \rightarrow (\sim \phi \rightarrow \sim \sim \phi)$                                                                                                                                       | PL1      |
| 2.  | $(\sim \phi \rightarrow \sim \sim \phi) \rightarrow ((\sim \phi \rightarrow \sim \phi) \rightarrow \phi)$                                                                                                 | PL2      |
| 3.  | $\sim \sim \phi$                                                                                                                                                                                          | premise  |
| 4.  | $\sim \phi \rightarrow \sim \sim \phi$                                                                                                                                                                    | 1,3, MP  |
| 5.  | $(\sim \phi \rightarrow \sim \phi) \rightarrow \phi$                                                                                                                                                      | 2,4, MP  |
| 6.  | $(\sim \phi \rightarrow ((\sim \phi \rightarrow \sim \phi) \rightarrow \sim \phi) \rightarrow ((\sim \phi \rightarrow (\sim \phi \rightarrow \sim \phi)) \rightarrow (\sim \phi \rightarrow \sim \phi)))$ | PL2      |
| 7.  | $\sim \phi \rightarrow ((\sim \phi \rightarrow \sim \phi) \rightarrow \sim \phi)$                                                                                                                         | PL1      |
| 8.  | $(\sim \phi \rightarrow (\sim \phi \rightarrow \sim \phi)) \rightarrow (\sim \phi \rightarrow \sim \phi)$                                                                                                 | 6,7, MP  |
| 9.  | $\sim \phi \rightarrow (\sim \phi \rightarrow \sim \phi)$                                                                                                                                                 | PL1      |
| 10. | $\sim \phi \rightarrow \sim \phi$                                                                                                                                                                         | 8,9, MP  |
| 11. | $\phi$                                                                                                                                                                                                    | 5,10, MP |

By DT,  $\vdash \sim \sim \phi \rightarrow \phi$

■



9. Double Negation II:  $\vdash \phi \rightarrow \sim \sim \phi$

*Proof.*

- |    |                                             |                   |
|----|---------------------------------------------|-------------------|
| 1. | $\sim \sim \sim \phi \rightarrow \sim \phi$ | double Negation I |
| 2. | $\phi \rightarrow \sim \sim \phi$           | contraposition II |

■

10. Permutation:  $\vdash [\phi \rightarrow (\psi \rightarrow \chi)] \rightarrow [\psi \rightarrow (\phi \rightarrow \chi)]$

*Proof.*

- |    |                                            |                |
|----|--------------------------------------------|----------------|
| 1. | $\phi$                                     | premise        |
| 2. | $\psi$                                     | premise        |
| 3. | $\phi \rightarrow (\psi \rightarrow \chi)$ | premise        |
| 4. | $\chi$                                     | 1,3, MP, 2, MP |

By DT, we move from  $\phi, \psi, \phi \rightarrow (\psi \rightarrow \chi) \vdash \chi$  to  $\vdash [\phi \rightarrow (\psi \rightarrow \chi)] \rightarrow [\psi \rightarrow (\phi \rightarrow \chi)]$

■

11. Modus Ponens:  $\vdash \phi \rightarrow [(\phi \rightarrow \psi) \rightarrow \psi]$

*Proof.*

- |    |                         |         |
|----|-------------------------|---------|
| 1. | $\phi$                  | premise |
| 2. | $\phi \rightarrow \psi$ | premise |
| 3. | $\psi$                  | 1,2, MP |

By DT, we move from  $\phi, \phi \rightarrow \psi \vdash \psi$  to  $\vdash \phi \rightarrow [(\phi \rightarrow \psi) \rightarrow \psi]$

■

12.  $\psi \vdash \phi \rightarrow \psi$

*Proof.*

- |    |        |         |
|----|--------|---------|
| 1. | $\psi$ | premise |
|----|--------|---------|

By DT, we move from  $\phi, \psi \vdash \psi$  to  $\psi \vdash \phi \rightarrow \psi$

■

13.  $\sim \phi \vdash \phi \rightarrow \psi$

*Proof.* Follows from ex falso quod libet, via DT. We move from  $\phi, \sim \phi \vdash \psi$  to  $\sim \phi \vdash \phi \rightarrow \psi$

■

Further PL-tautologies (stated without proof)

1. Syllogism:  $\vdash_{\text{PL}} ((\phi \rightarrow \psi) \wedge (\psi \rightarrow \chi)) \rightarrow (\phi \rightarrow \chi)$
2. Import/Export:  $\vdash_{\text{PL}} (\phi \rightarrow (\psi \rightarrow \chi)) \leftrightarrow ((\phi \wedge \psi) \rightarrow \chi)$
3. Composition:  $\vdash_{\text{PL}} ((\phi \rightarrow \psi) \wedge (\phi \rightarrow \chi)) \leftrightarrow (\phi \rightarrow (\psi \wedge \chi))$
4. Dilemma:  $\vdash_{\text{PL}} ((\phi \rightarrow \chi) \wedge (\psi \rightarrow \chi)) \leftrightarrow ((\phi \vee \psi) \rightarrow \chi)$
5. Biconditional:  $\vdash_{\text{PL}} ((\phi \rightarrow \psi) \wedge (\psi \rightarrow \phi)) \leftrightarrow (\phi \leftrightarrow \psi)$
6. Negated conjunction:  $\vdash_{\text{PL}} (\phi \rightarrow \sim \psi) \leftrightarrow \sim (\phi \wedge \psi)$

### MPL Analog Axiom proofs

$$1. \vdash_K \Box(\phi \rightarrow \psi) \rightarrow (\Diamond\phi \rightarrow \Diamond\psi)$$

*Proof.*

- |    |                                                                                                 |                        |
|----|-------------------------------------------------------------------------------------------------|------------------------|
| 1. | $(\phi \rightarrow \psi) \rightarrow (\sim \psi \rightarrow \sim \phi)$                         | PL (contraposition)    |
| 2. | $\Box(\phi \rightarrow \psi) \rightarrow \Box(\sim \psi \rightarrow \sim \phi)$                 | 1, Becker              |
| 3. | $\Box(\sim \psi \rightarrow \sim \phi) \rightarrow (\Box \sim \psi \rightarrow \Box \sim \phi)$ | K                      |
| 4. | $\Box(\phi \rightarrow \psi) \rightarrow (\Box \sim \psi \rightarrow \Box \sim \phi)$           | 2,3, PL (syllogism)    |
| 5. | $\Box(\phi \rightarrow \psi) \rightarrow (\sim \Box \sim \phi \rightarrow \sim \Box \sim \psi)$ | 4, PL (contraposition) |
| 6. | $\Box(\phi \rightarrow \psi) \rightarrow (\Diamond\phi \rightarrow \Diamond\psi)$               | def of $\Diamond$      |

■

$$2. \vdash_T \phi \rightarrow \Diamond\phi$$

*Proof.*

- |    |                                        |       |
|----|----------------------------------------|-------|
| 1. | $\Box \sim \phi \rightarrow \sim \phi$ | T     |
| 2. | $\phi \rightarrow \sim \Box \sim \phi$ | 1, PL |

■

$$3. \vdash_B \phi \rightarrow \Box\Diamond\phi$$

*Proof.*

- |    |                                                       |                              |
|----|-------------------------------------------------------|------------------------------|
| 1. | $\sim \Box \sim \Box \sim \phi \rightarrow \sim \phi$ | B (given def of $\Diamond$ ) |
| 2. | $\phi \rightarrow \Box \sim \Box \sim \phi$           | 1, PL                        |
| 3. | $\phi \rightarrow \Box\Diamond\phi$                   | def of $\Diamond$            |

■

$$4. \vdash_{S4} \Diamond\Diamond\phi \rightarrow \Diamond\phi$$

*Proof.*

- |    |                                                                      |                        |
|----|----------------------------------------------------------------------|------------------------|
| 1. | $\Box \sim \phi \rightarrow \Box\Box \sim \phi$                      | S4                     |
| 2. | $\Box \sim \phi \rightarrow \sim \sim \Box \sim \phi$                | PL                     |
| 3. | $\Box\Box \sim \phi \rightarrow \Box \sim \sim \Box \sim \phi$       | 2, Becker              |
| 4. | $\Box \sim \phi \rightarrow \Box \sim \sim \Box \sim \phi$           | 1,3, Pl (syllogism)    |
| 5. | $\sim \Box \sim \sim \Box \sim \phi \rightarrow \sim \Box \sim \phi$ | 4, PL (contraposition) |

■

$$5. \vdash_{S5} \Diamond\phi \rightarrow \Box\Diamond\phi$$

*Proof.*

- |    |                                                            |                         |
|----|------------------------------------------------------------|-------------------------|
| 1. | $\sim \Box \sim \Box \sim \phi \rightarrow \Box \sim \phi$ | S5 (def of $\Diamond$ ) |
| 2. | $\sim \Box \sim \phi \rightarrow \Box \sim \Box \sim \phi$ | 1, PL                   |
| 3. | $\Diamond\phi \rightarrow \Box\Diamond\phi$                | def of $\Diamond$       |

■

**Remark:** Any weaker system's axiom can be proven with a stronger system's axioms – this is because the stronger system's set of valid wffs is a superset of a weaker system's.

### 3.4 Worked Examples

#### PL Worked Examples

1.  $\vdash P \rightarrow (P \rightarrow P)$

*Proof.*

1.  $P \rightarrow (P \rightarrow P)$  PL1

■

2.  $\vdash P \rightarrow P$

*Proof.*

- |    |                                                                                                                                   |         |
|----|-----------------------------------------------------------------------------------------------------------------------------------|---------|
| 1. | $(P \rightarrow ((P \rightarrow P) \rightarrow P)) \rightarrow ((P \rightarrow (P \rightarrow P)) \rightarrow (P \rightarrow P))$ | PL2     |
| 2. | $P \rightarrow ((P \rightarrow P) \rightarrow P)$                                                                                 | PL1     |
| 3. | $(P \rightarrow (P \rightarrow P)) \rightarrow (P \rightarrow P)$                                                                 | 1,2, MP |
| 4. | $P \rightarrow (P \rightarrow P)$                                                                                                 | PL1     |
| 5. | $P \rightarrow P$                                                                                                                 | 3,4, PL |

■

3.  $(\sim P \rightarrow P) \rightarrow P$

*Proof.*

- |    |                                                                                                                                                                                |         |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1. | $(\sim P \rightarrow \sim P) \rightarrow ((\sim P \rightarrow P) \rightarrow P)$                                                                                               | PL3     |
| 2. | $(\sim P \rightarrow ((\sim P \rightarrow \sim P) \rightarrow \sim P)) \rightarrow ((\sim P \rightarrow (\sim P \rightarrow \sim P)) \rightarrow (\sim P \rightarrow \sim P))$ | PL2     |
| 3. | $\sim P \rightarrow ((\sim P \rightarrow \sim P) \rightarrow \sim P)$                                                                                                          | PL1     |
| 4. | $(\sim P \rightarrow (\sim P \rightarrow \sim P)) \rightarrow (\sim P \rightarrow \sim P)$                                                                                     | 2,3, MP |
| 5. | $\sim P \rightarrow (\sim P \rightarrow \sim P)$                                                                                                                               | PL1     |
| 6. | $\sim P \rightarrow \sim P$                                                                                                                                                    | 4,5, MP |
| 7. | $(\sim P \rightarrow P) \rightarrow P$                                                                                                                                         | 1,6, MP |

■

#### MPL Worked Examples

1.  $\vdash_T \Box \Box P \rightarrow P$

*Proof.*

- |    |                                  |         |
|----|----------------------------------|---------|
| 1. | $\Box \Box P \rightarrow \Box P$ | T       |
| 2. | $\Box P \rightarrow P$           | T       |
| 3. | $\Box \Box P \rightarrow P$      | 1,2, PL |

■

2.  $\vdash_{S4} \Diamond \Diamond \Diamond P \rightarrow \Diamond P$

*Proof.* This is the same as  $\vdash_{S4} \sim \Box \sim \sim \Box \sim \sim \Box \sim P \rightarrow \sim \Box \sim P$ .

1.  $\Box \sim \sim \Box \sim P \rightarrow \Box \Box \sim \sim \Box \sim P$  S4
2.  $\Box \sim \sim \Box \sim P \rightarrow \sim \sim \Box \sim \sim \Box \sim P$  PL
3.  $\Box \Box \sim \sim \Box \sim P \rightarrow \Box \sim \sim \Box \sim \sim \Box \sim P$  Becker
4.  $\Box \sim \sim \Box \sim P \rightarrow \Box \sim \sim \Box \sim \sim \Box \sim P$  1,3, PL (syllogism)
5.  $\sim \Box \sim \sim \Box \sim \sim \Box \sim P \rightarrow \sim \Box \sim \sim \Box \sim P$  4, PL (contraposition)
6.  $\Box \sim P \rightarrow \Box \Box \sim P$  S4
7.  $\Box \sim P \rightarrow \sim \sim \Box \sim P$  PL
8.  $\Box \Box \sim P \rightarrow \Box \sim \sim \Box \sim P$  Becker
9.  $\Box \sim P \rightarrow \Box \sim \sim \Box \sim P$  6,8, PL
10.  $\sim \Box \sim \sim \Box \sim P \rightarrow \sim \Box \sim P$  10, PL
11.  $\sim \Box \sim \sim \Box \sim \sim \Box \sim P \rightarrow \sim \Box \sim P$  7,12, PL
12.  $\Diamond \Diamond \Diamond P \rightarrow \Diamond P$  def of  $\Diamond$

Alternatively, we have

1.  $\Diamond \Diamond \Diamond P \rightarrow \Diamond \Diamond P$  S4 $\Diamond$
2.  $\Diamond \Diamond P \rightarrow \Diamond P$  S4 $\Diamond$
3.  $\Diamond \Diamond \Diamond P \rightarrow \Diamond P$  1,2, PL

■

3.  $\vdash_{S4} \Box P \rightarrow \Box \Diamond \Box P$

*Proof.* We want to prove an equivalent  $\vdash_{S4} \Box P \rightarrow \Box \sim \Box \sim \Box P$ . We need to prove the following lemma:

**Lemma 3.7.** The  $T\Diamond$  axiom is provable in S4 i.e.  $\vdash_{S4} \phi \rightarrow \Diamond \phi$

*Proof.*

1.  $\Box \sim \phi \rightarrow \sim \phi$  T
2.  $\phi \rightarrow \sim \Box \sim \phi$  1, PL
3.  $\phi \rightarrow \Diamond \phi$  def of  $\Diamond$

■

We now proceed with the main proof.

1.  $\Box P \rightarrow \Diamond \Box P$  T $\Diamond$
2.  $\Box \Box P \rightarrow \Box \Diamond \Box P$  Becker
3.  $\Box P \rightarrow \Box \Box P$  S4
4.  $\Box P \rightarrow \Box \Diamond \Box P$  3,2, PL

■

4.  $\vdash_{S5} \Box(\Box P \rightarrow \Box Q) \vee \Box(\Box Q \rightarrow \Box P)$

*Proof.* We require two lemmas for this.

**Lemma 3.8.** The  $B\Diamond$ -axiom is provable in S5 i.e.  $\vdash_{S5} \phi \rightarrow \Box \Diamond \phi$

*Proof.*

- |    |                                                |                        |
|----|------------------------------------------------|------------------------|
| 1. | $\Box \sim \phi \rightarrow \sim \phi$         | T                      |
| 2. | $\phi \rightarrow \Diamond \phi$               | 1, PL (contraposition) |
| 3. | $\Diamond \phi \rightarrow \Box \Diamond \phi$ | S5                     |
| 4. | $\phi \rightarrow \Box \Diamond \phi$          | PL                     |

■

**Lemma 3.9.** The S4-axiom is provable in S5 i.e.  $\vdash_{S5} \Box \phi \rightarrow \Box \Box \phi$

*Proof.*

- |    |                                                      |              |
|----|------------------------------------------------------|--------------|
| 1. | $\Box \phi \rightarrow \Box \Diamond \Box \phi$      | B $\Diamond$ |
| 2. | $\Diamond \Box \phi \rightarrow \Box \phi$           | S5           |
| 3. | $\Box \Diamond \Box \phi \rightarrow \Box \Box \phi$ | Becker       |
| 4. | $\Box \phi \rightarrow \Box \Box \phi$               | PL           |

■

We now proceed with the main proof

- |     |                                                                                    |               |
|-----|------------------------------------------------------------------------------------|---------------|
| 1.  | $\sim \Box P \rightarrow (\Box P \rightarrow \Box Q)$                              | PL            |
| 2.  | $\Box \sim \Box P \rightarrow \Box(\Box P \rightarrow \Box Q)$                     | 1, Becker     |
| 3.  | $\sim \Box(\Box P \rightarrow \Box Q) \rightarrow \Diamond \Box P$                 | 2, PL         |
| 4.  | $\Diamond \Box P \rightarrow \Box P$                                               | S5            |
| 5.  | $\sim \Box(\Box P \rightarrow \Box Q) \rightarrow \Box P$                          | 3,4, PL       |
| 6.  | $\Box P \rightarrow \Box \Box P$                                                   | S4            |
| 7.  | $\sim \Box(\Box P \rightarrow \Box Q) \rightarrow \Box \Box P$                     | 5,6, PL       |
| 8.  | $\Box P \rightarrow (\Box Q \rightarrow \Box P)$                                   | PL1           |
| 9.  | $\Box \Box P \rightarrow \Box(\Box Q \rightarrow \Box P)$                          | 8, Becker     |
| 10. | $\sim \Box(\Box P \rightarrow \Box Q) \rightarrow \Box(\Box Q \rightarrow \Box P)$ | 7,9, PL       |
| 11. | $\Box(\Box P \rightarrow \Box Q) \vee \Box(\Box Q \rightarrow \Box P)$             | def of $\vee$ |

■

## 4 Metatheory

### 4.1 Propositional Logic Metatheory

We have two meta-rules: Cut and the Deduction Theorem. We want to prove that they hold.

**Theorem 4.1 (Cut).** If  $\Gamma_1 \vdash_S \delta_1, \dots, \Gamma_n \vdash_S \delta_n$  and  $\Sigma, \delta_1, \dots, \delta_n \vdash_S \phi$ , then  $\Gamma_1, \dots, \Gamma_n, \Sigma \vdash_S \phi$

*Proof.* We are given that there exists a proof  $A_i$  of  $\delta_i$  from  $\Gamma_i$  for  $i = 1, \dots, n$  and that there exists a proof  $B$  of  $\phi$  from  $\Sigma, \delta_1, \dots, \delta_n$ . Let  $C$  be the result of concatenating all these proofs, in that order. That is,  $C$  begins with a first phase, consisting of the formulas of proof  $A_1$ , followed by the formulas of proof  $A_2$ , and so on, finishing with the formulas of proof  $A_n$ . Then, in the second phase,  $C$  concludes with the formulas of proof  $B$ . The last formula of  $C$  counts as a proof from  $\Gamma_1, \dots, \Gamma_n, \Sigma$ —that is, that each of  $C$  is either an axiom, a member of  $\Gamma_1$  or of  $\Gamma_2, \dots$ , or of  $\Gamma_n$ , or of  $\Sigma$ , or following from earlier lines in  $C$  by MP. For short, we must show that each line of  $C$  is “legit”. Clearly, each line  $j$  of the first phase  $C$  is legit:  $j$  is one of the  $A_i$  segments;  $A_i$  is a proof from  $\Gamma_i$ ; so the formula on line  $j$  is either an axiom, a member of  $\Gamma_i$ , or what

follows from earlier lines in that  $A_i$  segment by MP. Consider, finally, the second phase of  $C$ , namely the  $B$  portion. Since  $B$  is a proof from  $\Sigma, \delta_1, \dots, \delta_n$ , the formula on any line  $j$  here is either (i) an axiom, (ii) a member of  $\Sigma$ , (iii) one of the  $\delta_i$ s, or (iv) what follows from earlier lines of the  $B$  portion by MP. Line  $j$  is clearly legit in cases (i), (ii) and (iv). In case (iii), the formula on line  $j$  is some  $\delta_i$ . But  $\delta_i$  also occurred in the first phase of  $C$ , as the last line,  $k$ , of the  $A_i$  portion. So  $\delta_i$  is either an axiom or member of  $\Gamma_i$ , or follows from earlier lines in the  $A_i$  portion – which are before  $k$  – by MP. In either of the first two cases, line  $j$  is legit; and it's also legit in the last case because lines before  $k$  in  $C$  are also lines before  $j$ . ■

**Remark:** We can also just use Cut1 to prove Cut generally by iteratively using Cut1 on  $\Gamma_1, \Gamma_2, \dots, \Gamma_n, \Sigma \vdash_S \phi$ .

**Theorem 4.2 (Deduction Theorem).** If  $\Gamma, \phi \vdash_{PL} \psi$ , then  $\Gamma \vdash_{PL} \phi \rightarrow \psi$

*Proof.* Suppose  $\Gamma \cup \{\phi\} \vdash \psi$ . Thus, there exists some proof  $A$  from  $\Gamma \cup \{\phi\} \vdash \psi$ . Each line  $\alpha_i$  of  $A$  is either a member of  $\Gamma \cup \{\phi\}$ , an axiom, or follows from earlier lines in the proof by MP; the last line of  $A$  is  $\psi$ . Our strategy will be to establish that:

$$\text{for each } \alpha_i \text{ in proof } A, \Gamma \vdash \phi \rightarrow \alpha_i \quad (*)$$

We already know that each line of proof  $A_i$  is provable from  $\Gamma \cup \{\phi\}$ ; what  $(*)$  says is that if you stick “ $\phi \rightarrow$ ” in front of any of those lines, the result is provable from  $\Gamma$  all by itself. Once we succeed in establishing  $(*)$ , we will have proved the deduction theorem. For since the last line of the proof  $A$  is  $\psi$ ,  $(*)$  tells us that  $\phi \rightarrow \psi$  is provable from  $\Gamma$ .

$(*)$  says that each line of proof  $A$  has the property of being provable from  $\Gamma$  when prefixed with “ $\phi \rightarrow$ ”. We prove this by induction on the length of  $\psi$ 's proof.

Base Case: we must show that all starting points i.e. axioms and premises of  $A$  have the property. Suppose that line  $\alpha_i$  in  $A$  is an axiom, and we want to show that  $\alpha_i$  has the property of  $\Gamma \vdash \phi \rightarrow \alpha_i$ . Consider this:

1.  $\alpha_i$                       Axiom
2.  $\phi \rightarrow \alpha_i$             PL1, 1, MP

We do not need to actually use members from  $\Gamma$ . Next suppose that line  $\alpha_i$  is a premise – that is a member of  $\Gamma \cup \{\phi\}$ . This subdivides into two cases. The first is where  $\alpha_i$  is  $\phi$  itself. Here,  $\phi \rightarrow \alpha_i$  is just  $\phi \rightarrow \phi$  which can be proved from no premises. So,  $\Gamma \vdash \phi \rightarrow \phi$ . The second subcase is where  $\alpha_i \in \Gamma$ . But here we can prove  $\phi \rightarrow \alpha_i$  from  $\Gamma$  as follows:

1.  $\alpha_i$                       Premise
2.  $\phi \rightarrow \alpha_i$             PL1, 1, MP

Base Case proved.

Inductive Step: We assume that all earlier lines of the proof have the property we're interested in and we show that the result of applying MP to any of those earlier lines has the property as well. So suppose that  $\alpha_i$  follows from earlier lines in the proof by MP. That means that the earlier lines have to have the forms  $\chi \rightarrow \alpha_i$  and  $\chi$ . Furthermore, the inductive hypothesis tells us that the result of prefixing either of these lines with “ $\phi \rightarrow$ ”. Thus  $\Gamma \vdash \phi \rightarrow (\chi \rightarrow \alpha_i)$  and  $\Gamma \vdash \phi \rightarrow \chi$ . But then, given the MP technique and Cut,  $\Gamma \vdash \phi \rightarrow \alpha_i$ . So, by induction  $(*)$  is established, and the DT follows as a special case. ■

**Theorem 4.3 (Soundness for PL).** If  $\Gamma \vdash_{PL} \phi$ , then  $\Gamma \models_{PL} \phi$ .

*Proof.* (Sketch). We want to show that PL1-3 and any member of  $\Gamma$  are PL-valid (Lemma 1). Then, we want to show that PL-rules (MP) preserves PL-validity (Lemma 2). With these two lemmas, we can prove soundness by strong induction on the length of a proof, denoted  $n$ .

*Base Case.*  $n = 0$ . Suppose  $\Gamma \vdash_0 \phi$ . Then  $\phi$  is a PL-axiom or a member of  $\Gamma$ . So,  $\Gamma \models_{PL} \phi$ .

Let  $n$  be given.

*Induction Hypothesis.* Suppose for  $m < n$  :  $\Gamma \vdash_m \phi$  implies  $\Gamma \models_{\text{PL}} \phi$ .

*Induction Step.* Suppose  $\Gamma \vdash_n \phi$ . Consider how the last line,  $\phi$ , is obtained. There are three cases:

1.  $\phi$  is a PL-axiom. Then  $\Gamma \models_{\text{PL}} \phi$
2.  $\phi$  is a member of  $\Gamma$ . Then  $\Gamma \models_{\text{PL}} \phi$ .
3.  $\phi$  is obtained by MP. Then  $\Gamma \vdash_{m_1} \psi$  and  $\Gamma \vdash_{m_2} \psi \rightarrow \phi$  with  $m_1, m_2 < n$ . By IH,  $\Gamma \models_{\text{PL}} \psi$  and  $\Gamma \models_{\text{PL}} \psi \rightarrow \phi$ . So  $\Gamma \models_{\text{PL}} \phi$  (by Lemma 2)

■

**Theorem 4.4 (Completeness for PL).** If  $\Gamma \models_{\text{PL}} \phi$ , then  $\Gamma \vdash_{\text{PL}} \phi$

### Facts about Consistent Sets in PL

**Proposition 4.1.** Let  $\perp$  abbreviate a canonical contradiction,  $\sim (P \rightarrow P)$ . In the context of PL, we say that  $\Gamma$  is consistent if  $\Gamma \vdash_{\text{PL}} \perp$ ; otherwise, if  $\Gamma \not\vdash_{\text{PL}} \perp$ ,  $\Gamma$  is said to be consistent. We write  $\Gamma \vdash_{\text{PL}}$  to mean that  $\Gamma$  is inconsistent. Then,

- (a)  $\Gamma \vdash_{\text{PL}}$  iff for every formula  $\phi$ ,  $\Gamma \vdash_{\text{PL}} \phi$
- (b)  $\Gamma, \sim \phi \vdash_{\text{PL}}$  iff  $\Gamma \vdash_{\text{PL}} \phi$
- (c)  $\Gamma, \phi \vdash_{\text{PL}}$  iff  $\Gamma \vdash_{\text{PL}} \sim \phi$

*Proof.* We prove each statement in turn

- (a)  $\Gamma \vdash_{\text{PL}}$  iff for every formula  $\phi$ ,  $\Gamma \vdash_{\text{PL}} \phi$ .  
 $(\Rightarrow)$  Assume  $\Gamma \vdash_{\text{PL}}$ . Then, we know that  $\vdash_{\text{PL}} \sim \phi \rightarrow (\phi \rightarrow \psi)$ . So,  $\vdash_{\text{PL}} \sim (P \rightarrow P) \rightarrow ((P \rightarrow P) \rightarrow \phi)$ , where  $\phi$  is any wff. By Modus Ponens,  $\sim (P \rightarrow P) \vdash_{\text{PL}} (P \rightarrow P) \rightarrow \phi$ . Now, we also have  $\vdash_{\text{PL}} P \rightarrow P$ . By Cut,  $\perp \vdash_{\text{PL}} \phi$ . By Cut again,  $\Gamma \vdash_{\text{PL}} \phi$ .  
 $(\Leftarrow)$  Assume for any  $\phi$ ,  $\Gamma \vdash_{\text{PL}} \phi$ . Then trivially, we can let  $\phi$  be  $\sim (P \rightarrow P)$ . Hence,  $\Gamma \vdash_{\text{PL}} \perp \implies \Gamma \vdash_{\text{PL}}$
- (b)  $\Gamma, \sim \phi \vdash_{\text{PL}}$  iff  $\Gamma \vdash_{\text{PL}} \phi$   
 $(\Rightarrow)$  Assume  $\Gamma, \sim \phi \vdash_{\text{PL}}$ . By DT,  $\Gamma \vdash_{\text{PL}} \sim \phi \rightarrow \sim (P \rightarrow P)$ . Contraposition holds for PL, so  $\sim \psi \rightarrow \sim \phi \vdash_{\text{PL}} \phi \rightarrow \psi$ . Then, substituting gives us

$$\sim \phi \rightarrow \sim (P \rightarrow P) \vdash_{\text{PL}} (P \rightarrow P) \rightarrow \phi$$

By a twice application of Cut,  $\Gamma \vdash_{\text{PL}} \phi$   $(\Leftarrow)$  Assume that  $\Gamma \vdash_{\text{PL}} \phi$ . We want to show that  $\phi, \sim \phi \vdash_{\text{PL}} \sim (P \rightarrow P)$ , which immediately follows from the ex falso quodlibet rule by substituting  $\sim (P \rightarrow P)$  for  $\psi$ . By Cut,  $\Gamma, \sim \phi \vdash_{\text{PL}}$

- (c)  $\Gamma, \phi \vdash_{\text{PL}}$  iff  $\Gamma \vdash_{\text{PL}} \sim \phi$   
 $(\Rightarrow)$  Assume  $\Gamma, \phi \vdash_{\text{PL}}$ . By DT,  $\Gamma \vdash_{\text{PL}} \phi \rightarrow \sim (P \rightarrow P)$ . Again, we can use contraposition to show that  $\sim \sim \phi \rightarrow \sim (P \rightarrow P) \vdash_{\text{PL}} (P \rightarrow P) \rightarrow \sim \phi$ . We also know that  $\phi \rightarrow \sim (P \rightarrow P) \vdash_{\text{PL}} \sim \sim \phi \rightarrow \sim (P \rightarrow P)$ , by transitivity of the conditional and  $\vdash_{\text{PL}} \phi \rightarrow \sim \sim \phi$ . Now, we can assert the following:

$$\begin{aligned} \phi \rightarrow \sim (P \rightarrow P) \vdash_{\text{PL}} \sim \sim \phi \rightarrow \sim (P \rightarrow P) & \quad (\text{PL}) \\ \phi \rightarrow \sim (P \rightarrow P) \vdash_{\text{PL}} (P \rightarrow P) \rightarrow \sim \phi & \quad (\text{Cut}) \\ \Gamma \vdash_{\text{PL}} (P \rightarrow P) \rightarrow \sim \phi & \quad (\text{Assumption, Cut}) \\ \Gamma \vdash_{\text{PL}} \sim \phi & \quad (\text{PL, Cut}) \end{aligned}$$

$(\Leftarrow)$  Assume  $\Gamma \vdash_{\text{PL}} \sim \phi$ . From the ex falso quodlibet rule,  $\phi, \sim \phi \vdash_{\text{PL}}$ . By Cut,  $\Gamma, \phi \vdash_{\text{PL}}$

■

## 4.2 MPL metatheory

Let  $S$  be any one of the systems K, D, T, B, S4 or S5. We want to yield

**Theorem 4.5 (Adequacy for MPL).**  $\vdash_S \phi$  iff  $\models_S \phi$

This follows from soundness and completeness. The strategy to prove each is largely similar. For each theorem, we prove 2 systems as an example.

**Theorem 4.6 (Soundness for MPL).** If  $\vdash_S \phi$ , then  $\models_S \phi$

*Rough Sketch of the Proof for any system S.* Recall that  $\vdash_S \phi$  means that there is a finite sequence of wffs terminating in  $\phi$ , each member of which is an S-axiom of the result of applying an S-rule to earlier members. But each S-axiom is S-valid. And each S-rule preserves S-validity. Consequently, each member of the proof sequence – in particular,  $\phi$  – is valid. So  $\models_S \phi$ .

This strategy applies to K, D, T, B, S4 and S5. We will prove soundness for S5.

**Soundness for S5.**

**Proposition 4.2.** If  $\vdash_{S5} \phi$ , then  $\models_{S5} \phi$

We first prove two auxiliary lemmas.

**Lemma 4.7.** S5 axioms are S5 valid i.e.

$$\models_{S5} \phi \rightarrow (\psi \rightarrow \phi) \quad (\text{PL1})$$

$$\models_{S5} (\phi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \chi)) \quad (\text{PL2})$$

$$\models_{S5} (\sim \psi \rightarrow \sim \phi) \rightarrow ((\sim \psi \rightarrow \phi) \rightarrow \psi) \quad (\text{PL3})$$

$$\models_{S5} \Box(\phi \rightarrow \psi) \rightarrow (\Box\phi \rightarrow \Box\psi) \quad (\text{K})$$

$$\models_{S5} \Box\phi \rightarrow \phi \quad (\text{T})$$

$$\models_{S5} \Diamond\Box\phi \rightarrow \Box\phi \quad (\text{S5})$$

*Proof.* The K, T and S5 axioms have been proven earlier. The PL1-3 axioms have also been proven earlier to be PL-valid. To generalise to S5-validity, we just need to relativise the valuation functions to some  $w$ . This obviously generalises because there are no modal operators so  $\mathcal{R}$  does not come into play ■

**Lemma 4.8.** S5 rules preserve S5 validity i.e.

$$\text{if } \models_{S5} \phi, \models_{S5} \phi \rightarrow \psi, \text{ then } \models_{S5} \psi \quad (\text{MP})$$

$$\text{if } \models_{S5} \phi, \text{ then } \models_{S5} \Box\phi \quad (\text{NEC})$$

*Proof.* We prove both rules preserve validity:

- MP: Assume  $\models_{S5} \phi, \models_{S5} \phi \rightarrow \psi$ . Then,  $V_{\mathcal{M}}(\phi, w) = 1$  and  $V_{\mathcal{M}}(\phi \rightarrow \psi, w) = 1$  for every  $w \in \mathcal{W}$  and for every  $\mathcal{M}$ . Then, by the semantic definition of  $\rightarrow$ , in any model  $\mathcal{M}$  and world  $w$ , if  $V_{\mathcal{M}}(\phi, w) = 1$ ,  $V_{\mathcal{M}}(\psi, w) \neq 0$  since otherwise,  $V_{\mathcal{M}}(\phi \rightarrow \psi, w) = 0$ . Hence,  $V_{\mathcal{M}}(\psi, w) = 1$ . Thus,  $\models_{S5} \psi$
- NEC: Assume  $\models_{S5} \phi$ . This means that for all models  $\mathcal{M}$  and for all  $w \in \mathcal{W}$  for any  $\mathcal{M}$ ,  $V_{\mathcal{M}}(\phi, w) = 1$ . Then, since for any  $\mathcal{M}$ ,  $V_{\mathcal{M}}(\phi, w) = 1$  for all  $w \in \mathcal{W}$ , by the semantic clause for  $\Box$ ,  $V_{\mathcal{M}}(\Box\phi, w) = 1$ . Since this fact holds for all  $\mathcal{M}$ ,  $\models_{S5} \Box\phi$

■

We now begin the main proof by induction on the length of the proof.



*Proof.* Let  $\vdash_n \phi$  means that there is a proof of  $\phi$  (in S5) in which there are  $n$  applications of S5-axioms or S5-rules (MP or NEC). We want to show that  $\vdash_n \phi$  implies  $\models_{S5} \phi$ .

Base Case:  $n = 1$ . Suppose  $\vdash_1 \phi$ . By def,  $\phi$  is an S5-axiom. By the first lemma,  $\models_{S5} \phi$ .

Let  $n$  be given.

IH: Suppose for all  $m < n$  :  $\vdash_m \phi$  implies  $\models_{S5} \phi$ . We want to show given IH, if  $\vdash_n \phi$ , then  $\models_{S5} \phi$ .

Suppose  $\vdash_n \phi$ . Then, the last line  $\phi$  occurs in three cases:

1.  $\phi$  is an S5-axiom: By the first lemma,  $\models_{S5} \phi$
2.  $\phi$  is obtained by NEC: Then,  $\phi = \Box\psi$  and  $\vdash_m \psi$  with  $m < n$ . By IH,  $\models_{S5} \psi$ . By the second lemma,  $\models_{S5} \Box\psi$  and hence  $\models_{S5} \phi$
3.  $\phi$  is obtained by MP: Then  $\vdash_{m_1} \psi, \vdash_{m_2} \psi \rightarrow \phi$  with  $m_1, m_2 < n$ . By IH,  $\models_{S5} \psi, \models_{S5} \psi \rightarrow \phi$ . By the second lemma,  $\models_{S5} \phi$ .

So, the IH is established and for any proof of length  $n$ , if  $\vdash_{S5} \phi$ , then  $\models_{S5} \phi$  ■

**Theorem 4.9 (Completeness for MPL).** If  $\models_S \phi$ , then  $\vdash_S \phi$ .

While this proof is non-examinable, we take interest in the new notion of provability, how we can prove the deduction theorem for MPL and the properties of maximally consistent sets.

**Definition 4.1 (Consistency of a Set).** Let  $\perp$  abbreviate the canonical contradiction  $\sim (P \rightarrow P)$ . A set of wffs  $\Gamma$  is:

- S-consistent iff  $\Gamma \not\vdash_S \perp$
- S-inconsistent iff  $\Gamma \vdash_S \perp$

We write  $\Gamma \not\vdash_S$  when  $\Gamma$  is S-consistent and  $\Gamma \vdash_S$  when  $\Gamma$  is S-inconsistent

*Remark:* Notice that  $\perp$  abbreviates a specific wff, and NOT  $\sim (\phi \rightarrow \phi)$ . Therefore, when you prove that  $\Gamma \vdash_S \perp$ , you must prove that  $\Gamma \vdash_S \sim (P \rightarrow P)$ .

**Definition 4.2 (Maximal Consistent Set).** A set  $\Theta$  is maximally consistent iff (i) for any  $\phi$ , either  $\phi \in \Theta$  or  $\sim \phi \in \Theta$  and (ii)  $\Theta \not\vdash_S$

In other words, a set  $\Theta$  is maximally consistent iff it includes either the truth or falsity of every statement and is consistent.

Notice that our definition of provability does not license proofs from a set of premises i.e.  $\Gamma \vdash_S \phi$ , and only proofs from the empty set i.e.  $\vdash_S \phi$ . So, we need a new definition of provability from a set.

**Definition 4.3 (Provability from a Set).** A wff  $\phi$  is provable from a set  $\Gamma$  iff  $\vdash_S (\gamma_1 \wedge \dots \wedge \gamma_n) \rightarrow \phi$  for some  $\gamma_1, \dots, \gamma_n \in \Gamma$ .

This new definition allows for the Deduction Theorem to work for MPL and for Cut to work.

**Theorem 4.10 (Deduction Theorem for MPL).** Under the definition of provability from a set, if  $\Gamma, \phi \vdash_S \psi$ ,  $\Gamma \vdash_S \phi \rightarrow \psi$

*Proof.* Suppose  $\Gamma \cup \{\phi\} \vdash_S \psi$ . So, for some  $\alpha_1, \dots, \alpha_n \vdash_S (\alpha_1 \dots, \alpha_n) \rightarrow \psi$ , where perhaps one of the  $\alpha_i$ s is  $\phi$  and the others are members of  $\Gamma$ . If  $\phi$  is one of the  $\alpha_i$ s, say  $\alpha_k$ , then  $(\alpha_1 \wedge \dots \wedge \alpha_{k-1} \wedge \alpha_{k+1} \wedge \dots \wedge \alpha_n) \rightarrow (\phi \rightarrow \psi)$  is an MPL-tautological consequence  $(\alpha_1 \wedge \dots \wedge \alpha_n) \rightarrow \psi$ , and so is a theorem of S, whence  $\Gamma \vdash_S \phi \rightarrow \psi$ . And if none of the  $\alpha_i$ s is  $\phi$ , then each is in  $\Gamma$ ; but  $(\alpha_1 \wedge \dots \wedge \alpha_n) \rightarrow (\phi \rightarrow \psi)$  is an MPL-tautological consequence of  $(\alpha_1 \wedge \dots \wedge \alpha_n) \rightarrow \psi$ , whence again  $\Gamma \vdash_S \phi \rightarrow \psi$  ■

**Theorem 4.11 (Cut for MPL).** Under the definition of provability from a set, if  $\Gamma_1 \vdash_S \delta_1, \dots, \Gamma_n \vdash_S \delta_n$  and  $\Sigma, \delta_1, \dots, \delta_n \vdash_S \phi$ , then  $\Gamma_1, \dots, \Gamma_n, \Sigma \vdash_S \phi$ .

*Proof.* This is just an application of transitivity of the material conditional. Consider the two-case:  $\Gamma \vdash_S \delta, \Sigma, \delta \vdash_S \phi$ . Then, by definition,  $\vdash_S (\gamma_1 \wedge \dots \wedge \gamma_n) \rightarrow \delta$ , for some  $\gamma_1, \dots, \gamma_n \in \Gamma$  and  $\vdash_S (\sigma_1 \wedge \dots \wedge \sigma_n \wedge \delta) \rightarrow \phi$ , for  $\sigma_1, \dots, \sigma_n \in \Sigma$ . By PL,  $\vdash_S (\sigma_1 \wedge \dots \wedge \sigma_n \wedge \gamma_1 \wedge \dots \wedge \gamma_n) \rightarrow \phi$ , which implies  $\Sigma, \Gamma \vdash_S \phi$ . Now, we can generalise this to the  $n$  case by an iterative application of the two-case. ■

**Proposition 4.3 (Properties of S Provability from a set).** The new definition of provability from a set, the following features hold

- (a) Excluded Middle MP:  $\phi \rightarrow \psi, \sim \phi \rightarrow \psi \vdash_S \psi$
- (b) Ex Falso Quodlibet:  $\phi, \sim \phi \vdash_S \psi$
- (c) Modus Ponens:  $\phi, \phi \rightarrow \psi \vdash_S \psi$
- (d) Negated Conditional:  $\sim (\phi \rightarrow \psi) \vdash_S \phi$  and  $\sim (\phi \rightarrow \psi) \vdash_S \sim \psi$
- (e) if  $\phi \in \Gamma$ , then  $\Gamma \vdash_S \phi$

*Proof.* We prove each in turn

- (a) Excluded Middle MP:  $\phi \rightarrow \psi, \sim \phi \rightarrow \psi \vdash_S \psi$   
First,  $\phi \rightarrow \psi, \sim \phi \rightarrow \psi \vdash_{PL} \psi$ , which is the same as  $\phi \rightarrow \psi \wedge \sim \phi \rightarrow \psi \vdash_{PL} \psi$  By DT,  $\vdash_{PL} ((\phi \rightarrow \psi) \wedge (\sim \phi \rightarrow \psi)) \rightarrow \psi$  By completeness of PL,  $\models_{PL} ((\phi \rightarrow \psi) \wedge (\sim \phi \rightarrow \psi)) \rightarrow \psi$ . So, it is an MPL tautology. Since all MPL-tautologies are S-theorems, we conclude  $\vdash_S ((\phi \rightarrow \psi) \wedge (\sim \phi \rightarrow \psi)) \rightarrow \psi$ , and the result follows by definition.
- (b) Ex Falso Quodlibet:  $\phi, \sim \phi \vdash_S \psi$   
First,  $\phi, \sim \phi \vdash_{PL} \psi$ , which is the same as  $\phi \wedge \sim \phi \vdash_S \psi$  Then, by DT,  $\vdash_{PL} (\phi \wedge \sim \phi) \rightarrow \psi$ . By PL completeness,  $\models_{PL} (\phi \wedge \sim \phi) \rightarrow \psi$ . Since all MPL tautologies are S-theorems,  $\vdash_S (\phi \wedge \sim \phi) \rightarrow \psi$ , and the desired result follows immediately by definition.
- (c) Modus Ponens:  $\phi, \phi \rightarrow \psi \vdash_S \psi$   
First,  $\phi, \phi \rightarrow \psi \vdash_{PL} \psi$ , which is the same as  $\phi \wedge \phi \rightarrow \psi \vdash_{PL} \psi$ . By DT,  $\vdash_{PL} (\phi \wedge (\phi \rightarrow \psi)) \rightarrow \psi$ . By completeness of PL,  $\models_{PL} (\phi \wedge (\phi \rightarrow \psi)) \rightarrow \psi$ . Hence,  $\vdash_S (\phi \wedge (\phi \rightarrow \psi)) \rightarrow \psi$ . This follows as an MPL-tautology and hence a theorem of S. The result follows by definition.
- (d) Negated Conditional:  $\sim (\phi \rightarrow \psi) \vdash_S \phi$  and  $\sim (\phi \rightarrow \psi) \vdash_S \sim \psi$   
Same strategy as above
- (e) if  $\phi \in \Gamma$ , then  $\Gamma \vdash_S \phi$   
Assume  $\phi \in \Gamma$ . Then, by definition, since  $\vdash_S (\gamma_1 \wedge \dots \wedge \gamma_n) \rightarrow \phi$ , for some  $\gamma_1, \dots, \gamma_n \in \Gamma$ , choose  $\gamma := \phi$ , and hence  $\vdash_S \phi \rightarrow \phi$ . This last step is justified since it is a MPL-tautology and hence a theorem of S.
- (f) if  $\Gamma \vdash_S \phi$ , then  $\gamma_1, \dots, \gamma_n \vdash_S \phi$ , for some  $\gamma_1, \dots, \gamma_n \in \Gamma$  (or else  $\vdash_S \phi$ ).  
Just follows by definition of provability from a set. ■

Now we can arrive at the properties of maximal consistent sets  $\Theta$ .

**Lemma 4.12 (Properties of consistent sets).** Let  $\Gamma$  be a set of wffs.

- (a)  $\Gamma \vdash_S$  iff, for every  $\phi$ ,  $\Gamma \vdash_S \phi$
- (b)  $\Gamma, \sim \phi \vdash_S$  iff  $\Gamma \vdash_S \phi$
- (c)  $\Gamma, \phi \vdash_S$  iff  $\Gamma \vdash_S \sim \phi$

*Proof.* Given Cut and Deduction Theorem for MPL (proven above), we can just prove it in the exact same way as in Proposition 4.1. ■

**Lemma 4.13 (Properties of maximal consistent sets).** A maximal S-consistent set  $\Theta$  has the following properties:

1.  $\phi \in \Theta$  iff  $\Theta \vdash_S \phi$
2. (a)  $\sim \phi \in \Theta$  iff  $\phi \notin \Theta$   
 (b)  $\phi \rightarrow \psi \in \Theta$  iff  $\phi \notin \Theta$  or  $\psi \in \Theta$   
 (c)  $\Box \phi \in \Theta$  iff  $\phi \in \Sigma$  for every maximally S-consistent  $\Sigma$  with  $\mathcal{R}\Theta, \Sigma$

*Remark:* The relation over maximally consistent words is defined as  $\mathcal{R}\Theta, \Sigma$  iff  $\Box^-(\Theta) := \{\phi : \Box \phi \in \Theta\} \subseteq \Sigma$ .

*Proof.* We prove each in turn:

1. Claim:  $\phi \in \Theta$  iff  $\Theta \vdash_S \phi$   
 ( $\Rightarrow$ ) Assume  $\phi \in \Theta$ . Then,  $\models_{PL} \phi \rightarrow \phi$  which implies  $\vdash_{PL} \phi \rightarrow \phi$ , which implies  $\vdash_S \phi \rightarrow \phi$ , which by assumption implies  $\Theta \vdash_S \phi$ .  
 ( $\Leftarrow$ ) Assume  $\Theta \vdash_S \phi$ . From definition, for any  $\phi$ , either  $\phi \in \Theta$  or  $\sim \phi \in \Theta$ . Assume for contradiction that  $\sim \phi \in \Theta$ . Then, again,  $\models_{PL} \sim \phi \rightarrow \sim \phi$ , implying  $\vdash_{PL} \sim \phi \rightarrow \sim \phi$ , implying  $\vdash_S \sim \phi \rightarrow \sim \phi$ , which by definition implies that  $\Theta \vdash_S \sim \phi$ . So,  $\Theta \vdash_S \phi$  and  $\Theta \vdash_S \sim \phi$ , which we can show contradicts consistency as it follows that  $\Theta \vdash_S \phi \wedge \sim \phi$ . Then, we know that  $\models_{PL} (\phi \wedge \sim \phi) \rightarrow \sim (P \rightarrow P)$ , which implies  $\vdash_S (\phi \wedge \sim \phi) \rightarrow \sim (P \rightarrow P)$ , so  $\Theta \vdash_S \perp$  by transitivity of the conditional. For more detail, consider that we can yield the following:

$$\begin{aligned} \vdash_S (\gamma_1 \wedge \dots \wedge \gamma_n) &\rightarrow (\phi \wedge \sim \phi) \\ \vdash_S (\gamma_1 \wedge \dots \wedge \gamma_n) &\rightarrow ((\phi \wedge \sim \phi) \rightarrow \sim (P \rightarrow P)) \\ \vdash_S (\gamma_1 \wedge \dots \wedge \gamma_n) &\rightarrow \sim (P \rightarrow P) \end{aligned}$$

The second line follows from the property that if  $\vdash_S \phi$ , then  $\vdash_S \psi \rightarrow \phi$  (we could do away with this assumption and apply Cut instead)

2. We then prove the sub-lemmas

- (a) Claim:  $\sim \phi \in \Theta$  iff  $\phi \notin \Theta$   
 ( $\Rightarrow$ ) Assume  $\sim \phi \in \Theta$ . AFC that  $\phi \in \Theta$ . Then by (1),  $\Theta \vdash_S \phi$  and  $\Theta \vdash_S \sim \phi$ , which violates consistency as shown earlier. So,  $\phi \notin \Theta$ .  
 ( $\Leftarrow$ ) Assume  $\phi \notin \Theta$ . By def of  $\Theta$ , for any  $\phi$ , either  $\phi \in \Theta$  or  $\sim \phi \in \Theta$ . So,  $\sim \phi \in \Theta$
- (b) Claim:  $\phi \rightarrow \psi \in \Theta$  iff  $\phi \notin \Theta$  or  $\psi \in \Theta$   
 ( $\Rightarrow$ ) Assume  $\phi \rightarrow \psi \in \Theta$ . Then  $\Theta \vdash_S \phi \rightarrow \psi$  by (1). AFC that  $\phi \in \Theta$  and  $\psi \notin \Theta$ . By the definition of  $\Theta$ ,  $\sim \psi \in \Theta$ . Since  $\vdash_S (\phi \wedge \sim \psi) \rightarrow \sim (\phi \rightarrow \psi)$  which follows because it is a MPL tautology and hence an S theorem. So  $\Theta \vdash_S \sim (\phi \rightarrow \psi)$ . As usual, we can show that  $\Theta$  is inconsistent.

An alternative strategy would be to AFC that  $\phi \in \Theta$  and  $\psi \notin \Theta$ , show that  $\Theta \vdash_S \sim (\phi \rightarrow \psi)$ , define  $\Theta := \Gamma \cup \{\phi \rightarrow \psi\}$ . Then, since  $\Gamma \vdash_S \sim (\phi \rightarrow \psi)$ ,  $\Theta := \Gamma \cup \{\phi \rightarrow \psi\} \vdash_S \perp$ .

( $\Leftarrow$ ) Assume  $\phi \notin \Theta$  or  $\psi \in \Theta$ . Then, suppose for reductio that  $\phi \rightarrow \psi \notin \Theta$ . So,  $\sim (\phi \rightarrow \psi) \in \Theta$ . Since  $\models_{PL} \sim (\phi \rightarrow \psi) \rightarrow (\phi \wedge \sim \psi)$ , implying  $\vdash_S \sim (\phi \rightarrow \psi) \rightarrow (\phi \wedge \sim \psi)$ . By definition,  $\Theta \vdash_S \phi \wedge \sim \psi$ . Since  $\vdash_S (\phi \wedge \sim \psi) \rightarrow \phi$  and  $\vdash_S (\phi \wedge \sim \psi) \rightarrow \sim \psi$ , then  $\Theta \vdash_S \phi$  and  $\Theta \vdash_S \sim \psi$  due to transitivity. Then, by (1), we have  $\phi \in \Theta, \sim \psi \in \Theta$ , a contradiction.

- (c) Claim:  $\Box \phi \in \Theta$  iff  $\phi \in \Sigma$  for every maximally S-consistent  $\Sigma$  with  $\mathcal{R}\Theta, \Sigma$ .  
 ( $\Rightarrow$ ) This direction is trivial. Assume  $\Box \phi \in \Theta$ . Then for some arbitrary  $\Sigma$  where  $\mathcal{R}\Theta, \Sigma$ , since  $\{\phi : \Box \phi \in \Theta\} \subseteq \Sigma$ , this means that  $\phi \in \Sigma$ .  
 ( $\Leftarrow$ ) Assume for every maximally consistent set  $\Sigma$  s.t.  $\mathcal{R}\Theta, \Sigma$ ,  $\phi \in \Sigma$ . Assume for contradiction that  $\Box \phi \notin \Theta$ . Then, since  $\Box^-(\Theta) = \{\phi : \Box \phi \in \Theta\} \subseteq \Sigma$ , it is possible to construct a maximally consistent set  $\Sigma$  s.t.  $\sim \phi \in \Sigma$  and thus  $\phi \notin \Sigma$  for some maximally consistent  $\Sigma$ , which contradicts our assumption that  $\phi \in \Sigma$  for every maximal consistent set  $\Sigma$  s.t.  $\mathcal{R}\Theta, \Sigma$ .

■

Another thing maximal consistent sets are used in is in what Sider calls canonical models.

**Definition 4.4 (Canonical Models for S).** Define a canonical model for S as  $\mathcal{M}_S = \langle \mathcal{W}, \mathcal{R}, \mathcal{I} \rangle$  as follows:

- $\mathcal{W} = \{\Theta : \Theta \text{ is maximally S-consistent}\}$
- $\mathcal{R}\Theta, \Sigma$  iff  $\phi \in \Sigma$  whenever  $\Box\phi \in \Theta$
- $\mathcal{I}(\alpha, \Theta) = 1$  iff  $\alpha \in \Theta$ , for each sentence letter  $\alpha$

Now we want to give the proof of completeness, which relies on two key lemmas

**Lemma 4.14 (Lindenbaum's Lemma).** Every S-consistent set has a maximally S-consistent superset: i.e. if  $\Gamma$  is a consistent set, there is a maximally consistent set  $\Theta$ , with  $\Gamma \subseteq \Theta$

Next, we say that a set of wffs  $\Theta$  is satisfied at some world  $w$  of some Kripke model  $\mathcal{M} = \langle \mathcal{W}, \mathcal{R}, \mathcal{I} \rangle$  iff  $V_{\mathcal{M}}(\phi, w) = 1$  for each  $\phi \in \Theta$  – “ $\mathcal{M}$  is a model for  $\Theta$ ”.

**Lemma 4.15 (Canonical Model Lemma).** Every maximally S-consistent set  $\Theta$  is satisfied by some world  $w$  in some model  $\mathcal{M}$ . In fact, we can pick  $\mathcal{M}$  to be ‘the canonical model’ for S –  $\mathcal{M}_S$

Then, we proceed with the proof of completeness

*Proof.* First, assume  $\Gamma \not\vdash_S \phi$  for contraposition. Then,

- $\therefore \Gamma, \sim \phi \not\vdash_S$  (property of inconsistent sets)
- $\therefore$  There is some maximally S-consistent  $\Theta \supseteq \Gamma \cup \{\sim \phi\}$  (Lindenbaum's Lemma)
- $\therefore \Theta$  is satisfied by some  $w$  in  $\mathcal{M}_S$  (Canonical Model Lemma)
- $\therefore \Gamma \not\vdash_S \phi$

By contraposition, if  $\Gamma \models_S \phi$ , then  $\Gamma \vdash_S \phi$ . Completeness is established

■

### Admissibility of Metarules

**Definition 4.5 (Admissibility of Metarules).** We say that a rule R of the form  $\frac{\phi_1 \dots \phi_n}{\psi} R$  is K-admissible iff it preserves K-derivability i.e.  $\vdash_K \psi$  holds if  $\vdash_K \phi_1 \dots \vdash_K \phi_n$ .

**Example 4.1.** Show that

$$\frac{\phi_1 \leftrightarrow \phi_2}{\chi(\phi_1) \leftrightarrow \chi(\phi_2)} \text{Subst, where } \chi(\phi_i) \text{ uniformly substitutes } \phi_i \text{ for } \alpha \text{ in } \chi(\alpha)$$

*Proof.* We prove this by induction on the complexity of  $\chi$ .

- Base Case:  $\chi(\alpha)$  is a sentence letter, either the sentence letter  $\alpha$  or not the sentence letter  $\alpha$ .
  - If  $\chi(\alpha)$  is the sentence letter  $\alpha$ , then  $\chi(\phi_1) = \phi_1$  and  $\chi(\phi_2) = \phi_2$ . So, since  $\vdash_K \phi_1 \leftrightarrow \phi_2$ , also  $\vdash_K \chi(\phi_1) \leftrightarrow \chi(\phi_2)$
  - If  $\chi(\alpha)$  is not the sentence letter  $\alpha$ , then  $\chi(\alpha) = \chi(\phi_1) = \chi(\phi_2)$  i.e. the substitution has no effect. So, by PL,  $\vdash_K \chi(\phi_1) \leftrightarrow \chi(\phi_2)$
- Inductive case: we assume all wffs of smaller complexity than  $\chi$  has the desired property of K-admissibility
  - $\chi = \sim \psi$  and the inductive hypothesis holds for  $\psi$ . Then
    1.  $\vdash_K \phi_1 \leftrightarrow \phi_2$  i.e. Antecedent of the substitution rule
    2.  $\vdash_K \psi(\phi_1) \leftrightarrow \psi(\phi_2)$  using IH and 1

- 3.  $\vdash_K \sim \psi(\phi_1) \leftrightarrow \sim \psi(\phi_2)$  using PL and 2
- 4.  $\vdash_K \chi(\phi_1) \leftrightarrow \chi(\phi_2)$  using def of  $\chi$  and 3
- $\chi = \psi \rightarrow \eta$  and the IH holds for  $\psi$  and  $\eta$ . Then, we want to show that

$$\frac{\phi_1 \leftrightarrow \phi_2}{(\psi(\phi_1) \rightarrow \eta(\phi_1)) \leftrightarrow (\psi(\phi_2) \rightarrow \eta(\phi_2))}$$

We have three theorems given by assumption

- 1.  $\vdash_K \phi_1 \leftrightarrow \phi_2$  by Assumption
- 2.  $\vdash_K \psi(\phi_1) \leftrightarrow \psi(\phi_2)$  by IH and 1
- 3.  $\vdash_K \eta(\phi_1) \leftrightarrow \eta(\phi_2)$

We want to claim that the following is a PL-validity, and thus a K-validity and a K-theorem.

$$(\phi_1 \leftrightarrow \phi_2) \rightarrow ((\psi(\phi_1) \leftrightarrow \psi(\phi_2)) \rightarrow [(\eta(\phi_1) \leftrightarrow \eta(\phi_2)) \rightarrow ((\psi(\phi_1) \rightarrow \eta(\phi_1)) \leftrightarrow (\psi(\phi_2) \rightarrow \eta(\phi_2)))]]) \quad (1)$$

We now want to show that (1) is PL-valid

- 1. AFC that for some PL-interpretation  $\mathcal{J}, V_{\mathcal{J}}$  of (1) was 0. Since (1) is a chain of conditionals, we know that this means that the consequent  $V_{\mathcal{J}}((\psi(\phi_1) \rightarrow \eta(\phi_1)) \leftrightarrow (\psi(\phi_2) \rightarrow \eta(\phi_2))) = 0$  and all the antecedents in (1) are true i.e.
  - (a)  $V_{\mathcal{J}}(\phi_1 \leftrightarrow \phi_2) = 1$
  - (b)  $V_{\mathcal{J}}(\psi(\phi_1) \leftrightarrow \psi(\phi_2)) = 1$
  - (c)  $V_{\mathcal{J}}(\eta(\phi_1) \leftrightarrow \eta(\phi_2)) = 1$
- 2. WLOG, suppose that  $V_{\mathcal{J}}((\psi(\phi_1) \rightarrow \eta(\phi_1)) \leftrightarrow (\psi(\phi_2) \rightarrow \eta(\phi_2))) = 0$  because  $V_{\mathcal{J}}((\psi(\phi_1) \rightarrow \eta(\phi_1))) = 1$  but  $V_{\mathcal{J}}((\psi(\phi_2) \rightarrow \eta(\phi_2))) = 0$  (the converse direction is symmetrical). Then,
  - (a)  $V_{\mathcal{J}}((\psi(\phi_1) \rightarrow \eta(\phi_1))) = 1$
  - (b)  $V_{\mathcal{J}}(\eta(\phi_2)) = 0$  since  $V_{\mathcal{J}}((\psi(\phi_2) \rightarrow \eta(\phi_2))) = 0$
  - (c)  $V_{\mathcal{J}}(\psi(\phi_2)) = 1$  since  $V_{\mathcal{J}}((\psi(\phi_2) \rightarrow \eta(\phi_2))) = 0$
- 3. We now reason as follows
  - (a)  $V_{\mathcal{J}}(\psi(\phi_1)) = 1$  since  $V_{\mathcal{J}}(\psi(\phi_2)) = 1$  (2c) and  $V_{\mathcal{J}}(\phi_1 \leftrightarrow \phi_2) = 1$  (1a)
  - (b)  $V_{\mathcal{J}}(\eta(\phi_1)) = 0$  since  $V_{\mathcal{J}}(\eta(\phi_1) \leftrightarrow \eta(\phi_2)) = 1$  (1c) and  $V_{\mathcal{J}}(\eta(\phi_2)) = 0$  (2b)
  - (c) But 3a and 3b together violates that  $V_{\mathcal{J}}((\psi(\phi_1) \rightarrow \eta(\phi_1))) = 1$  from 2a

So,

$$\begin{aligned} &\vdash_K (\psi(\phi_1) \rightarrow \eta(\phi_1)) \leftrightarrow (\psi(\phi_2) \rightarrow \eta(\phi_2)) && \text{(due to (1))} \\ &\vdash_K \chi(\phi_1) \leftrightarrow \chi(\phi_2) \end{aligned}$$

- $\chi = \Box\psi$  and the inductive hypothesis holds for  $\psi$ . Then

- 1.  $\vdash_K \phi_1 \leftrightarrow \phi_2$  due to antecedent of substitution rule
- 2.  $\vdash_K \psi(\phi_1) \leftrightarrow \psi(\phi_2)$  by IH and 1
- 3.  $\vdash_K \Box\psi(\phi_1) \leftrightarrow \Box\psi(\phi_2)$  by Becker in both directions
- 4.  $\vdash_K \chi(\phi_1) \leftrightarrow \chi(\phi_2)$

■

## 5 Classical predicate logic, extensions, and deviations

### 5.1 Classical Predicate Logic

**Definition 5.1 (Primitive vocabulary of PC).** Predicate Calculus has the following vocabulary:

- connectives:  $\rightarrow, \sim, \forall$
- variables:  $x, y, \dots$  (with or without numerical subscripts)
- $n$ -place predicates:  $F, G, \dots$  (with or without numerical subscripts)
- individual constants (names):  $a, b, \dots$  (with or without numerical subscripts)
- parentheses

**Remark:**  $\exists \alpha \phi$  is short for  $\sim \forall \alpha \sim \phi$

**Definition 5.2 (Definition of a PC-term).** If  $\alpha$  is a variable or an individual constant,  $\alpha$  is a term

**Definition 5.3 (Definition of a PC-wff).** We can formulate a PC-wff recursively:

- If  $\Pi$  is an  $n$ -place predicate and  $\alpha_1, \dots, \alpha_n$  are terms,  $\Pi \alpha_1, \dots, \alpha_n$  is a wff
- If  $\phi$  and  $\psi$  are wffs, and  $\alpha$  is a variable,  $\sim \phi$ ,  $(\phi \rightarrow \psi)$ , and  $\forall \alpha \phi$  are wffs
- Only strings that can be shown to be terms and wffs using these clauses are wffs.

**Definition 5.4 (Free variable occurrence).** An occurrence of variable  $\alpha$  in wff  $\phi$  is bound in  $\phi$  iff it occurs in a wff of the form  $\forall \alpha \psi$ . Otherwise it is free.

In other words, all and only the occurrences of  $\alpha$  in the scope of a quantifier  $\forall \alpha$  are bound.

**Definition 5.5 (PC-model).** A PC-model is a pair  $\langle \mathcal{D}, \mathcal{I} \rangle$  such that:

- $\mathcal{D}$  is a non-empty set i.e. the domain
- $\mathcal{I}$  is a function meeting the following condition:
  - $\mathcal{I}(\alpha) \in \mathcal{D}$  for any constant  $\alpha$
  - $\mathcal{I}(\Pi)$  is an  $n$ -place relation over  $\mathcal{D}$  for an  $n$ -place predicate  $\Pi$

i.e. an interpretation function.

**Remark:** an  $n$ -place relation over  $\mathcal{D}$  is a set of  $n$ -tuples of members of  $\mathcal{D}$

**Definition 5.6 (Variable assignment).** A variable assignment  $g$  for a PC-model  $\mathcal{M} = \langle \mathcal{D}, \mathcal{I} \rangle$  is a function  $g(\alpha) \in \mathcal{D}$  for each variable  $\alpha$

We now have variant assignments that is the same as the original assignment but changes the semantic value for just one variable.

**Definition 5.7 (Variant assignment).** When  $g$  is a variable assignment for  $\langle \mathcal{D}, \mathcal{I} \rangle$  and  $u \in \mathcal{D}$ , we define  $g_u^\alpha$  as follows:

$$g_u^\alpha(\beta) = \begin{cases} g(\beta) & \text{if } \beta \neq \alpha \\ u & \text{if } \beta = \alpha \end{cases}$$

**Remark:** Basically, we read  $g_u^\alpha$  as sending  $\alpha$  to  $u$ . In other words, it keeps the semantic value of all other variables the same except  $\alpha$ , and changes  $\alpha$  to  $u$ .

**Definition 5.8 (Term Denotation).** Let  $\mathcal{M} = \langle \mathcal{D}, \mathcal{I} \rangle$  be a PC-model,  $g$  a assignment for  $\mathcal{M}$ :

- $[\alpha]_{\mathcal{M},g} = \mathcal{I}(\alpha)$  if  $\alpha$  is a constant
- $[\alpha]_{\mathcal{M},g} = g(\alpha)$  if  $\alpha$  is a variable

**Definition 5.9 (PC-valuation).** The valuation function  $V_{\mathcal{M},g}$  for PC-model  $\mathcal{M} = \langle \mathcal{D}, \mathcal{I} \rangle$  and variable assignment  $g$  is the unique function that assigns 0 or 1 to each wff and satisfies the following conditions:

- $V_{\mathcal{M},g}(\Pi\alpha_1, \dots, \alpha_n) = 1$  iff  $\langle [\alpha_1]_{\mathcal{M},g}, \dots, [\alpha_n]_{\mathcal{M},g} \rangle \in \mathcal{I}(\Pi)$ , for terms  $\alpha_i$ ,  $n$ -ary predicate  $\Pi$ .
- $V_{\mathcal{M},g}(\sim \phi) = 1$  iff  $V_{\mathcal{M},g}(\phi) = 0$ , for wff  $\phi$
- $V_{\mathcal{M},g}(\phi \rightarrow \psi) = 1$  iff  $V_{\mathcal{M},g}(\phi) = 0$  or  $V_{\mathcal{M},g}(\psi) = 1$ , for wffs  $\phi$  and  $\psi$
- $V_{\mathcal{M},g}(\forall \alpha \phi) = 1$  iff, for every  $u \in \mathcal{D}$ ,  $V_{\mathcal{M},g_u^\alpha}(\phi) = 1$ , for wff  $\phi$  and variable  $\alpha$

We now can define truth in a model (truth under all assignments in that model)

**Definition 5.10 (Truth in a PC-model).** A wff  $\phi$  is true in a PC-model  $\mathcal{M} = \langle \mathcal{D}, \mathcal{I} \rangle$  iff  $V_{\mathcal{M},g}(\phi) = 1$  for every assignment  $g$  for  $\mathcal{M}$

We have two facts about truth values of PC-wffs.

The first says that the truth value of a sentence is unaffected by the assignment.

**Proposition 5.1.** If  $g$  and  $h$  agree on the free variables in  $\phi$ ,  $V_{\mathcal{M},g}(\phi) = V_{\mathcal{M},h}(\phi)$

*Proof.* Suppose  $g$  and  $h$  agree on free variables. We prove that the consequent holds on the complexity of  $\phi$  i.e.  $V_{\mathcal{M},g}(\phi) = V_{\mathcal{M},h}(\phi)$ .

BC:  $\phi = \Pi\alpha_1 \dots \alpha_n$ . Then, each term  $\alpha_i$  is either a constant or a freely occurring variable. If  $\alpha_i$  is a constant, then for any pair  $g, h$

$$[\alpha_i]_g = [\alpha_i]_h$$

If  $\alpha_i$  is a variable, by assumption,

$$[\alpha_i]_g = [\alpha_i]_h$$

Then,

$$\langle [\alpha_1]_g, \dots, [\alpha_n]_g \rangle = \langle [\alpha_1]_h, \dots, [\alpha_n]_h \rangle$$

So, it follows that

$$\langle [\alpha_1]_g, \dots, [\alpha_n]_g \rangle \in \mathcal{I}(\Pi^n) \text{ iff } \langle [\alpha_1]_h, \dots, [\alpha_n]_h \rangle \in \mathcal{I}(\Pi^n)$$

Therefore,  $V_{\mathcal{M},g}(\phi) = V_{\mathcal{M},h}(\phi)$  for each  $w$  in the base case.

IH: we assume for any wff of lower complexity that  $\phi$ , call this  $\psi$ ,  $V_{\mathcal{M},g}(\psi) = V_{\mathcal{M},h}(\psi)$ .

Then, we have four cases:

- Case 1:  $\phi = \sim \psi$ . By IH,  $V_{\mathcal{M},g}(\psi) = V_{\mathcal{M},h}(\psi)$ . Hence  $V_{\mathcal{M},g}(\sim \psi) = V_{\mathcal{M},h}(\sim \psi)$  and  $V_{\mathcal{M},g}(\phi) = V_{\mathcal{M},h}(\phi)$ .
- Case 2:  $\phi = \psi_1 \rightarrow \psi_2$ . By IH,  $V_{\mathcal{M},g}(\psi_1) = V_{\mathcal{M},h}(\psi_1)$  and  $V_{\mathcal{M},g}(\psi_2) = V_{\mathcal{M},h}(\psi_2)$ , implying that  $V_{\mathcal{M},g}(\psi_1 \rightarrow \psi_2) = V_{\mathcal{M},h}(\psi_1 \rightarrow \psi_2)$ . Hence,  $V_{\mathcal{M},g}(\phi) = V_{\mathcal{M},h}(\phi)$
- Case 3:  $\phi = \forall \alpha \psi$ . By IH,  $V_{\mathcal{M},g}(\psi) = V_{\mathcal{M},h}(\psi)$ .  
We know that  $V_{\mathcal{M},g}(\forall \alpha \psi) = 1$  iff  $\forall u \in \mathcal{D}$ ,  $V_{\mathcal{M},g_u^\alpha}(\psi) = 1$  iff  $\forall u \in \mathcal{D}$ ,  $V_{\mathcal{M},h}(\psi) = 1$  iff  $V_{\mathcal{M},h}(\forall \alpha \psi) = 1$  where  $g_u^\alpha$  and  $h_u^\alpha$  agree on free variables. By IH, we conclude that  $V_{\mathcal{M},g}(\phi) = V_{\mathcal{M},h}(\phi)$

IH established, and the claim immediately follows. ■

The second says that substitution of variables does not matter as long as the variable assignment assigns both variables the same object in the domain.

**Proposition 5.2.** For any model  $\mathcal{M}$ , if  $[\alpha]_g = [\beta]_g$ , then  $V_g(\phi) = V_g(\phi(\beta/\alpha))$

*Proof.* Suppose  $[\alpha]_{\mathcal{M},g} = [\beta]_{\mathcal{M},g}$ . We prove that  $V_{\mathcal{M},g}(\phi) = V_{\mathcal{M},g}(\phi(\beta/\alpha))$  by induction on  $\phi$ 's complexity.

BC: WLOG,  $\phi = \Pi\alpha, \gamma_1, \dots, \gamma_n$  (it does not matter how many  $\alpha$ s there are or where  $\alpha$  is). Then, by assumption, we have

$$\begin{aligned} & \langle [\alpha]_{\mathcal{M},g}, [\gamma_1]_{\mathcal{M},g}, \dots, [\gamma_n]_{\mathcal{M},g} \rangle \in \mathcal{I}(\Pi^n) \\ & \text{iff } \langle [\beta]_{\mathcal{M},g}, [\gamma_1]_{\mathcal{M},g}, \dots, [\gamma_n]_{\mathcal{M},g} \rangle \in \mathcal{I}(\Pi^n) \\ & \implies \langle [\alpha]_{\mathcal{M},g}, [\gamma_1]_{\mathcal{M},g}, \dots, [\gamma_n]_{\mathcal{M},g} \rangle \in \mathcal{I}(\Pi^n) \\ & \text{iff } \langle [\beta]_{\mathcal{M},g}, [\gamma_1]_{\mathcal{M},g}, \dots, [\gamma_n]_{\mathcal{M},g} \rangle \in \mathcal{I}(\Pi^n) \end{aligned}$$

Hence,  $V_{\mathcal{M},g}(\phi, w) = V_{\mathcal{M},g}(\phi(\beta/\alpha), w)$ .

IH: Let  $\phi_n$  denote that  $\phi$  has complexity  $n$ . Then, we assume that for all  $n < m$ ,  $V_{\mathcal{M},g}(\phi_n) = V_{\mathcal{M},g}(\phi_n(\beta/\alpha))$ . We have four cases to consider:

- Case 1:  $\phi_m = \sim \phi_n$ . Then,  $V_{\mathcal{M},g}(\phi_n) = V_{\mathcal{M},g}(\phi_n(\beta/\alpha))$  (by IH)  $\iff V_{\mathcal{M},g}(\sim \phi_n) = V_{\mathcal{M},g}(\sim \phi_n(\beta/\alpha))$   $\iff V_{\mathcal{M},g}(\phi_m) = V_{\mathcal{M},g}(\phi_m(\beta/\alpha))$
- Case 2:  $\phi_m = \phi_{n_1} \rightarrow \phi_{n_2}$  with  $n_1, n_2 < m$ . Then  $V_{\mathcal{M},g}(\phi_{n_1}) = V_{\mathcal{M},g}(\phi_{n_1}(\beta/\alpha))$  and  $V_{\mathcal{M},g}(\phi_{n_2}) = V_{\mathcal{M},g}(\phi_{n_2}(\beta/\alpha))$   $\iff V_{\mathcal{M},g}(\phi_{n_1} \rightarrow \phi_{n_2}) = V_{\mathcal{M},g}((\phi_{n_1} \rightarrow \phi_{n_2})(\beta/\alpha))$   $\iff V_{\mathcal{M},g}(\phi_m) = V_{\mathcal{M},g}(\phi_m(\beta/\alpha))$
- Case 3:  $\phi_m = \forall \gamma \phi_n$ . By IH,  $V_{\mathcal{M},g}(\phi_n) = V_{\mathcal{M},g}(\phi_n(\beta/\alpha))$ . Then, we need to consider three cases:
  1.  $\gamma = \alpha$ : Consider that  $V_{\mathcal{M},g}(\forall \alpha \phi_n) = 1$   $\iff$  for all  $u \in \mathcal{D}$ ,  $V_{\mathcal{M},g_u^\alpha}(\phi_n) = 1$ . But by IH,  $V_{\mathcal{M},g}(\phi_n) = V_{\mathcal{M},g}(\phi_n(\beta/\alpha))$ . Hence, for all  $u \in \mathcal{D}$ ,  $V_{\mathcal{M},g_u^\alpha}(\phi_n) = 1$  holds  $\iff \forall u \in \mathcal{D}$ ,  $V_{\mathcal{M},g}(\phi_n(\beta/\alpha)) = 1$   $\iff V_{\mathcal{M},g}(\forall \alpha \phi(\beta/\alpha)) = 1$ .
  2.  $\gamma = \beta$ : Then,  $\alpha$  is not substitutable for  $\beta$ , since in  $\beta$  is freely occurring in some formula beginning with  $\forall \beta$ . So this case cannot happen, on the assumption that we respect the rules of substitutability.
  3.  $\gamma \neq \alpha, \beta$ : Consider  $V_{\mathcal{M},g}(\forall \gamma \phi_n) = 1$   $\iff \forall u \in \mathcal{D}$ ,  $V_{\mathcal{M},g_u^\gamma}(\phi_n) = 1$ . By IH and assumption, for each  $u \in \mathcal{D}$  this holds  $\iff V_{\mathcal{M},g_u^\gamma}(\phi_n(\beta/\alpha)) = 1$  which holds  $\iff V_{\mathcal{M},g}(\forall \gamma \phi_n(\beta/\alpha)) = 1$ .

Therefore,  $V_{\mathcal{M},g}(\phi_m) = V_{\mathcal{M},g}(\phi_m(\beta/\alpha))$

Claim proven ■

**Remark:** We use these facts about truth-in-a-model to establish validity.

**Definition 5.11 (PC-validity).** A wff  $\phi$  is PC-valid – in symbols:  $\models_{\text{PC}} \phi$  –  $\iff \phi$  is true in all PC-models

We usually assume  $V_{\mathcal{M},g}(\phi) = 0$  for reductio to show  $\models_{\text{PC}} \phi$ .

**Definition 5.12 (Semantic Consequence).** A wff  $\phi$  is a semantic consequence of a set of wffs  $\Gamma$  –  $\Gamma \models_{\text{PC}} \phi$  –  $\iff V_{\mathcal{M},g}(\phi) = 1$  for every PC-model  $\mathcal{M}$  and assignment  $g$  with  $V_{\mathcal{M},g}(\gamma) = 1$  for each  $\gamma \in \Gamma$

**Example 5.1 (Week 5 Q1b).** Show that:

1.  $\models_{\text{PC}} \forall \alpha \phi \rightarrow \phi(\beta/\alpha)$

*Proof.* AFC that  $V_{\mathcal{M},g}(\forall \alpha \phi \rightarrow \phi(\beta/\alpha)) = 0$  for some  $\mathcal{M}$ . Then, we have

- (a)  $V_g(\forall \alpha \phi) = 1$
- (b)  $V_g(\phi(\beta/\alpha)) = 0$

Let  $g$  be some variable assignment such that

$$[\alpha]_{g_{[\beta]_g}^\alpha} = [\beta]_g$$

**Remark:** This is so that  $\beta$ 's denotation is arbitrary

If  $[\alpha]_g = [\beta]_g$ , then  $V_g(\phi) = V_g(\phi(\beta/\alpha))$ . So,  $V_{g_{[\beta]_g}^\alpha}(\phi(\beta/\alpha)) = V_{g_{[\beta]_g}^\alpha}(\phi) = 0$ .

From **Proposition 5.1**,  $V_g(\forall \alpha \phi) = V_{g_{[\beta]_g}^\alpha}(\forall \alpha \phi) = 1$ , implying that  $V_{g_{[\beta]_g}^\alpha}(\phi) = 1$ , a contradiction. ■

2.  $\models_{\text{PC}} \forall \alpha (\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \forall \alpha \psi)$  whenever  $\alpha$  does not occur free in  $\phi$

*Proof.* AFC that there is an  $\mathcal{M}$  s.t.  $V_g(\forall \alpha (\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \forall \alpha \psi)) = 0$  where  $\alpha$  does not occur freely in  $\phi$ . Then, we have

- (a)  $V_g(\forall \alpha \phi \rightarrow \psi) = 1$ , then  $\forall x \in \mathcal{D}$ , either  $V_{g_x^\alpha}(\phi) = 0$  or  $V_{g_x^\alpha}(\psi) = 1$



- (b)  $V_g(\phi \rightarrow \forall \alpha \psi) = 0$ , then  $V_g(\phi) = 1$  and  $V_g(\forall \alpha \psi) = 0$ . Hence, there exists some  $u \in \mathcal{D}$  s.t.  $V_{g_u^\alpha}(\phi) = 1$  and there exists some  $v \in \mathcal{D}$  s.t.  $V_{g_v^\alpha}(\psi) = 0$ .

Since  $\alpha$  does not occur free in  $\phi$ , we can conclude that from (1), for all  $x \in \mathcal{D}$ ,  $V_{g_x^\alpha}(\psi) = 1$  and from (2), there exists some  $v \in \mathcal{D}$  s.t.  $V_{g_v^\alpha}(\psi) = 0$ , a contradiction. ■

3. If  $\models_{PC} \phi$ , then  $\models_{PC} \forall \alpha \phi$

*Proof.* Assume  $\models_{PC} \phi$ . Then for any  $\mathcal{M}$  and assignment  $g$  for  $\mathcal{M}$ ,  $V_{\mathcal{M},g}(\phi) = 1$ .

AFC that  $\not\models_{PC} \forall \alpha \phi$ . So there is some  $\mathcal{M}$  s.t.  $V_{\mathcal{M},g}(\forall \alpha \phi) = 0$ . So, there is some  $u \in \mathcal{D}$  s.t.  $V_{g_u^\alpha}(\phi) = 0$  under that  $\mathcal{M}$ . But as established, for any  $\mathcal{M}$  and assignment  $g$  for  $\mathcal{M}$ ,  $V_{\mathcal{M},g}(\phi) = 1$ . So,  $V_{g_u^\alpha}(\phi) = 1$ , a contradiction. ■

## 5.2 Extensions of PC: identity, descriptor and lambda operators, SOL, Generalised Quantifiers, Functions

For each of the extensions of PC, we touch on semantics but the emphasis is on motivation and application.

### 5.2.1 Identity

The additional clause is added to the syntax of PC.

**Definition 5.13 (Additional Syntax Clause for Identity).** If  $\alpha$  and  $\beta$  are terms, then  $\alpha = \beta$  is a wff

**Definition 5.14 (Additional Valuation Clause for Identity).**  $V_{\mathcal{M},g}(\alpha = \beta) = 1$  iff  $[\alpha]_{\mathcal{M},g} = [\beta]_{\mathcal{M},g}$

**Motivation:** Some people think that numerical identity is a logical operator and hence should be given its own syntax in PC.

**Example 5.2.** The following statements are symbolised in PC with identity in the following way:

- There is exactly one  $F$   
Formalisation:  $\exists x(Fx \wedge \forall y(Fx \rightarrow y = x))$
- There is at least one  $F$   
Formalisation:  $\exists x Fx$
- There is at most one  $F$   
Formalisation:  $\forall x \forall y (Fx \wedge Fy \rightarrow x = y)$
- There are exactly two  $F$ s  
Formalisation:  $\exists x \exists y (Fx \wedge Fy \wedge \neg x = y) \wedge \forall x \forall y \forall z (Fx \wedge Fy \wedge Fz \rightarrow x = y \vee y = z \vee x = z)$
- There are at least two  $F$ s  
Formalisation:  $\exists x \exists y (Fx \wedge Fy \wedge \neg x = y)$  (8.2)
- There are at most two  $F$ s  
Formalisation:  $\forall x \forall y \forall z (Px \wedge Py \wedge Pz \rightarrow x = y \vee y = z \vee x = z)$
- There are at least three  $F$ s  
Formalisation:  $\exists x \exists y \exists z (Fx \wedge Fy \wedge Fz \wedge \neg x = y \wedge \neg x = z \wedge \neg y = z)$

### 5.2.2 The descriptor operator

**Definition 5.15 (Additional clause for  $\iota$ ).** If  $\phi$  is a wff and  $\alpha$  is a variable, then  $\iota\alpha\phi$  is a term

#### Motivation

The  $\iota$  operator is meant to capture definite descriptions in PC without doing violence to our grammar i.e. picking out a unique object/only object in the predicate's extension.

**Example 5.3.** Formalise “The author of Logic for Philosophy likes metaphysics” in

- $\mathcal{L}_=$ :  $\exists x(Ax \wedge \forall y(Ay \rightarrow y = x) \wedge Lx)$
- PC with  $\iota$ :  $L\iota x Ax$

It seems unlikely that a simple sentence embeds so much logical structure (having a conditional etc) and so the  $\iota$  makes more sense.

**Definition 5.16 (Additional denotation clause for  $\iota$ ).** The term denotation for some term  $\iota\beta\phi$  is

$$[\iota\beta\phi]_{\mathcal{M},g} = \begin{cases} \text{the } u \text{ in } \mathcal{D} \text{ such that } V_{\mathcal{M},g_u^\beta}(\phi) = 1 & \text{if there is a unique such } u \\ \text{undefined} & \text{otherwise} \end{cases}$$

**Definition 5.17 (Modified valuation clause for  $\iota$ ).**  $V_{\mathcal{M},g}(\Pi\alpha_1, \dots, \alpha_n) = 1$  iff  $[\alpha_1]_{\mathcal{M},g}, \dots, [\alpha_n]_{\mathcal{M},g}$  are all defined and  $\langle [\alpha_1]_{\mathcal{M},g}, \dots, [\alpha_n]_{\mathcal{M},g} \rangle \in \mathcal{I}(\Pi)$

So, all formulas of the form  $\Pi\alpha_1, \dots, \alpha_n$  with undefined  $\iota$ -terms are false.

### 5.2.3 The lambda operator

#### Motivation

We want to formalise “Logic is such that it is good and necessary”. Normally, we formalise it as  $Ga \wedge Na$ . But, we might want to have a complex predicate that is something is ‘good and necessary’. In which case, we use a lambda operator that forms a one place predicate:  $\lambda x(Gx \wedge Nx)a$ .

**Definition 5.18 (Complex Predicate).** If  $\alpha$  is a variable and  $\phi$  is a wff, then  $\lambda\alpha\phi$  is a one place predicate.

**Remark:** When  $F$  is unary, we might write  $(\lambda xFx)(y)$  for ease of reading.

The complex predicate reads “...is an  $\alpha$  such that  $\phi(\alpha)$ ”.

To formulate the valuation in the extended PC language with the  $\lambda$  operator, we must outline what the extension of the complex predicate is

**Definition 5.19 (Extension of  $\lambda\alpha\phi$ ).**  $\phi^{\mathcal{M},g,\alpha} = \{u \in \mathcal{D} : V_{\mathcal{M},g_u^\alpha}(\phi) = 1\}$

The basic idea is that  $\phi(\alpha)^{\mathcal{M},g,\alpha}$  is the set of things that make  $\phi(\alpha)$  true when assigned to  $\alpha$ . We need to fix the value of other freely occurring variables that are not  $\alpha$ .

**Definition 5.20 (Valuation for lambda operator).**  $V_{\mathcal{M},g}((\lambda\alpha\phi)(\beta)) = 1$  iff  $[\beta]_{\mathcal{M},g} \in \phi^{\mathcal{M},g,\alpha}$

### 5.2.4 Second Order Logic

#### Motivation

We want to add  $n$ -place predicate variables and second order quantifiers to first order logic.

**Definition 5.21 (Definition of an SOL-wff).** We add the following clauses to the FOL definition of a wff

- If  $\pi$  is an  $n$ -place predicate variable, and  $\alpha_1, \dots, \alpha_n$  are (individual) terms, then  $\pi\alpha_1 \dots \alpha_n$  is a wff

- If  $\pi$  is an  $n$ -place predicate variable and  $\phi$  is a wff, then  $\forall\pi\phi$  is a wff

For instance  $\forall X(X_a \vee \sim X_a)$  is a wff.

With predicate variables we now need to include

**Definition 5.22 (SOL Variable Assignments).** a variable assignment  $g$  for a PC-model  $\langle \mathcal{D}, \mathcal{I} \rangle$  is a function which assigns a member of  $\mathcal{D}$  to each individual variable  $\alpha$  and an  $n$ -place relation over  $\mathcal{D}$  to each  $n$ -place predicate variable  $\pi$

**Definition 5.23 (SOL Variant Assignment).** When  $g$  is a variable assignment for  $\langle \mathcal{D}, \mathcal{I} \rangle$  and  $U$  is an  $n$ -place relation over  $\mathcal{D}$  we define  $g_U^\pi$  as follows:

$$g_U^\pi = \begin{cases} g(\sigma) & \text{if } \sigma \neq \pi \\ U & \text{if } \sigma = \pi \end{cases}$$

**Definition 5.24 (SOL Valuation).** We add the following clauses to PC for SOL:

- $V_{\mathcal{M},g}(\pi\alpha_1, \dots, \alpha_n) = 1$  iff  $\langle [\alpha_1]_{\mathcal{M},g}, \dots, [\alpha_n]_{\mathcal{M},g} \rangle \in g(\pi)$
- $V_{\mathcal{M},g}(\forall\pi\phi) = 1$  iff, for every  $n$ -place relation  $U$  over  $\mathcal{D}$ ,  $V_{\mathcal{M},g_U^\pi}(\phi) = 1$

### 5.2.5 Function Symbols

**Motivation:** Usually, we formalise “George W. Bush’s father was a politician” as  $Pc$ . But the predicate “George W. Bush’s father” as the constant  $c$ , we ignore the semantic complexity of both “George W. Bush” and “father”, since the sentence implies “Someone’s father was a politician”. To capture the semantic complexity, we add *function symbols* to PC. For instance, we symbolise “George W. Bush’s father” as  $f(a)$  where  $a$  is George W. Bush, and  $f(\cdot)$  is like “ $\_$ ’s father”. We can formalise someone’s father was a politician as  $\exists xPf(x)$

We can have multiplace functions like  $c = s(a, b)$  as “3 is the sum of 2 and 1”.

**Remark:** the function does not just take in *names* i.e. it can be  $f(f(a))$ .

**Definition 5.25 (Terms for function symbols).** We add the following clauses to the definitions of terms in PC:

- Names and variables are terms
- if  $f$  is an  $n$ -place function symbol and  $\alpha_1, \dots, \alpha_n$  are terms, then  $f(\alpha_1, \dots, \alpha_n)$  is a term
- Only strongs that can be shown to be terms by the preceding clauses are terms

We must add the following to our definition of a PC-model

**Definition 5.26 (PC+FS Model for function symbols).** We add the following clause to the definition clauses for PC-FS models

- If  $f$  is an  $n$ -place symbol, then  $\mathcal{I}(f)$  is an  $n$ -place (total) function defined on  $\mathcal{D}$

**Remark:** A ‘total’ function “defined on  $\mathcal{D}$ ” means that the function must have a well-defined output (which is a member of  $\mathcal{D}$ ) whenever it is given as inputs any  $n$  members of  $\mathcal{D}$

**Definition 5.27 (Denotations for PC+FS).** For any model  $\mathcal{M} = \langle \mathcal{D}, \mathcal{I} \rangle$ , the variable assignment  $g$  for  $\mathcal{M}$ , and term  $\alpha$ , we define  $[\alpha]_{\mathcal{M},g}$  as

$$[\alpha]_{\mathcal{M},g} = \begin{cases} \mathcal{I}(\alpha) & \text{if } \alpha \text{ is a constant} \\ g(\alpha) & \text{if } \alpha \text{ is a variable} \\ \mathcal{I}(f)([\alpha_1]_{\mathcal{M},g} \dots [\alpha_n]_{\mathcal{M},g}) & \text{if } \alpha \text{ is a complex term } f(\alpha_1, \dots, \alpha_n) \end{cases}$$

### 5.2.6 Generalised Quantifiers

Here, we extend the standard quantifiers to include both generalised monadic and binary quantifiers. First we introduce the definition of the set of members of  $\mathcal{M}$ 's domain which renders  $\phi$  true

**Definition 5.28.** We denote the set of members of  $\mathcal{M}$  which satisfy  $\phi$  as

$$\phi^{\mathcal{M},g,\alpha} = \{u : u \in \mathcal{D} \text{ and } V_{\mathcal{M},g_u^\alpha}(\phi) = 1\}$$

**Remark:** Roughly,  $\phi^{\mathcal{M},g,\alpha}$  is the set of things  $u$  in  $\mathcal{D}$  s.t.  $\phi$  is true, relative to the v.a.  $g_u^\alpha$ .

#### Generalised Monadic Quantifiers

With our new terminology, we can write the following truth conditions:

$$\begin{aligned} V_{\mathcal{M},g}(\forall\alpha\phi) &= 1 \text{ iff } \phi^{\mathcal{M},g,\alpha} = \mathcal{D} \\ V_{\mathcal{M},g}(\exists\alpha\phi) &= 1 \text{ iff } \phi^{\mathcal{M},g,\alpha} \neq \emptyset \end{aligned}$$

Then, say we want to express that ‘there are at least  $n$   $\phi$ ’s’.

**Definition 5.29 (Generalised Monadic Quantifiers).** We add the following clauses to PC:

- if  $\alpha$  is a variable and  $\phi$  is a wff, then  $\exists_n\alpha\phi$  is a wff
- $V_{\mathcal{M},g}(\exists_n\alpha\phi) = 1$  iff  $|\phi^{\mathcal{M},g,\alpha}| \geq n$

We can also add other quantifiers such as  $Q$  that acts grammatically like  $\forall, \exists$ . So, we will add  $Q\alpha\phi$ . Each of the new quantifiers,  $Q$ , would be associated with a relation between sets,  $R_Q$ , such that  $Q\alpha\phi$  would be true in a PC-model,  $\mathcal{M}$ , with domain  $\mathcal{D}$ , relative to variable assignment  $g$ , iff  $\phi^{\mathcal{M},g,\alpha}$  bears  $R_Q$  to  $\mathcal{D}$ .

**Restrictions on  $R_Q$ :** It shouldn’t “concern particular objects” in the way that the following symbol concerns particular object

$$V_{\mathcal{M},g}(\exists_{\text{Ted-loved}}\alpha\phi) = 1 \text{ iff } \phi^{\mathcal{M},g,\alpha} \cap \{u : u \in \mathcal{D} \text{ and Ted loves } u\} \neq \emptyset$$

We should require the following of  $R_Q$ : if a subset  $X$  of some set  $D$  bears  $R_Q$  to  $D$ , and  $f$  is a one-to-one function with domain  $D$  and range  $D'$ , then  $f[X]$  must bear  $R_Q$  to  $D'$ .

#### Generalised Binary Quantifiers

While monadic quantifiers like  $\forall\alpha$  attach to a single open sentence  $\phi$ , naturally occurring monadic quantifiers are atypical. Consider the sentence “Every student is happy” vs “Everything is material”. The quantifier “Every” attaches to *two* predicates i.e. “[is a] student” and “is happy”. So, our notation for “every  $\phi$  is a  $\psi$ ” becomes  $(\forall\alpha : \phi)\psi$ .

**Definition 5.30 (Generalised Binary Quantifiers wffs).** We add the following clauses to the definition of wffs for Binary Quantifiers

- if  $\phi$  and  $\psi$  are wffs and  $\alpha$  is a variable, then  $(\forall\alpha : \phi)\psi$  and  $(\exists\alpha : \phi)\psi$  are wffs
- $V_{\mathcal{M},g}((\forall\alpha : \phi)\psi) = 1$  iff  $\phi^{\mathcal{M},g,\alpha} \subseteq \psi^{\mathcal{M},g,\alpha}$
- $V_{\mathcal{M},g}((\exists\alpha : \phi)\psi) = 1$  iff  $\phi^{\mathcal{M},g,\alpha} \cap \psi^{\mathcal{M},g,\alpha} \neq \emptyset$

A further important binary quantifier is the:

- if  $\phi$  and  $\psi$  are wffs and  $\alpha$  is a variable, then  $(\text{the } \alpha : \phi)\psi$  is a wff
- $V_{\mathcal{M},g}((\text{the } \alpha : \phi)\psi) = 1$  iff  $|\phi^{\mathcal{M},g,\alpha}| = 1$  and  $\phi^{\mathcal{M},g,\alpha} \subseteq \psi^{\mathcal{M},g,\alpha}$

That is,  $(\text{the } \alpha : \phi)\psi$  is true iff i) there is exactly one  $\phi$ , and ii) every  $\phi$  is a  $\psi$ . This truth condition, notice, is exactly the truth condition for Russell’s symbolisation of the  $\phi$  is a  $\psi$ .

### 5.2.7 Worked Examples

**Example 5.4 (PC with identity vs descriptor and lambda operators).** Consider the following sentence and dictionary:

(\*) The king of France is not bald

K:...is king of France, B:...is bald.

- (a) Using this dictionary, apply Russell's theory of descriptions to obtain two semantically non-equivalent symbolisations of (\*) in  $\mathcal{L}_=$ .

**Formalisation:**  $\sim \exists x(Kx \wedge \forall y(Ky \rightarrow y = x) \wedge Bx)$

- (b) Using the same dictionary, symbolise (\*) in  $\mathcal{L}_\iota$ . Show that this symbolisation is semantically equivalent to one of the symbolisations from part (a)/

**Formalisation:**  $\sim B\iota x Kx$ . The semantic argument is

$$\begin{aligned} V_{\mathcal{M},g}(\exists x(Kx \wedge \forall y(Ky \rightarrow y = x) \wedge \sim Bx)) &= 1 \text{ iff} \\ V_{\mathcal{M},g_u^x}(Kx \wedge \forall y(Ky \rightarrow y = x) \wedge \sim Bx) &= 1 \text{ for some } u \in \mathcal{D} \text{ iff} \\ V_{\mathcal{M},g_u^x}(Kx) = 1 \text{ and } V_{\mathcal{M},g_u^x}(\forall y(Ky \rightarrow y = x)) &= 1 \text{ and } V_{\mathcal{M},g_u^x}(\sim Bx) = 1 \text{ iff} \\ V_{\mathcal{M},g_{u u'}^x y}(Ky \rightarrow y = x) &= 1 \text{ for all } u' \in \mathcal{D}, V_{\mathcal{M},g_{u u'}^x y}(Kx) = 1, V_{\mathcal{M},g_{u u'}^x y}(Bx) = 0 \text{ iff} \\ \{\langle u \rangle\} = \mathcal{I}(K), V_{\mathcal{M},g_{u u'}^x y}(Kx) &= 1, V_{\mathcal{M},g_{u u'}^x y}(Bx) = 0 \text{ iff} \\ \iota x Kx \text{ is defined and } \langle [\iota x Kx]_{g_{u u'}^x y} \rangle &\notin \mathcal{I}(B) \text{ iff} \\ V_{\mathcal{M},g}(\sim B\iota x Kx) &= 1 \end{aligned}$$

- (c) Using the same dictionary, show that the other symbolisation may be captured (up to semantic equivalence) in  $\mathcal{L}_{\iota,\lambda}$ . Demonstrate the semantic equivalence with a semantic argument.

**Formalisation:**  $(\lambda x \sim Bx)(\iota x Kx)$ . The semantic argument is

$$\begin{aligned} V_{\mathcal{M},g}(\exists x(Kx \wedge \forall y(Ky \rightarrow y = x) \wedge \sim Bx)) &= 1 \text{ iff} \\ V_{\mathcal{M},g_u^x}(Kx \wedge \forall y(Ky \rightarrow y = x) \wedge \sim Bx) &= 1 \text{ for some } u \in \mathcal{D} \text{ iff} \\ V_{\mathcal{M},g_u^x}(Kx) = 1 \text{ and } V_{\mathcal{M},g_u^x}(\forall y(Ky \rightarrow y = x)) &= 1 \text{ and } V_{\mathcal{M},g_u^x}(\sim Bx) = 1 \text{ iff} \\ V_{\mathcal{M},g_{u u'}^x y}(Ky \rightarrow y = x) &= 1 \text{ for all } u' \in \mathcal{D}, V_{\mathcal{M},g_{u u'}^x y}(Kx) = 1, V_{\mathcal{M},g_{u u'}^x y}(Bx) = 0 \text{ iff} \\ \{\langle u \rangle\} = \mathcal{I}(K), V_{\mathcal{M},g_{u u'}^x y}(Kx) &= 1, V_{\mathcal{M},g_{u u'}^x y}(Bx) = 0 \text{ iff} \\ \{\langle u \rangle\} = \mathcal{I}(K), V_{\mathcal{M},g_{u u'}^x y}(Kx) &= 1, \langle [\iota x Kx]_{g_{u u'}^x y} \rangle \in \{u \in \mathcal{D} : V_{\mathcal{M},g_u^x}(\sim Bx) = 1\} \text{ iff} \\ \{\langle u \rangle\} = \mathcal{I}(K), V_{\mathcal{M},g_{u u'}^x y}(Kx) &= 1, \langle [\iota x Kx]_{g_{u u'}^x y} \rangle \in (\sim Bx)^{\mathcal{M},g,\alpha} \text{ iff} \\ V_{\mathcal{M},g}((\lambda x \sim Bx)(\iota x Kx)) &= 1 \end{aligned}$$

**Solution.**

The below is probably the most difficult exercise.

**Example 5.5 (Generalised quantifiers).** Determine which of the following binary generalised quantifiers can be symbolised in  $\mathcal{L}_=$

$$\begin{aligned} V_{\mathcal{M},g}((\text{No } \alpha : \phi)\psi) &= 1 \text{ iff } |\phi^{\mathcal{M},g,\alpha} \cap \psi^{\mathcal{M},g,\alpha}| = 0 \\ V_{\mathcal{M},g}((\text{At least two } \alpha : \phi)\psi) &= 1 \text{ iff } |\phi^{\mathcal{M},g,\alpha} \cap \psi^{\mathcal{M},g,\alpha}| \geq 2 \\ V_{\mathcal{M},g}((\text{Finitely many } \alpha : \phi)\psi) &= 1 \text{ iff } |\phi^{\mathcal{M},g,\alpha} \cap \psi^{\mathcal{M},g,\alpha}| < \aleph_0 \\ V_{\mathcal{M},g}((\text{Most } \alpha : \phi)\psi) &= 1 \text{ iff } |\phi^{\mathcal{M},g,\alpha} \cap \psi^{\mathcal{M},g,\alpha}| > |\phi^{\mathcal{M},g,\alpha} - \psi^{\mathcal{M},g,\alpha}| \end{aligned}$$

Justify your answers with suitable semantic arguments.

**Solution.**

- $V_{\mathcal{M},g}((\text{No } \alpha : \phi)\psi) = 1$  iff  $|\phi^{\mathcal{M},g,\alpha} \cap \psi^{\mathcal{M},g,\alpha}| = 0$ .

**Formalisation:**  $\sim \exists x(\phi \wedge \psi)$ . We now show this equivalence via semantic argument:

$$\begin{aligned} & V_{\mathcal{M},g}(\sim \exists x(\phi \wedge \psi)) = 1 \\ \text{iff } & V_{\mathcal{M},g}(\exists x(\phi \wedge \psi)) = 0 \\ \text{iff there is no } & u \in \mathcal{D} \text{ s.t. } V_{\mathcal{M},g_u^\alpha}(\phi \wedge \psi) = 1 \\ \text{iff } & (\phi \wedge \psi)^{\mathcal{M},g,\alpha} = \emptyset \\ \text{iff } & |\phi^{\mathcal{M},g,\alpha} \cap \psi^{\mathcal{M},g,\alpha}| = 0 \end{aligned}$$

- $V_{\mathcal{M},g}((\text{At least two } \alpha : \phi)\psi) = 1$  iff  $|\phi^{\mathcal{M},g,\alpha} \cap \psi^{\mathcal{M},g,\alpha}| \geq 2$ .

**Formalisation:**  $\exists x \exists y(\phi \wedge \phi(y/x) \wedge \psi \wedge \psi(y/x) \wedge \sim x = y)$ . Our semantic argument is as follows:

$$\begin{aligned} & V_{\mathcal{M},g}(\exists x \exists y(\phi \wedge \phi(y/x) \wedge \psi \wedge \psi(y/x) \wedge \sim x = y)) = 1 \\ \text{iff there is some } & u \in \mathcal{D} \text{ s.t. } V_{\mathcal{M},g}(\exists y(\phi \wedge \phi(y/x) \wedge \psi \wedge \psi(y/x) \wedge \sim x = y)) = 1 \\ \text{iff there is some } & u' \in \mathcal{D}' \text{ s.t. } V_{\mathcal{M},g}(\phi \wedge \phi(y/x) \wedge \psi \wedge \psi(y/x) \wedge \sim x = y) = 1 \\ \text{where } \mathcal{D}' = \{u : u \in \mathcal{D} \text{ s.t. } & V_{\mathcal{M},g}(\exists y(\phi \wedge \phi(y/x) \wedge \psi \wedge \psi(y/x) \wedge \sim x = y)) = 1\} \\ \text{iff } & (\phi \wedge \phi(y/x) \wedge \psi \wedge \psi(y/x) \wedge \sim x = y)^{\mathcal{M},g,\alpha} \supseteq \{u, u'\} \\ \text{iff } & |\phi^{\mathcal{M},g,\alpha} \cap \psi^{\mathcal{M},g,\alpha}| \geq 2 \end{aligned}$$

For the last two, see **Examples 5.7-5.8**.

**Example 5.6 (SOL).** Which of these generalised quantifiers, from **Example 5.5** can be captured in the language of second-order logic? Explain.

**Solution.** The first two generalised quantifiers can be trivially symbolised in SOL since they can be symbolised in  $\mathcal{L}_=$ . So, we focus on the last two which cannot.

The intuitive strategy is to formalise the notion of infinity via (i) (Dedekind) infinity, (ii) a function, (iii) an injective function, and negating this symbolisation (apparently, this is better than forming a bijection between the natural numbers and the set, and negating that)

- $V_{\mathcal{M},g}((\text{Finitely many } \alpha : \phi)\psi) = 1$  iff  $|\phi^{\mathcal{M},g,\alpha} \cap \psi^{\mathcal{M},g,\alpha}| < \aleph_0$ .

**IDEA:** Two sets have exactly the same cardinality if there is a bijection between 2 sets. A set is countably infinite if there is a bijection with the Natural Numbers.

A set is Dedekind infinite holds if and only if it can be put into a bijection with a proper subset i.e. for a bijective function  $f : A \rightarrow E$  s.t.  $E \subset A$ , then  $A$  is dedekind infinite.

It suffices to have an injection  $f : A \rightarrow E$  where  $E \subset A$ . So, we need to formalise 3 notions: (i) a function, (ii) one-to-one, (iii) proper subset.

– Function:

$$\text{Func}^{R:\phi \rightarrow \psi} \iff \forall \alpha(\phi(\alpha) \rightarrow \exists \beta(\psi(\beta/\alpha)) \wedge \forall \gamma(R\alpha\gamma \rightarrow \beta = \gamma))$$

– Injective function:

$$\text{Inj}^{R:\phi \rightarrow \psi} \iff \text{Func}^{R:\phi \rightarrow \psi} \wedge \forall \alpha \forall \beta \forall \gamma((\phi(\alpha) \wedge \phi(\beta/\alpha)) \wedge R\alpha\gamma \wedge R\beta\gamma \rightarrow \alpha = \beta)$$

i.e. if two objects mapping to the same objects, then the two objects are the same

– Proper subset and we reach the final solution:

$$((\text{Inf } \alpha : \phi)\psi) \iff \exists X \exists R(\forall \alpha(X\alpha \rightarrow (\phi(\alpha) \wedge \psi(\alpha)) \wedge \sim \forall \alpha(\phi(\alpha) \wedge \psi(\alpha)) \rightarrow X\alpha) \wedge \text{Inj}^{R:(\phi \wedge \psi) \rightarrow X})$$

Then, we have

$$(\text{Fin } \alpha : \phi)\psi \iff \sim ((\text{Inf } \alpha : \phi)\psi)$$

- $V_{\mathcal{M},g}((\text{Most } \alpha : \phi)\psi) = 1$  iff  $|\phi^{\mathcal{M},g,\alpha} \cap \psi^{\mathcal{M},g,\alpha}| > |\phi^{\mathcal{M},g,\alpha} - \psi^{\mathcal{M},g,\alpha}|$   
**IDEA:**  $|A| \leq |B| \iff \exists f : A \rightarrow B$  where  $f$  is an injection. Then,

$$|\phi - \psi| \leq |\phi \cap \psi|$$

Therefore,

$$|\phi \cap \psi| \not\leq |\phi - \psi| \iff \sim \exists f : A \rightarrow B$$

So, the solution

$$(\text{Most } \alpha : \phi)\psi \iff \exists R \text{ Inj}^{R:(\phi \wedge \sim \psi) \rightarrow (\phi \wedge \psi)} \wedge \sim \exists R \text{ Inj}^{R:(\phi \wedge \psi) \rightarrow (\phi \wedge \sim \psi)}$$

### 5.3 Basic Metatheory

Standardly, we have our two metatheorems that connect provability and semantic validity.

**Theorem 5.1 (Soundness of PC).** When  $\Gamma$  is a set of PC-sentences and  $\phi$  is a PC-sentence (none of which contain free variables), if  $\vdash_{\text{PC}} \phi$  then  $\models_{\text{PC}} \phi$

and

**Theorem 5.2 (Completeness of PC).** When  $\Gamma$  is a set of PC-sentences and  $\phi$  is a PC-sentence (none of which contain free variables), if  $\models_{\text{PC}} \phi$  then  $\vdash_{\text{PC}} \phi$

More interesting to do with the expressive power of our predicate language are the following two theorems, where satisfiability means the ability for every member in a set to be true at once.

**Theorem 5.3 (The Compactness Theorem).** A set  $\Gamma$  is satisfiable iff every finite subset of  $\Gamma$  is satisfiable

**Theorem 5.4 (Downward Löwenheim-Skolem Theorem).** If a set of  $\mathcal{L}_{=}$  sentences has infinite models, then it has models whose domains are only countable

*Remark:* An infinite model refers to the size of the domain. So, this just means that for any domain  $|\phi^{\mathcal{M},g,\alpha}| \leq \aleph_0$ , where  $\aleph_0$  is the smallest infinite cardinal (i.e. the cardinality of natural numbers)

**Example 5.7.** Show that there is no  $\mathcal{L}_{=}$  symbolisation of  $V_{\mathcal{M},g}((\text{Finitely many } \alpha : \phi)\psi) = 1$  iff  $|\phi^{\mathcal{M},g,\alpha} \cap \psi^{\mathcal{M},g,\alpha}| < \aleph_0$

*Proof.* Let  $\phi_n$  be some  $\mathcal{L}_{=}$  symbolisation of “there are at least  $n$  things such that  $\phi$  and  $\psi$ ”. Now, assume for reductio that  $V_{\mathcal{M},g}(\sim \phi_{\infty}) = 1$  iff  $|\phi_{\infty}^{\mathcal{M},g,\alpha}| < \aleph_0$ . Hence,  $V_{\mathcal{M},g}(\phi_{\infty}) = 1$  iff  $|\phi_{\infty}^{\mathcal{M},g,\alpha}| \geq \aleph_0$ . Then, suppose that  $\Gamma = \{\phi_n : n \in \mathbb{N}\}$ , and  $\Gamma \models \phi_{\infty}$ . By compactness, there exists a finite  $\Gamma_0 \subset \Gamma$  s.t.  $\Gamma_0 \models \phi_{\infty}$ . But since  $\Gamma_0$  has a largest  $\phi_n$ , then  $\Gamma_0 \not\models \phi_{\infty}$ , which is a contradiction ■

**Example 5.8.** Show that there is no  $\mathcal{L}_{=}$  symbolisation of  $V_{\mathcal{M},g}((\text{Most } \alpha : \phi)\psi) = 1$  iff  $|\phi^{\mathcal{M},g,\alpha} \cap \psi^{\mathcal{M},g,\alpha}| > |\phi^{\mathcal{M},g,\alpha} - \psi^{\mathcal{M},g,\alpha}|$

*Proof.* We want to prove this via a counterexample i.e. that there is no *generalised* quantifier of this type. We do this by trying to express that most real numbers are not natural numbers i.e. P1:  $((\text{Most } x : Rx \sim Nx)$  where  $R$  : is a real number and  $N$  : is a natural number.

We now construct Peano arithmetic via the following axioms:

- P2:  $\forall x Rx$  (all numbers are real)
- P3:  $\forall x(Nx \rightarrow \exists y \forall z((Szx \leftrightarrow z = y) \wedge Ny))$  (every natural number has a unique successor)
- P4:  $\exists x(Nx \wedge \forall y(Ny \rightarrow \sim Syx))$  (there is a first natural number)
- P5:  $\forall x \forall y \forall z(Nx \wedge Ny \wedge Sxz \wedge Syz \rightarrow x = y)$

Let  $\Gamma = \{P1, P2, P3, P4, P5\}$ .  $\Gamma$  has an infinite model  $\mathcal{M}^*$  since  $\mathcal{D}^* = \mathbb{R}$ . Then,  $|R^{\mathcal{M},g,\alpha}| = \mathbb{R}$ ,  $|N^{\mathcal{M},g,\alpha}| = \mathbb{N}$ . By the downward Lowenheim-Skolem Theorem, any finite  $\mathcal{L}_=$  sentences with an infinite model has a countable model of  $\Gamma$ . Let us call this  $\mathcal{M}$ , which must be countably infinite. Now we can show the contradiction.

$$|\mathcal{I}(\sim N) \cap \mathcal{I}(R)| > |\mathcal{I}(R) - \mathcal{I}(\sim N)| \implies |(\mathcal{D} - \mathcal{I}(N)) \cap \mathcal{D}| > |\mathcal{D} - (\mathcal{D} - \mathcal{I}(N))|$$

Since  $\mathcal{D} - \mathcal{I}(N) \subset \mathcal{D}$ ,

$$|\mathcal{D} - \mathcal{I}(N)| > |\mathcal{I}(N)|$$

We know that  $|\mathcal{I}(N)| \geq \aleph_0$  and that  $|\mathcal{D}| = \aleph_0$  by construction. But

$$|\mathcal{D} - \mathcal{I}(N)| > \aleph_0 \implies |\mathcal{D}| > \aleph_0$$

a contradiction. ■

## 6 Simplified Quantified modal logic: constant domains

### 6.1 Syntax and Semantics

**Definition 6.1 (Vocabulary of QML).** We have the following primitive vocabulary for QML:

- Connectives/quantifiers:  $\rightarrow, \sim, \Box, \forall$
- Variables:  $x, y, \dots$  (with or without numerical subscripts)
- $n$ -place predicates:  $F, G, \dots$  (with or without numerical subscripts), for each  $n > 0$
- binary predicate:  $=$
- individual constants (names):  $a, b, \dots$  (with or without numerical subscripts)

There are no 0-place predicates i.e. no sentence letters

We can now define a QML-term or QML-wff recursively by adding a clause for  $\Box$  to PC with  $=$

**Definition 6.2 (Definition of a QML-term).** If  $\alpha$  is a variable or an individual constant,  $\alpha$  is a term.

**Definition 6.3 (Definition of a QML-wff).** We can define a QML-wff recursively:

- If  $\Pi^n$  is an  $n$ -place predicate and  $\alpha_1, \dots, \alpha_n$  are terms,  $\Pi^n \alpha_1, \dots, \alpha_n$  is a wff
- If  $\alpha$  and  $\beta$  are terms, then  $\alpha = \beta$  is a wff
- If  $\phi$  and  $\psi$  are wffs, and  $\alpha$  is a variable,  $\sim \phi, (\phi \rightarrow \psi), \Box \phi$  and  $\forall \alpha \phi$  are wffs

We define free and bound variable occurrences in the same way as in PC.

#### Ambiguity: De Dicto and De Re

There are differing interpretations of modal sentences such as “Every Polish logician is necessarily a logician”. The (Kripkean) interpretation of it as “Necessarily, every Polish logician is a logician” is a *de dicto* modal expression (prefixing modality to a closed sentence) while the interpretation as “Every Polish logician is necessarily a logician” is a *de re* modal claim (prefixing modality to a property). We can iron this down with Sider’s definition.

**Definition 6.4 (Sider’s Modality De Re and De Dicto ).** A QML-wff is said to be *de re* if it has a subformula of the form  $\Box \phi(\alpha)$  in which the variable  $\alpha$  occurs freely; otherwise it is *de dicto*

**Example 6.1.** Disambiguate “Every Polish logician is necessarily a logician” by giving two QML-symbolisations:

- De Dicto: “Necessarily, every Polish logician is a logician”  
 $\Box \forall x (Px \wedge Lx \rightarrow Lx)$



- De Re: “Every Polish logician is necessarily a logician”  
 $\forall x(Px \wedge Lx \rightarrow \Box Lx)$

An important feature of SQML semantics is that it has a constant domain  $\mathcal{D}$  that is the same across all possible worlds. Furthermore, there is no accessibility relation  $\mathcal{R}$  so all worlds are deemed accessible (basically an S5 system).

**Definition 6.5 (SQML-model).** A SQML-model is a triple  $\langle \mathcal{W}, \mathcal{D}, \mathcal{I} \rangle$  s.t.

- $\mathcal{W}$  is a non-empty set i.e. the set of possible worlds
- $\mathcal{D}$  is a non-empty set i.e. the constant domain
- $\mathcal{I}$  is a function such that
  - $\mathcal{I}(\alpha) \in \mathcal{D}$  for each constant  $\alpha$
  - $\mathcal{I}(\Pi^n)$  is a set of  $n+1$ -tuples of the form  $\langle u_1, \dots, u_n, w \rangle$ , where  $u_1, \dots, u_n$  are members of  $\mathcal{D}$  and  $w \in \mathcal{W}$ , for each  $n$ -place predicate  $\Pi^n$

**Intensions and Extensions:** SQML-models relativise an extension of predicates to a specific world  $w$ . This is different from PC-models, which assigns an extension to predicates i.e. for some unary predicate  $F$ ,  $\mathcal{I}(F)$  is just a set of members of  $\mathcal{D}$ . In contrast, SQML-models assigns an ‘intension’ to predicates i.e. an extension that is relativised to some world. So, for the same unary  $F$ ,  $\mathcal{I}(F)$  is not just some set of  $\mathcal{D}$ -members, but a set of pairs  $\langle d, w \rangle$  with  $d \in \mathcal{D}$  and  $w \in \mathcal{W}$ .

For ease, we want to localise the interpretation function  $\mathcal{I}$  to a specific world so we can yield their  $w$ -extensions.

**Definition 6.6 ( $w$ -extensions).** Given a SQML-model  $\mathcal{M} = \langle \mathcal{W}, \mathcal{D}, \mathcal{I} \rangle$ , the extension of an  $n$ -place predicate  $\Pi^n$  at world  $w$ ,  $\mathcal{I}_w(\Pi^n)$ , is defined as follows:

$$\mathcal{I}_w(\Pi^n) = \{ \langle u_1, \dots, u_n \rangle : \langle u_1, \dots, u_n, w \rangle \in \mathcal{I}(\Pi^n) \}$$

**Remark:** this just means that to get the  $w$ -extension at some world  $w$ , take all the  $n+1$ -tuples with that specific  $w$  in the intension of  $\mathcal{I}(\Pi^n)$  i.e. the main interpretation function, and strip off the last element  $w$  to yield a set of  $n$ -tuples that are the extensions of  $\mathcal{I}_w(\Pi^n)$  relativised to that world  $w$ .

**Remark:** All the  $w$ -extensions for  $w \in \mathcal{W}$  uniquely determine the intension and vice versa.

**Definition 6.7 (SQML term denotations).** Let  $\mathcal{M} = \langle \mathcal{W}, \mathcal{D}, \mathcal{I} \rangle$  be an SQML model:

- An assignment  $g$  for  $\mathcal{M}$  is a function that maps each variable to a member of  $\mathcal{D}$
- For term  $\alpha$ , we define its denotation in  $\mathcal{M}$  relative to assignment  $g$ :

$$[\alpha]_{\mathcal{M},g} = \begin{cases} \mathcal{I}(\alpha) & \text{if } \alpha \text{ is a constant} \\ g(\alpha) & \text{if } \alpha \text{ is a variable} \end{cases}$$

**Remark:** The definition of a variant assignment  $g_d^\alpha$  is defined as before.

**Definition 6.8 (SQML Valuation).** The valuation function,  $V_{\mathcal{M},g}$ , for an SQML model  $\mathcal{M} = \langle \mathcal{W}, \mathcal{D}, \mathcal{I} \rangle$  and variable assignment  $g$  is the unique function that assigns 0 or 1 to each wff at each world and satisfies the following conditions:

*Atomic Formulas:* for terms  $\alpha, \beta, \dots$ , and  $n$ -ary predicate,  $\Pi^n$

- $V_{\mathcal{M},g}(\alpha = \beta, w) = 1$  iff  $[\alpha]_{\mathcal{M},g} = [\beta]_{\mathcal{M},g}$
- $V_{\mathcal{M},g}(\Pi^n \alpha_1, \dots, \alpha_n, w) = 1$  iff  $\langle [\alpha_1]_{\mathcal{M},g}, \dots, [\alpha_n]_{\mathcal{M},g}, w \rangle \in \mathcal{I}(\Pi^n)$  or equivalently  $\langle [\alpha_1]_{\mathcal{M},g}, \dots, [\alpha_n]_{\mathcal{M},g} \rangle \in \mathcal{I}_w(\Pi^n)$

*Connectives:* for formulas  $\phi$  and  $\psi$

- $V_{\mathcal{M},g}(\phi \rightarrow \psi, w) = 1$  iff  $V_{\mathcal{M},g}(\phi, w) = 0$  or  $V_{\mathcal{M},g}(\psi, w) = 1$
- $V_{\mathcal{M},g}(\sim \phi, w) = 1$  iff  $V_{\mathcal{M},g}(\phi, w) = 0$

*Modal Operators:* for formula  $\phi$

- $V_{\mathcal{M},g}(\Box \phi, w) = 1$  iff, for every  $v \in \mathcal{W}$ ,  $V_{\mathcal{M},g}(\phi, v) = 1$

*Quantifiers:* for formula  $\phi$  and variable  $\alpha$

- $V_{\mathcal{M},g}(\forall \alpha \phi, w) = 1$  iff, for every  $d \in \mathcal{D}$ ,  $V_{\mathcal{M},g_d^\alpha}(\phi, w) = 1$

**Definition 6.9 (Truth in an SQML model).** a wff  $\phi$  is true in a SQML-model  $\mathcal{M}$  iff  $V_{\mathcal{M},g}(\phi) = 1$  for every assignment  $g$  for  $\mathcal{M}$

**Definition 6.10 (SQML validity).** a wff  $\phi$  is SQML valid, i.e.  $\models_{\text{SQML}} \phi$ , iff  $\phi$  is true in all SQML models

## 6.2 Showing Validity and Invalidity

We present ways in which to generate semantic proofs for validity (reductio arguments) and Sider's method of generating counterexamples via drawing boxes and worlds.

**Example 6.2.** Determine which of the following schemas are SQML-valid:

1.  $\Box \forall \alpha \phi \rightarrow \forall \alpha \Box \phi$

**Claim:**  $\models_{\text{SQML}} \Box \forall \alpha \phi \rightarrow \forall \alpha \Box \phi$

*Proof.* AFC that  $V_{\mathcal{M},g}(\Box \forall \alpha \phi \rightarrow \forall \alpha \Box \phi, w) = 0$ . Then,  $V_{\mathcal{M},g}(\Box \forall \alpha \phi, w) = 1$  and  $V_{\mathcal{M},g}(\forall \alpha \Box \phi, w) = 0$ . From  $V_{\mathcal{M},g}(\Box \forall \alpha \phi, w) = 1$ , then  $V_{\mathcal{M},g}(\forall \alpha \phi, v) = 1$  for every  $v \in \mathcal{W}$ . Then, for each world  $v \in \mathcal{W}$ , and for every  $d \in \mathcal{D}$ , we have  $V_{\mathcal{M},g_d^\alpha}(\phi, v) = 1$ . From  $V_{\mathcal{M},g}(\forall \alpha \Box \phi, w) = 0$ , we have  $V_{\mathcal{M},g_u^\alpha}(\Box \phi, w) = 0$ , for some  $u \in \mathcal{D}$  at world  $w$ . Then,  $V_{\mathcal{M},g_u^\alpha}(\phi, v') = 0$  for some  $u \in \mathcal{D}$  at some world  $v' \in \mathcal{W}$ . Hence, we can choose some world  $v = v'$  and  $d = u$  s.t.  $V_{\mathcal{M},g_u^\alpha}(\phi, v') = 1$  and  $V_{\mathcal{M},g_u^\alpha}(\phi, v') = 0$  ■

2.  $\forall \alpha \Box \phi \rightarrow \Box \forall \alpha \phi$

**Claim:**  $\models_{\text{SQML}} \forall \alpha \Box \phi \rightarrow \Box \forall \alpha \phi$

*Proof.* AFC that  $V_{\mathcal{M},g}(\forall \alpha \Box \phi \rightarrow \Box \forall \alpha \phi, w) = 0$ . Then,  $V_{\mathcal{M},g}(\forall \alpha \Box \phi, w) = 1$ ,  $V_{\mathcal{M},g}(\Box \forall \alpha \phi, w) = 0$ . From  $V_{\mathcal{M},g}(\forall \alpha \Box \phi, w) = 1$ , for every  $d \in \mathcal{D}$ ,  $V_{\mathcal{M},g_d^\alpha}(\Box \phi, w) = 1$ . Then, for each  $d \in \mathcal{D}$ ,  $V_{\mathcal{M},g_d^\alpha}(\phi, v) = 1$  for every  $v \in \mathcal{W}$ . From  $V_{\mathcal{M},g}(\Box \forall \alpha \phi, w) = 0$ ,  $V_{\mathcal{M},g}(\forall \alpha \phi, v) = 0$  for some  $v' \in \mathcal{W}$ . Hence  $V_{\mathcal{M},g}(\phi, v') = 0$  for some  $u \in \mathcal{D}$ . Now, we can choose a  $v = v'$  and a  $d = u$  s.t.  $V_{\mathcal{M},g_u^\alpha}(\phi, v') = 1$  and  $V_{\mathcal{M},g_u^\alpha}(\phi, v') = 0$  ■

3.  $\Box \exists \alpha \phi \rightarrow \exists \alpha \Box \phi$

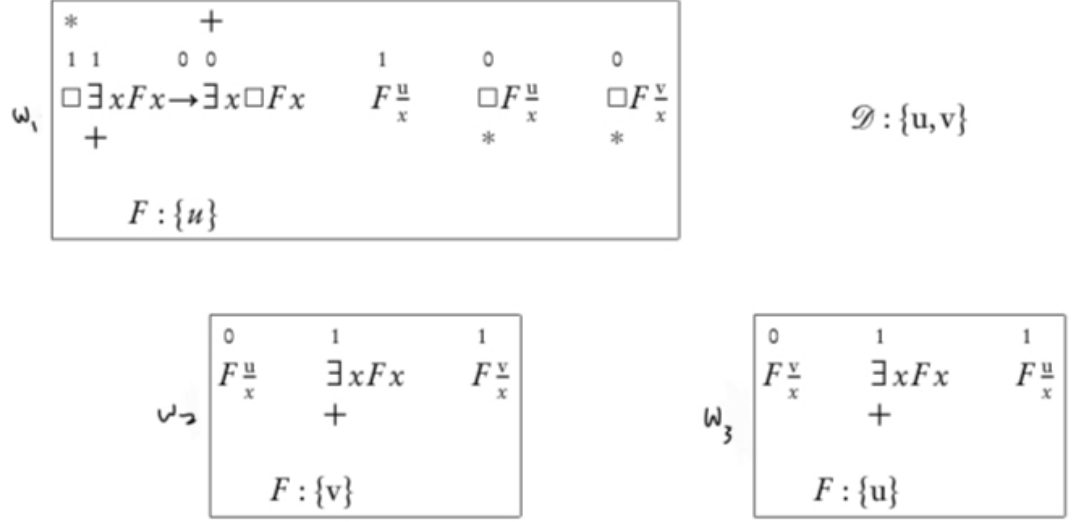
**Claim:**  $\not\models_{\text{SQML}} \Box \exists \alpha \phi \rightarrow \exists \alpha \Box \phi$

*Proof.* We let  $\phi := Fx$ .

Our official countermodel is

$$\begin{aligned}\mathcal{W} &= \{w_1, w_2, w_3\} \\ \mathcal{D} &= \{u, v\} \\ \mathcal{J}(F) &= \{\langle u, w_1 \rangle, \langle u, w_3 \rangle, \langle v, w_2 \rangle\}\end{aligned}$$

The illustration is



**Remark:** A symbol above the connective implies a commitment to *all* instances of a given type while a symbol below the connective implies a commitment to *some* instances of a given type.

We will talk through how this was constructed

- We begin with  $w_1$ : we are committed to the antecedent being true i.e. in all worlds,  $F$  is satisfied by some  $x$  from the domain; we are also committed to the consequent being false i.e. in all worlds, we are committed to there being no  $x$  in the domain satisfying  $\Box Fx$
- Then, in world  $w_1$ , we indicate assigning some object from the domain to  $F$  in place of the variable  $x$  i.e.  $g_u^x$  as  $F_x^u$ . We need one assignment to be true, so the extension of  $F$  must be at least  $\{u\}$
- Now, we know we need to make  $\Box Fx$  false for every object in the domain, so we say that  $\Box F_x^u$  is false in  $w_1$ , meaning that there is some world in which  $F_x^u$  is false. So, we need some new world
- So, we construct  $w_2$  where  $F_x^u$  is false; we need  $\exists x Fx$  to be true in this world, so we add a new object  $v$  into the domain, and make the extension of  $F$  in  $w_2$  be  $\{v\}$ . Then,  $F_x^v$  is true
- Notice that because we are committed to  $\Box Fx$  being false for every object in the domain, we need to go back to  $w_1$  and render  $\Box F_x^v$  false.
- Then, there needs to be a world in which  $F_x^v$  is false. So, we add a world  $w_3$  in which this is the case i.e. the extension of  $F$  is just  $\{u\}$ . Since  $\exists x Fx$  must hold,  $F_x^u$  must be true

**Remark:** We could just have done  $F_x^v$  is false in  $w_1$  and need not have created  $w_3$  ■

4.  $\exists \alpha \Box \phi \rightarrow \Box \exists \alpha \phi$

**Claim:**  $\models_{\text{SQML}} \exists \alpha \Box \phi \rightarrow \Box \exists \alpha \phi$

*Proof.* AFC that  $V_{\mathcal{M},g}(\exists \alpha \Box \phi \rightarrow \Box \exists \alpha \phi, w) = 0$ . Then,  $V_{\mathcal{M},g}(\exists \alpha \Box \phi, w) = 1, V_{\mathcal{M},g}(\Box \exists \alpha \phi, w) = 0$ . From  $V_{\mathcal{M},g}(\exists \alpha \Box \phi, w) = 1$ ,  $V_{\mathcal{M},g_u^\alpha}(\Box \phi, w) = 1$  for some  $u \in \mathcal{D}$ . Hence,  $V_{\mathcal{M},g_u^\alpha}(\phi, v) = 1$  for all  $v \in \mathcal{W}$ . From  $V_{\mathcal{M},g}(\Box \exists \alpha \phi, w) = 0$ ,  $V_{\mathcal{M},g}(\exists \alpha \phi, v') = 0$  for some  $v' \in \mathcal{W}$ . Then,  $V_{\mathcal{M},g_d^\alpha}(\phi, v') = 0$  for all  $d \in \mathcal{D}$ . So, we can choose a  $d = u$  s.t.  $V_{\mathcal{M},g_u^\alpha}(\phi, v') = 0$  from the second and  $v = v'$  s.t.  $V_{\mathcal{M},g_u^\alpha}(\phi, v') = 1$  from the first ■

## 7 PC and SQML: Proof Theory

### 7.1 PC and SQML Axiomatic Proofs

**Definition 7.1 (PC-instance).** A PC-instance of a schema is the result of uniformly replacing each schematic formula letter  $\phi, \psi, \dots$  with a PC-wff, and each schematic term  $\alpha, \beta, \dots$  with a PC-term

**Definition 7.2 (QML-instance).** A QML-instance of a schema is the result of uniformly replacing each schematic formula letter  $\phi, \psi, \dots$  with a QML-wff, and each schematic term  $\alpha, \beta, \dots$  with a QML-term

#### Axiomatic system for SQML.

- *Rule:* All QML-instances of MP, UG and NEC (where, in UG,  $\alpha$  is a variable) are QML-rules:

$$\frac{\phi \rightarrow \psi \quad \phi}{\psi} \text{MP} \quad \frac{\phi}{\Box \phi} \text{NEC} \quad \frac{\phi}{\forall \alpha \phi} \text{UG}$$

- *Axioms:* All QML-instances of the following PL-axiom schemas are SQML-axioms:

$$\phi \rightarrow (\psi \rightarrow \phi) \quad (\text{PL1})$$

$$(\phi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \chi)) \quad (\text{PL2})$$

$$(\sim \psi \rightarrow \sim \phi) \rightarrow ((\sim \psi \rightarrow \phi) \rightarrow \psi) \quad (\text{PL3})$$

- All QML-instances of PC-schema meeting the side-conditions are SQML-axioms:

$$\forall \alpha \phi \rightarrow \phi(\beta/\alpha) \quad (\text{PC1})$$

$$\forall \alpha (\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \forall \alpha \psi) \quad (\text{PC2})$$

- All QML-instances of (RX) and (II) are SQML-axioms:

$$\alpha = \alpha \quad (\text{RX})$$

$$\alpha = \beta \rightarrow (\phi(\alpha) \rightarrow \phi(\beta)) \quad (\text{II})$$

where, in (II),  $\beta$  is substitutable for  $\alpha$  and  $\phi(\beta)$  results from replacing **zero or more free occurrences** of  $\alpha$  with  $\beta$  in  $\phi(\alpha)$

- All QML-instances of the S5-schemas are SQML-axioms:

$$\Box(\phi \rightarrow \psi) \rightarrow (\Box \phi \rightarrow \Box \psi) \quad (\text{K})$$

$$\Box \phi \rightarrow \phi \quad (\text{T})$$

$$\Diamond \Box \phi \rightarrow \Box \phi \quad (\text{S5})$$

#### Side Conditions on (PC1) and (PC2):

- (PC1) is subject to the constraint that  $\alpha$  is a variable, and  $\phi(\beta/\alpha)$  results from  $\phi$  by correct substitution of  $\beta$  for  $\alpha$
- (PC2) is subject to the constraint that  $\alpha$  is a variable that does not occur freely in  $\phi$

It is clear we need a definition of correct substitution

**Definition 7.3 (Correct Substitution).** We specify substitution and correct substitution separately:

- We say that  $\beta$  is substitutable for  $\alpha$  in  $\phi$  if  $\alpha$  does not occur free in any subformula of  $\phi$  beginning with  $\forall \beta$

- When  $\beta$  is substitutable for  $\alpha$  in  $\phi$ , the formula which results from  $\phi$  by correct substitution of  $\beta$  for  $\alpha$ ,  $\phi(\beta/\alpha)$ , is the formula that results from replacing all and only free occurrences of  $\alpha$  in  $\phi$  with  $\beta$

**Remark:** Notice that (II) only requires the constraint of substitutability to be respected and NOT correct substitution.

**Example 7.1 (Substitution).** Compute the following substitutions: (i)  $(Px \wedge \Box Qy)(x/y)$ , (ii)  $(\Box \forall x Py)(x/y)$ , (iii)  $(\forall x Px \wedge Rxy)(y/x)$

**Solution.**

(i)  $(Px \wedge \Box Qy)(x/y)$ .

$$\begin{aligned} (Px \wedge \Box Qy)(x/y) &= (\sim (Px \rightarrow \sim \Box Qy))(x/y) \\ &= \sim ((Px \rightarrow \sim \Box Qy)(x/y)) \\ &= \sim ((Px)(x/y) \rightarrow \sim \Box((Qy)(x/y))) \\ &= \sim (Px \rightarrow \sim \Box Qx) \\ &= Px \wedge \Box Qx \end{aligned}$$

(ii)  $(\Box \forall x Py)(x/y)$

$$\begin{aligned} (\Box \forall x Py)(x/y) &= \Box((\forall x Py)(x/y)) \\ &= \text{undefined, since } y \text{ occurs free in subformula } \forall x Py \text{ beginning with } \forall x \end{aligned}$$

(iii)  $(\forall x Px \wedge Rxy)(y/x)$

$$\begin{aligned} (\forall x Px \wedge Rxy)(y/x) &= (\forall x Px)(y/x) \wedge (Rxy)(y/x) \\ &= \forall x Px \wedge Ryy \end{aligned}$$

In an exam, we would express  $\wedge$  in terms of Sider's official language of  $\rightarrow, \sim$  especially since the substitution rules make no provision for  $\wedge$ .

**Motivation for side-constraints of PC1 and PC2:**

- Without the constraint, PC1 generates non-valid instances for instance  $\not\models_{PC} \forall x \exists y Rxy \rightarrow \exists y Ryy$ . The countermodel to this is easy

$$\begin{aligned} \mathcal{D} &= \{1, 2\} \\ R &= \{\langle 1, 2 \rangle, \langle 2, 1 \rangle\} \end{aligned}$$

- Without the constraint, then there would be freely occurring variables in a PC-tautology, which cannot be right

Suppose that  $\phi_1 \rightarrow (\phi_2 \rightarrow \dots (\phi_n \rightarrow \psi) \dots)$  is a PC-tautology. Then we can help ourselves to the following meta-rule in abbreviated proofs:

$$\frac{\phi_1, \dots, \phi_n}{\psi} \text{PC}$$

**Example 7.2.**  $\vdash_{PC} \forall x (Fx \wedge Gx) \rightarrow \forall x Fx$

*Proof.*

- |                                                                                                                          |                   |
|--------------------------------------------------------------------------------------------------------------------------|-------------------|
| 1. $\forall x (Fx \wedge Gx) \rightarrow Fx \wedge Gx$                                                                   | PC1               |
| 2. $\forall x (Fx \wedge Gx) \rightarrow Fx$                                                                             | PL (syllogism), 1 |
| 3. $\forall x (\forall x (Fx \wedge Gx) \rightarrow Fx)$                                                                 | UG, 2             |
| 4. $\forall x (\forall x (Fx \wedge Gx) \rightarrow Fx) \rightarrow (\forall x (Fx \wedge Gx) \rightarrow \forall x Fx)$ | PC2               |
| 5. $(\forall x (Fx \wedge Gx) \rightarrow \forall x Fx)$                                                                 | MP, 3, 4          |

■

## 7.2 PC and SQML Toolkit and worked examples

### 7.2.1 PC: Tautologies and Worked Examples

**Tautologies** The PC tautologies we use centre around  $\forall\alpha$ .

1.  $\vdash_{PC} \forall\alpha\phi \rightarrow \forall\beta\phi(\beta/\alpha)$ .

*Proof.*

- |                                                                                                                                                |        |
|------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1. $\forall\alpha\phi \rightarrow \phi(\beta/\alpha)$                                                                                          | PC1    |
| 2. $\forall\beta(\forall\alpha\phi \rightarrow \phi(\beta/\alpha))$                                                                            | UG     |
| 3. $\forall\beta(\forall\alpha\phi \rightarrow \phi(\beta/\alpha)) \rightarrow (\forall\alpha\phi \rightarrow \forall\beta\phi(\beta/\alpha))$ | PC2    |
| 4. $\forall\alpha\phi \rightarrow \forall\beta\phi(\beta/\alpha)$                                                                              | 2,3,MP |

■

2. Distribution:  $\vdash_{PC} \forall\alpha(\phi \rightarrow \psi) \rightarrow (\forall\alpha\phi \rightarrow \forall\alpha\psi)$

*Proof.*

- |                                                                                                                                                                                                                                        |          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1. $\forall\alpha(\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \psi)$                                                                                                                                                          | PC1      |
| 2. $\forall\phi \rightarrow \phi$                                                                                                                                                                                                      | PC1      |
| 3. $(\forall\alpha(\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \psi)) \rightarrow ((\forall\alpha\phi \rightarrow \phi) \rightarrow (\forall\alpha(\phi \rightarrow \psi) \rightarrow (\forall\alpha\phi \rightarrow \psi)))$ | PL       |
| 4. $\forall\alpha(\phi \rightarrow \psi) \rightarrow (\forall\alpha\phi \rightarrow \psi)$                                                                                                                                             | 1,2,3,MP |
| 5. $\forall\alpha(\forall\alpha(\phi \rightarrow \psi) \rightarrow (\forall\alpha\phi \rightarrow \psi))$                                                                                                                              | 4, UG    |
| 6. $\forall\alpha(\forall\alpha(\phi \rightarrow \psi) \rightarrow (\forall\alpha\phi \rightarrow \psi)) \rightarrow (\forall\alpha(\phi \rightarrow \psi) \rightarrow \forall\alpha(\forall\phi \rightarrow \psi))$                   | PC2      |
| 7. $\forall\alpha(\phi \rightarrow \psi) \rightarrow \forall\alpha(\forall\alpha\phi \rightarrow \psi)$                                                                                                                                | 5,6, MP  |
| 8. $\forall\alpha(\forall\alpha\phi \rightarrow \psi) \rightarrow (\forall\alpha\phi \rightarrow \forall\alpha\psi)$                                                                                                                   | PC2      |
| 9. $\forall\alpha(\phi \rightarrow \psi) \rightarrow (\forall\alpha\phi \rightarrow \forall\alpha\psi)$                                                                                                                                | 7,8, MP  |

■

**Remarks:**

- Line 3 follows from the PL tautology  $(P \rightarrow (Q \rightarrow R)) \rightarrow ((S \rightarrow Q) \rightarrow (P \rightarrow (S \rightarrow R)))$
- Step 1 and 2 are legal instances of PC1, regardless of what  $\phi, \psi$  look like since we do not replace  $\alpha$  with any other variable
- Line 8 is a legal instance of PC2, since  $\alpha$  does not occur free in  $\forall\alpha\phi$

**Worked Example** Our worked example shows (i) how to prove  $\exists$  and (ii) how to prove from premises in PC

1.  $\exists x\forall yRxy \vdash_{PC} \forall y\exists xRxy$  i.e.  $\sim \forall x \sim \forall yRxy \vdash_{PC} \forall y \sim \forall x \sim Rxy$

*Proof.*

- |                                                                 |                          |
|-----------------------------------------------------------------|--------------------------|
| 1. $\sim \forall x \sim \forall yRxy$                           | premise                  |
| 2. $\forall yRxy \rightarrow Rxy$                               | PC1                      |
| 3. $\sim Rxy \rightarrow \sim \forall yRxy$                     | 2, PL (contraposition)   |
| 4. $\forall x(\sim Rxy \rightarrow \sim \forall yRxy)$          | 3, UG                    |
| 5. $\forall x \sim Rxy \rightarrow \forall x \sim \forall yRxy$ | Distribution, 4, MP      |
| 6. $\sim \forall x \sim Rxy$                                    | 1,5, PL (contraposition) |
| 7. $\forall y \sim \forall x \sim Rxy$                          | 6, UG                    |

■

### 7.2.2 SQML: Theorems and Worked Examples

#### SQML-theorems

What is provable in SQML is of philosophical interest: predominantly, the necessity of existence, the Barcan Formula and the Converse Barcan Formula.

**Theorem 7.1 (Necessity of Existence).**  $\vdash_{\text{SQML}} \Box \forall \alpha \Box \exists \beta (\alpha = \beta)$

*Proof.*

- |    |                                                                      |                        |
|----|----------------------------------------------------------------------|------------------------|
| 1. | $\alpha = \alpha$                                                    | RX                     |
| 2. | $\forall \beta \sim \beta = \alpha \rightarrow \sim \alpha = \alpha$ | PC1                    |
| 3. | $\alpha = \alpha \rightarrow \sim \forall \beta \sim \beta = \alpha$ | PL (contraposition)    |
| 4. | $\sim \forall \beta \sim \beta = \alpha$                             | 1,3, MP                |
| 5. | $\Box \forall \alpha \Box \sim \forall \beta \sim \beta = \alpha$    | 4, NEC, UG, NEC        |
| 6. | $\Box \forall \alpha \Box \exists \beta (\beta = \alpha)$            | def of $\exists \beta$ |

■

**Theorem 7.2 (Barcan Formula).**  $\vdash_{\text{SQML}} \forall \alpha \Box \phi(\alpha) \rightarrow \Box \forall \alpha \phi(\alpha)$

We must first prove

**Lemma 7.3.** The B axiom is provable in SQML i.e.  $\vdash_{\text{SQML}} \Diamond \Box \phi \rightarrow \phi$

*Proof.*

- |    |                                            |    |
|----|--------------------------------------------|----|
| 1. | $\Box \phi \rightarrow \phi$               | T  |
| 2. | $\Diamond \Box \phi \rightarrow \Box \phi$ | S5 |
| 3. | $\Diamond \Box \phi \rightarrow \phi$      | PL |

■

We now are ready to prove it

*Proof.*

- |    |                                                                               |                           |
|----|-------------------------------------------------------------------------------|---------------------------|
| 1. | $\forall \alpha \Box \phi \rightarrow \Box \phi$                              | PC1                       |
| 2. | $\Diamond \forall \alpha \Box \phi \rightarrow \Diamond \Box \phi$            | 1, NEC, K $\Diamond$ , MP |
| 3. | $\Diamond \Box \phi \rightarrow \phi$                                         | B                         |
| 4. | $\Diamond \forall \alpha \Box \phi \rightarrow \phi$                          | 2,3, PC                   |
| 5. | $\Diamond \forall \alpha \Box \phi \rightarrow \forall \alpha \phi$           | 4, UG, PC2, MP            |
| 6. | $\Box \Diamond \forall \alpha \Box \phi \rightarrow \Box \forall \alpha \phi$ | 5, Becker                 |
| 7. | $\forall \alpha \Box \phi \rightarrow \Box \Diamond \forall \alpha \Box \phi$ | B $\Diamond$              |
| 8. | $\forall \alpha \Box \phi \rightarrow \Box \forall \alpha \phi$               | 6,7, PL                   |

■

**Corollary 7.4 (Existential Barcan Formula).**  $\vdash_{\text{SQML}} \Diamond \exists \alpha \phi(\alpha) \rightarrow \exists \alpha \Diamond \phi(\alpha)$

**Theorem 7.5 (Converse Barcan Formula).**  $\vdash_{\text{SQML}} \Box \forall \alpha \phi(\alpha) \rightarrow \forall \alpha \Box \phi(\alpha)$

*Proof.*

- |                                                                                                                                      |           |
|--------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 1. $\forall\alpha\phi \rightarrow \phi(\alpha/\alpha)$                                                                               | PC1       |
| 2. $\Box\forall\alpha\phi \rightarrow \Box\phi$                                                                                      | 1, Becker |
| 3. $\forall\alpha(\Box\forall\alpha\phi \rightarrow \Box\phi)$                                                                       | UG2       |
| 4. $\forall\alpha(\Box\forall\alpha\phi \rightarrow \Box\phi) \rightarrow (\Box\forall\alpha\phi \rightarrow \forall\alpha\Box\phi)$ | PC2       |
| 5. $\Box\forall\alpha\phi \rightarrow \forall\alpha\Box\phi$                                                                         | 3,4, MP   |

■

**Corollary 7.6 (Existential Converse Barcan Formula).**  $\vdash_{\text{SQML}} \exists\alpha\Diamond\phi(\alpha) \rightarrow \Diamond\exists\alpha\phi(\alpha)$

The next corollary follows immediately from soundness:

**Corollary 7.7.** We have the following SQML-validities:

$$\begin{aligned}
&\models_{\text{SQML}} \Box\forall\alpha\exists\beta(\beta = \alpha) && \text{(Necessity of Existence)} \\
&\models_{\text{SQML}} \forall\alpha\Box\phi(\alpha) \rightarrow \Box\forall\alpha\phi(\alpha) && \text{(BF)} \\
&\models_{\text{SQML}} \Box\forall\alpha\phi(\alpha) \rightarrow \forall\alpha\Box\phi(\alpha) && \text{(CBF)}
\end{aligned}$$

### SQML Worked Examples

**Example 7.3.** Show that  $\vdash_{\text{SQML}} (\Box\forall\alpha\phi \wedge \Diamond\exists\alpha\psi) \rightarrow \Diamond\exists\alpha(\phi \wedge \psi)$

*Proof.* First, we reduce the claim to

$$\vdash_{\text{SQML}} \sim (\Box\forall\alpha\phi \rightarrow \sim \Diamond\exists\alpha\psi) \rightarrow \Diamond\exists\alpha \sim (\phi \rightarrow \sim \psi)$$

Then, we can show

$$(\Box\forall\alpha\phi \rightarrow \sim \Diamond\exists\alpha\psi) \leftrightarrow (\Box\forall\alpha\phi \rightarrow \sim \sim \Box \sim \sim \forall\alpha \sim \psi)$$

and

$$(\Box\forall\alpha\phi \rightarrow \sim \sim \Box \sim \sim \forall\alpha \sim \psi) \leftrightarrow (\Box\forall\alpha\phi \rightarrow \Box\forall\alpha \sim \psi)$$

By the substitution of equivalent formulas, we have

$$\vdash_{\text{SQML}} \sim (\Box\forall\alpha\phi \rightarrow \Box\forall\alpha \sim \psi) \rightarrow \sim \Box\forall\alpha(\phi \rightarrow \sim \psi)$$

Now, we start the main proof

- |                                                                                                                                                       |                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| 1. $\forall\alpha(\phi \rightarrow \sim \psi) \rightarrow (\forall\alpha\phi \rightarrow \forall\alpha \sim \psi)$                                    | Distribution                     |
| 2. $\Box\forall\alpha(\phi \rightarrow \sim \psi) \rightarrow \Box(\forall\alpha\phi \rightarrow \forall\alpha \sim \psi)$                            | 1, Becker                        |
| 3. $\Box(\forall\alpha\phi \rightarrow \forall\alpha \sim \psi) \rightarrow (\Box\forall\alpha\phi \rightarrow \Box\forall\alpha \sim \psi)$          | K                                |
| 4. $\Box\forall\alpha(\phi \rightarrow \sim \psi) \rightarrow (\Box\forall\alpha\phi \rightarrow \Box\forall\alpha \sim \psi)$                        | 2,3, PL (transitivity)           |
| 5. $\sim (\Box\forall\alpha\phi \rightarrow \Box\forall\alpha \sim \psi) \rightarrow \sim \Box\forall\alpha(\phi \rightarrow \sim \psi)$              | PL (contraposition)              |
| 6. $\sim \Box\forall\alpha(\phi \rightarrow \sim \psi) \rightarrow \sim \sim \Diamond \sim \sim \exists\alpha \sim (\phi \rightarrow \sim \psi)$      | def of $\exists\alpha, \Diamond$ |
| 7. $\sim \sim \Diamond \sim \sim \exists\alpha \sim (\phi \rightarrow \sim \psi) \rightarrow \Diamond\exists\alpha \sim (\phi \rightarrow \sim \psi)$ | PL                               |
| 8. $\sim (\Box\forall\alpha\phi \rightarrow \sim \Diamond\exists\alpha\psi) \rightarrow \Diamond\exists\alpha \sim (\phi \rightarrow \sim \psi)$      | 5,6,7, PL (transitivity)         |
| 9. $(\Box\forall\alpha\phi \wedge \Diamond\exists\alpha\psi) \rightarrow \Diamond\exists\alpha(\phi \wedge \psi)$                                     | def of $\wedge$                  |

■



### 7.3 Basic Metatheory of SQML

**Theorem 7.8 (Soundness of SQML).** If  $\vdash_{\text{SQML}} \phi$ , then  $\models_{\text{SQML}} \phi$

We have to prove 2 lemmas first

**Lemma 7.9.** SQML-axioms are SQML-valid

*Proof.* PL1-PL3 are all PL-valid as shown before, and hence S-valid and SQML valid. K, T, S5 are all S-valid as shown previously, and hence SQML valid. PC1-PC2 are PC-valid and hence SQML-valid. We can generalise S-validity to SQML since we can treat a MPL-wff as an n-place predicate with only constants as temrs. We can also generalise PC-validity since we can just fix a world  $w$  and show a contradiction in SQML semantics. It remains to show that RX and II are valid.

- RX:  $\alpha = \alpha$   
AFC that  $V_{\mathcal{M},g}(a = a, w) = 0$ . But for any  $g$ ,

$$[\alpha]_{\mathcal{M},g} = [\alpha]_{\mathcal{M},g}$$

Hence,  $V_{\mathcal{M},g}(a = a, w) = 1$

- II:  $\alpha = \beta \rightarrow (\phi(\alpha) \rightarrow \phi(\beta))$ .  
See Fabian Pregel's solution on validity of II. Alternatively, we can prove on induction.

■

**Lemma 7.10.** All SQML rules preserve SQML validity

*Proof.* Since MP preserves PL-validity, it preserves SQML validity. Since NEC preserves S5-validity, it preserves SQML validity. It remains to show that UG preserves SQML validity.

UG: if  $\models_{\text{SQML}} \phi$  then  $\models_{\text{SQML}} \forall \alpha \phi$ .

Assume  $\models_{\text{SQML}} \phi$ . Then  $V_{\mathcal{M},g}(\phi, w) = 1$  holds for any  $w \in \mathcal{W}$  and any  $g$  in each world, for each model  $\mathcal{M}$ . Consider some model  $\mathcal{M}$ . We know that for any world  $w$ ,  $V_{\mathcal{M},g}(\phi, w) = 1$  for any assignment  $g$ , so  $V_{\mathcal{M},g_u^a}(\phi, w) = 1$  for every  $u \in \mathcal{D}$  and thus  $V_{\mathcal{M},g}(\forall \alpha \phi, w) = 1$ .

Since this holds for any  $w \in \mathcal{W}$  then, it is true in a model  $\mathcal{M}$ .

Since it holds for all models  $\mathcal{M}$ ,  $\models_{\text{SQML}} \forall \alpha \phi$

■

We now engage in the main proof which is done by induction on complexity of  $\phi$ .

*Proof.* Let  $\vdash_n \phi$  denote a proof of length  $n$ .

BC:  $\vdash_1 \phi$ . By def,  $\phi$  is an SQML axiom. By lemma 1,  $\models_{\text{SQML}} \phi$ .

IH: Suppose  $\vdash_m \phi$  implies  $\models_{\text{SQML}} \phi$  for all  $m < n$ . We want to show that  $\vdash_n \phi$  implies  $\models_{\text{SQML}} \phi$ .

Suppose  $\vdash_n \phi$ . There are four cases for the last line of  $\phi$ .

1. Case 1:  $\phi$  is an SQML axiom. As in base case, by lemma 1,  $\models_{\text{SQML}} \phi$
2. Case 2:  $\phi$  is obtained by NEC. Then  $\phi = \Box \psi$ , with  $\vdash_m \psi$  with  $m < n$ . So,  $\models_{\text{SQML}} \psi$ . By lemma 2, if  $\models_{\text{SQML}} \psi$ , then  $\models_{\text{SQML}} \Box \psi$ . So  $\models_{\text{SQML}} \phi$
3. Case 3:  $\phi$  is obtained by UG. Then  $\phi = \forall \alpha \psi$ , with  $\vdash_m \psi$ , where  $m < n$ . By IH,  $\models_{\text{SQML}} \psi$ . By lemma 2,  $\models_{\text{SQML}} \forall \alpha \psi$ . Hence  $\models_{\text{SQML}} \phi$
4. Case 4:  $\phi$  is obtained by MP. Ten,  $\vdash_{m_1} \psi$  and  $\vdash_{m_2} \psi \rightarrow \phi$  with  $m_1, m_2 < n$ . So, by IH,  $\models_{\text{SQML}} \psi$  and  $\models_{\text{SQML}} \psi \rightarrow \phi$ . Hence, by lemma 2,  $\models_{\text{SQML}} \phi$ .

Claim proven.

■

## 8 Quantified modal logic: variable domains, 2D semantics

### 8.1 Variable Domain Quantified Modal Logic

**Motivation:** The reason why SQML generates controversial validities such as the Barcan formula or the necessity of existence is because it has a fixed domain in every world. But intuitively, we might think that each world has its own custom domain with its own objects peculiar to that world. This gives rise to variable domain semantics or variable domain quantified modal logic. More a more in-depth treatment of the rationale – see Williamson’s *Bare Possibilia*

**Definition 8.1 (VDQML model).** A VDQML-model is a quintuple  $\langle \mathcal{W}, \mathcal{R}, \mathcal{D}, \mathcal{Q}, \mathcal{I} \rangle$ :

- $\mathcal{W}$  is a non-empty set i.e. the set of worlds
- $\mathcal{R}$  is a binary relation over  $\mathcal{W}$  i.e. the ‘accessibility relation’
- $\mathcal{D}$  is a non-empty set i.e. the super-domain
- $\mathcal{Q}$  is a function that assigns each  $w \in \mathcal{W}$  a subset  $\mathcal{Q}(w)$  of  $\mathcal{D}$  i.e. the domain function
- $\mathcal{I}$  is a function such that
  - $\mathcal{I}(\alpha) \in \mathcal{D}$  for each constant  $\alpha$
  - $\mathcal{I}(\Pi^n)$  is a set of  $n+1$  tuples of the form  $\langle u_1, \dots, u_n, w \rangle$  where  $u_1, \dots, u_n$  are members of  $\mathcal{D}$  and  $w \in \mathcal{W}$ , for each  $n$ -place predicate  $\Pi^n$

i.e. the interpretation function

**Remark:**

1.  $\mathcal{W}$  and  $\mathcal{R}$  play the same role as in MPL
2.  $\mathcal{Q}$  provides domains for quantifiers to range over. We write  $\mathcal{Q}_w$  for  $\mathcal{Q}(w)$  where
  - $\mathcal{D}$  represents the super domain of all possible objects
  - $\mathcal{Q}_w$  represents the domain of  $w$  (the set of possible objects that exist in  $w$ )
  - $\mathcal{Q}_w$  can be an empty set
3.  $\mathcal{I}(\Pi^n)$  determines the extension of  $\Pi^n$  in each world  $w$ , which we may write  $\mathcal{I}_w(\Pi^n)$ :

$$\mathcal{I}_w(\Pi^n) = \{ \langle u_1, \dots, u_n \rangle : \langle u_1, \dots, u_n, w \rangle \in \mathcal{I}(\Pi^n) \}$$

**Definition 8.2 (VDQML Valuations).** The valuation  $V_{\mathcal{M},g}$  for a VDQML-model  $\mathcal{M} = \langle \mathcal{W}, \mathcal{R}, \mathcal{D}, \mathcal{Q}, \mathcal{I} \rangle$  and assignment  $g$  is the unique function that assigns 0 or 1 to each wff at each world and satisfies the following conditions:

*Atomic Formulas:* for terms  $\alpha, \beta, \dots$ , and  $n$ -ary predicate,  $\Pi^n$

- $V_{\mathcal{M},g}(\alpha = \beta, w) = 1$  iff  $[\alpha]_{\mathcal{M},g} = [\beta]_{\mathcal{M},g}$
- $V_{\mathcal{M},g}(\Pi^n \alpha_1, \dots, \alpha_n, w) = 1$  iff  $\langle [\alpha_1]_{\mathcal{M},g}, \dots, [\alpha_n]_{\mathcal{M},g}, w \rangle \in \mathcal{I}(\Pi^n)$  or equivalently  $\langle [\alpha_1]_{\mathcal{M},g}, \dots, [\alpha_n]_{\mathcal{M},g} \rangle \in \mathcal{I}_w(\Pi^n)$

*Connectives:* for formulas  $\phi$  and  $\psi$

- $V_{\mathcal{M},g}(\phi \rightarrow \psi, w) = 1$  iff  $V_{\mathcal{M},g}(\phi, w) = 0$  or  $V_{\mathcal{M},g}(\psi, w) = 1$
- $V_{\mathcal{M},g}(\sim \phi, w) = 1$  iff  $V_{\mathcal{M},g}(\phi, w) = 0$

*Modal Operators:* for formula  $\phi$

- $V_{\mathcal{M},g}(\Box \phi, w) = 1$  iff, for every  $v \in \mathcal{W}$  such that  $\mathcal{R}wv$ ,  $V_{\mathcal{M},g}(\phi, v) = 1$

*Quantifiers:* for formula  $\phi$  and variable  $\alpha$

- $V_{\mathcal{M},g}(\forall\alpha\phi, w) = 1$  iff, for every  $d \in \mathcal{D}_w$ ,  $V_{\mathcal{M},g_d^\alpha}(\phi, w) = 1$

**Remark:** The most important ways in which VDQML differs from SQML is the addition of the accessibility relation and the fact that the quantifier ranges over  $\mathcal{D}_w$  and not just  $\mathcal{D}$  i.e.  $\forall$  and  $\Box$  interact with one another more deeply.

**Definition 8.3 (VDQML validity).** a wff  $\phi$  is VDQML-valid i.e.  $\models_{\text{VDQML}} \phi$  iff  $V_{\mathcal{M},g}(\phi, w) = 1$  for every VDQML model  $\mathcal{M} = \langle \mathcal{W}, \mathcal{R}, \mathcal{D}, \mathcal{Q}, \mathcal{I} \rangle$ , for every  $w \in \mathcal{W}$  and every  $g$  for  $\mathcal{M}$

The following facts about PC-validity generalise to QML

**Proposition 8.1.** if  $g$  and  $h$  agree on the free variables in  $\phi$ , then  $V_{\mathcal{M},g}(\phi, w) = V_{\mathcal{M},h}(\phi, w)$  for each  $w \in \mathcal{W}$

*Proof.* Suppose  $g$  and  $h$  agree on free variables. We prove that the consequent holds on the complexity of  $\phi$  i.e.  $V_{\mathcal{M},g}(\phi, w) = V_{\mathcal{M},h}(\phi, w)$  for each  $w$ .

BC:  $\phi = \Pi\alpha_1 \dots \alpha_n$ . Then, each term  $\alpha_i$  is either a constant or a freely occurring variable. If  $\alpha_i$  is a constant, then for any pair  $g, h$

$$[\alpha_i]_g = [\alpha_i]_h$$

If  $\alpha_i$  is a variable, by assumption,

$$[\alpha_i]_g = [\alpha_i]_h$$

Then, for any world  $w \in \mathcal{W}$ ,

$$\langle [\alpha_1]_g, \dots, [\alpha_n]_g, w \rangle = \langle [\alpha_1]_h, \dots, [\alpha_n]_h, w \rangle$$

So, it follows that

$$\langle [\alpha_1]_g, \dots, [\alpha_n]_g, w \rangle \in \mathcal{I}(\Pi^n) \text{ iff } \langle [\alpha_1]_h, \dots, [\alpha_n]_h, w \rangle \in \mathcal{I}(\Pi^n)$$

Therefore,  $V_{\mathcal{M},g}(\phi, w) = V_{\mathcal{M},h}(\phi, w)$  for each  $w$  in the base case.

IH: we assume for any wff of lower complexity that  $\phi$ , call this  $\psi$ ,  $V_{\mathcal{M},g}(\psi, w) = V_{\mathcal{M},h}(\psi, w)$ .

Then, we have four cases:

- Case 1:  $\phi = \sim \psi$ . By IH,  $V_{\mathcal{M},g}(\psi, w) = V_{\mathcal{M},h}(\psi, w)$ . Hence  $V_{\mathcal{M},g}(\sim \psi, w) = V_{\mathcal{M},h}(\sim \psi, w)$  and  $V_{\mathcal{M},g}(\phi, w) = V_{\mathcal{M},h}(\phi, w)$ .
- Case 2:  $\phi = \psi_1 \rightarrow \psi_2$ . By IH,  $V_{\mathcal{M},g}(\psi_1, w) = V_{\mathcal{M},h}(\psi_1, w)$  and  $V_{\mathcal{M},g}(\psi_2, w) = V_{\mathcal{M},h}(\psi_2, w)$ , implying that  $V_{\mathcal{M},g}(\psi_1 \rightarrow \psi_2, w) = V_{\mathcal{M},h}(\psi_1 \rightarrow \psi_2, w)$ . Hence,  $V_{\mathcal{M},g}(\phi, w) = V_{\mathcal{M},h}(\phi, w)$
- Case 3:  $\phi = \forall\alpha\psi$ . By IH,  $V_{\mathcal{M},g}(\psi, w) = V_{\mathcal{M},h}(\psi, w)$ . We know that  $V_{\mathcal{M},g}(\forall\alpha\psi, w) = 1$  iff  $\forall u \in \mathcal{D}$ ,  $V_{\mathcal{M},g_u^\alpha}(\psi, w) = 1$  iff  $\forall u \in \mathcal{D}$ ,  $V_{\mathcal{M},h}(\psi, w) = 1$  iff  $V_{\mathcal{M},h}(\forall\alpha\psi, w) = 1$  where  $g_u^\alpha$  and  $h_u^\alpha$  agree on free variables. By IH, we conclude that  $V_{\mathcal{M},g}(\phi, w) = V_{\mathcal{M},h}(\phi, w)$
- Case 4:  $\phi = \Box\psi$ . By IH,  $V_{\mathcal{M},g}(\psi, w) = V_{\mathcal{M},h}(\psi, w)$  for all  $w \in \mathcal{W}$ . Then,  $V_{\mathcal{M},g}(\Box\psi, w) = 1$  iff  $\forall v \in \mathcal{W}$  s.t.  $\mathcal{R}wv$   $V_{\mathcal{M},g}(\psi, w) = 1$  iff  $\forall v \in \mathcal{W}$  s.t.  $\mathcal{R}wv$   $V_{\mathcal{M},h}(\psi, w) = 1$  iff  $V_{\mathcal{M},h}(\Box\psi, w) = 1$ . Hence,  $V_{\mathcal{M},g}(\phi, w) = V_{\mathcal{M},h}(\phi, w)$

IH established, and the claim immediately follows. ■

**Proposition 8.2.** if  $[\alpha]_{\mathcal{M},g} = [\beta]_{\mathcal{M},g}$ , then  $V_{\mathcal{M},g}(\phi, w) = V_{\mathcal{M},g}(\phi(\beta/\alpha), w)$ , for each  $w \in \mathcal{W}$

*Proof.* Suppose  $[\alpha]_{\mathcal{M},g} = [\beta]_{\mathcal{M},g}$ . We prove that  $V_{\mathcal{M},g}(\phi, w) = V_{\mathcal{M},g}(\phi(\beta/\alpha), w)$  for each  $w \in \mathcal{W}$  by induction on  $\phi$ 's complexity.

BC: WLOG,  $\phi = \Pi\alpha, \gamma_1, \dots, \gamma_n$  (it does not matter how many  $\alpha$ s there are or where  $\alpha$  is). Then, by assumption, for any world  $w \in \mathcal{W}$ , we have

$$\begin{aligned} & \langle [\alpha]_{\mathcal{M},g}, [\gamma_1]_{\mathcal{M},g}, \dots, [\gamma_n]_{\mathcal{M},g} \rangle \in \mathcal{I}_w(\Pi^n) \\ & \text{iff } \langle [\beta]_{\mathcal{M},g}, [\gamma_1]_{\mathcal{M},g}, \dots, [\gamma_n]_{\mathcal{M},g} \rangle \in \mathcal{I}_w(\Pi^n) \\ & \implies \langle [\alpha]_{\mathcal{M},g}, [\gamma_1]_{\mathcal{M},g}, \dots, [\gamma_n]_{\mathcal{M},g,w} \rangle \in \mathcal{I}(\Pi^n) \\ & \text{iff } \langle [\beta]_{\mathcal{M},g}, [\gamma_1]_{\mathcal{M},g}, \dots, [\gamma_n]_{\mathcal{M},g,w} \rangle \in \mathcal{I}(\Pi^n) \end{aligned}$$

Hence,  $V_{\mathcal{M},g}(\phi, w) = V_{\mathcal{M},g}(\phi(\beta/\alpha), w)$ .

IH: Let  $\phi_n$  denote that  $\phi$  has complexity  $n$ . Then, we assume that for all  $n < m$ ,  $V_{\mathcal{M},g}(\phi_n, w) = V_{\mathcal{M},g}(\phi_n(\beta/\alpha), w)$ . We have four cases to consider:

- Case 1:  $\phi_m = \sim \phi_n$ . Then,  $V_{\mathcal{M},g}(\phi_n, w) = V_{\mathcal{M},g}(\phi_n(\beta/\alpha), w)$  (by IH)  $\iff V_{\mathcal{M},g}(\sim \phi_n, w) = V_{\mathcal{M},g}(\sim \phi_n(\beta/\alpha), w) \iff V_{\mathcal{M},g}(\phi_m, w) = V_{\mathcal{M},g}(\phi_m(\beta/\alpha), w)$
- Case 2:  $\phi_m = \phi_{n_1} \rightarrow \phi_{n_2}$  with  $n_1, n_2 < m$ . Then  $V_{\mathcal{M},g}(\phi_{n_1}, w) = V_{\mathcal{M},g}(\phi_{n_1}(\beta/\alpha), w)$  and  $V_{\mathcal{M},g}(\phi_{n_2}, w) = V_{\mathcal{M},g}(\phi_{n_2}(\beta/\alpha), w) \iff V_{\mathcal{M},g}(\phi_{n_1} \rightarrow \phi_{n_2}, w) = V_{\mathcal{M},g}((\phi_{n_1} \rightarrow \phi_{n_2})(\beta/\alpha), w) \iff V_{\mathcal{M},g}(\phi_m, w) = V_{\mathcal{M},g}(\phi_m(\beta/\alpha), w)$
- Case 3:  $\phi_m = \forall \gamma \phi_n$ . By IH,  $V_{\mathcal{M},g}(\phi_n, w) = V_{\mathcal{M},g}(\phi_n(\beta/\alpha), w)$ . Then, we need to consider three cases:
  1.  $\gamma = \alpha$ : Consider that  $V_{\mathcal{M},g}(\forall \alpha \phi_n, w) = 1$  iff for all  $u \in \mathcal{D}$ ,  $V_{\mathcal{M},g_u^\alpha}(\phi_n, w) = 1$ . But by IH,  $V_{\mathcal{M},g}(\phi_n, w) = V_{\mathcal{M},g}(\phi_n(\beta/\alpha), w)$ . Hence, for all  $u \in \mathcal{D}$ ,  $V_{\mathcal{M},g_u^\alpha}(\phi_n, w) = 1$  holds iff  $\forall u \in \mathcal{D}$ ,  $V_{\mathcal{M},g}(\phi_n(\beta/\alpha), w) = 1$  iff  $V_{\mathcal{M},g}(\forall \alpha \phi(\beta/\alpha), w) = w$ .
  2.  $\gamma = \beta$ : Then,  $\alpha$  is not substitutable for  $\beta$ , since in  $\beta$  is freely occurring in some formula beginning with  $\forall \beta$ . So this case cannot happen, on the assumption that we respect the rules of substitutability.
  3.  $\gamma \neq \alpha, \beta$ : Consider  $V_{\mathcal{M},g}(\forall \gamma \phi_n, w) = 1$  iff  $\forall u \in \mathcal{D}$ ,  $V_{\mathcal{M},g_u^\gamma}(\phi_n, w) = 1$ . By IH and assumption, for each  $u \in \mathcal{D}$  this holds iff  $V_{\mathcal{M},g_u^\gamma}(\phi_n(\beta/\alpha), w) = 1$  which holds iff  $V_{\mathcal{M},g}(\forall \gamma \phi_n(\beta/\alpha), w) = 1$ .

Therefore,  $V_{\mathcal{M},g}(\phi_m, w) = V_{\mathcal{M},g}(\phi_m(\beta/\alpha), w)$

- Case 4:  $\phi_m = \Box \phi_n$  By IH,  $V_{\mathcal{M},g}(\phi_n, w) = V_{\mathcal{M},g}(\phi_n(\beta/\alpha), w)$  for each  $w \in \mathcal{W}$ . So, at any world  $v \in \mathcal{W}$  s.t.  $\mathcal{R}wv$ ,  $V_{\mathcal{M},g}(\phi_n, v) = V_{\mathcal{M},g}(\phi_n(\beta/\alpha), v)$ . So, we yield

$$\begin{aligned}
 &\therefore V_{\mathcal{M},g}(\phi_n, v) = 1, \forall v \in \mathcal{W} \text{ s.t. } \mathcal{R}wv \\
 &\text{iff } V_{\mathcal{M},g}(\phi_n, v) = 1, \forall v \in \mathcal{W} \text{ s.t. } \mathcal{R}wv \\
 &\therefore V_{\mathcal{M},g}(\Box \phi_n, w) = V_{\mathcal{M},g}(\Box \phi_n(\beta/\alpha), w) \\
 &\therefore V_{\mathcal{M},g}(\phi_m, w) = V_{\mathcal{M},g}(\phi_m(\beta/\alpha), w)
 \end{aligned}$$

Claim proven ■

In VDQML, the two formulas are invalid

**Example 8.1 (The necessity of existence).**  $\not\models_{\text{VDQML}} \Box \forall \alpha \Box \exists \beta (\alpha = \beta)$

*Proof.* We need to find a counter-model that satisfies  $V_{\mathcal{M},g}(\Box \forall \alpha \Box \exists \beta (\alpha = \beta), w) = 0$  which means  $V_{\mathcal{M},g}(\forall \alpha \Box \exists \beta (\alpha = \beta), v) = 0$  for some  $v \in \mathcal{W}$  s.t.  $\mathcal{R}wv$  and for some  $d \in \mathcal{D}_v$ ,  $V_{\mathcal{M},g_d^\alpha}(\Box \exists \beta (\alpha = \beta), v) = 0$ . Consequently, we must satisfy  $V_{\mathcal{M},g_d^\alpha}(\exists \beta (\alpha = \beta), u) = 0$  for some  $u \in \mathcal{W}$  s.t.  $\mathcal{R}vu$ . Then, for that  $u$ , for all  $d' \in \mathcal{D}_u$ ,  $V_{\mathcal{M},g_{d'}^\beta}(\alpha = \beta, u) = 0$ . Our countermodel is

$$\begin{aligned}
 \mathcal{W} &= \{r, a\} \\
 \mathcal{D} &= \{1\} \\
 \mathcal{D}_r &= \{1\} \\
 \mathcal{D}_a &= \emptyset \\
 \mathcal{R} &= \{\langle r, r, \rangle, \langle r, a \rangle, \langle a, r \rangle, \langle a, a \rangle\}
 \end{aligned}$$

Basically, we just need to find one world (in this case,  $a$ ) in which  $\exists \beta (\beta = \alpha)$  fails, where the assignment to  $\alpha$  comes from another world, in this case  $r$ . ■

**Example 8.2 (Barcan formula).**  $\not\models_{\text{VDQML}} \forall \alpha \Box \phi \rightarrow \Box \forall \alpha \phi$

*Proof.* Counter-model:

$$\begin{aligned}
\mathcal{W} &= \{r, a\} \\
\mathcal{R} &= \{\langle r, r \rangle, \langle r, a \rangle, \langle a, a \rangle\} \\
\mathcal{D} &= \{u, v\} \\
\mathcal{D}_r &= \{u\} \\
\mathcal{D}_a &= \{u, v\} \\
\mathcal{I}(F) &= \{\langle u, r \rangle, \langle u, a \rangle\}
\end{aligned}$$

■

Also, under VDQML, the PC1 axiom no longer holds and we need to revise it.

**Example 8.3 (PC1).**  $\not\models_{\text{VDQML}} \forall \alpha \phi \rightarrow \phi(\beta/\alpha)$

*Proof.* Let  $\phi := F\alpha$  The countermodel is

$$\begin{aligned}
\mathcal{W} &= \{w\} \\
\mathcal{D} &= \{u, v\} \\
\mathcal{D}_w &= \{u\} \\
\mathcal{I}(F) &= \{\langle u, w \rangle\} \\
\mathcal{R} &= \emptyset
\end{aligned}$$

then,  $g$  could be such that  $\langle [\beta]_{\mathcal{M},g}, w \rangle = \langle v, w \rangle$ . So,  $\langle [\beta]_{\mathcal{M},g}, w \rangle \notin \mathcal{I}(F)$ . So, there is some  $g$  such that  $V_{\mathcal{M},g}(\forall \alpha F\alpha, w) = 1$  while  $V_{\mathcal{M},g}(F\beta, w) = 0$  ■

**Example 8.4 (Revised PC1).**  $\models_{\text{VDQML}} \forall \alpha \phi \rightarrow (\exists \gamma(\gamma = \beta) \rightarrow \phi(\beta/\alpha))$  with  $\gamma \neq \alpha, \beta$

*Proof.* AFC that there is an  $\mathcal{M}$  s.t.  $V_{\mathcal{M},g}(\forall \alpha \phi \rightarrow (\exists \gamma(\gamma = \beta) \rightarrow \phi(\beta/\alpha)), w) = 0$ . Then we have the following

1.  $V_{\mathcal{M},g}(\forall \alpha \phi, w) = 1$
2.  $V_{\mathcal{M},g}(\exists \gamma(\gamma = \beta), w) = 1$
3.  $V_{\mathcal{M},g}(\phi(\beta/\alpha), w) = 0$

From (1):  $\forall d \in \mathcal{D}_w, V_{\mathcal{M},g_d^\alpha}(\phi, w) = 1$

From (2):  $\exists d' \in \mathcal{D}_w$  s.t.  $[\gamma]_{g_{d'}^\gamma} = [\beta]_{g_{d'}^\gamma}$

It is important to remember that  $g$  is fixed throughout the whole proof. Hence, from 1) we know that for any  $u \in \mathcal{D}_w$ ,  $V_{g_u^\alpha}(\phi, w) = 1$ . Now, since  $\alpha$  does not appear in  $\phi(\beta/\alpha)$  we see that  $V_{g_u^\alpha}(\phi(\beta/\alpha), w) = V_g(\phi(\beta/\alpha), w)$  as  $g$  and  $g_u^\alpha$  agree on all the free variables in  $\phi(\beta/\alpha)$ . So from  $V_{g_u^\alpha}(\phi, w) = 1$  and  $V_{g_u^\alpha}(\phi(\beta/\alpha), w) = 0$  we can conclude that  $u = [\alpha]_{g_u^\alpha} \neq [\beta]_{g_u^\alpha}$ . Changing  $u$  definitely changes the value of  $[\alpha]_{g_u^\alpha}$ , but since  $\alpha \neq \beta$  it does not change the value of  $[\beta]_{g_u^\alpha}$ . It is also important to note that  $[\beta]_{g_u^\alpha} = [\beta]_g$  for any  $u \in D_w$ . Hence, for any  $u \in D_w$  we have that  $u = [\alpha]_{g_u^\alpha} \neq [\beta]_g$ .

This is what contradicts 2, as from 2 we can conclude that there is a  $d' \in \mathcal{D}_w$  such that  $[\gamma]_{g_{d'}^\gamma} = [\beta]_{g_{d'}^\gamma}$ .  $[\gamma]_{g_{d'}^\gamma} = d'$ , and so we get that  $d' = [\beta]_{g_{d'}^\gamma} = [\beta]_g$ , which is our contradiction. ■

**Intuition:** In VDQML, because we can have an empty domain relative to a world  $w$ , with a non-empty superdomain, we need to instantiate the existence of at least one object in the world in order to make the axiom work.

**Example 8.5 (PC2).**  $\models_{\text{VDQML}} \forall \alpha(\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \forall \alpha \psi)$  whenever  $\alpha$  does not occur free in  $\phi$

*Proof.* AFC that  $V_{\mathcal{M},g}(\forall \alpha(\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \forall \alpha \psi), w) = 0$ . Hence,

1.  $V_{\mathcal{M},g}(\forall \alpha(\phi \rightarrow \psi), w) = 1$

$$2. V_{\mathcal{M},g}(\phi \rightarrow \forall\psi, w) = 0$$

From (1):  $\forall d \in \mathcal{D}_w, V_{\mathcal{M},g_d^\alpha}(\phi \rightarrow \psi, w) = 1$ . Hence,  $V_{\mathcal{M},g_d^\alpha}(\psi, w) = 1$  or  $V_{\mathcal{M},g_d^\alpha}(\phi, w) = 0$ .

From (2):  $V_{\mathcal{M},g}(\phi, w) = 1$  for some  $d' \in \mathcal{D}_w, V_{\mathcal{M},g_{d'}^\alpha}(\psi, w) = 0$ . Since  $g$  and  $g_{d'}^\alpha$  agree on the free variables as  $\alpha$  does not occur freely in  $\phi$ ,  $V_{\mathcal{M},g}(\phi, w) = V_{\mathcal{M},g_{d'}^\alpha}(\phi, w) = 1$ .

Since  $V_{\mathcal{M},g_{d'}^\alpha}(\phi, w) = 1$ , it must be the case that  $V_{\mathcal{M},g_{d'}^\alpha}(\psi, w) = 1$  due to (1) but for that  $d' \in \mathcal{D}, V_{\mathcal{M},g_{d'}^\alpha}(\psi, w) = 0$  ■

### Increasing/Decreasing domains:

Basically, we want to add an additional constraint on  $\mathcal{R}$  by saying that any world accessible from  $r$  will have a larger domain i.e. increasing domains

**Definition 8.4 (Increasing Domain).** A relation  $\mathcal{R}$  has an increasing domain constraint just when if  $\mathcal{R}uv$ , then  $\mathcal{D}_v \subseteq \mathcal{D}_u$

**Definition 8.5 (Decreasing Domain).** A relation  $\mathcal{R}$  has a decreasing domain constraint just when if  $\mathcal{R}uv$ , then  $\mathcal{D}_v \supseteq \mathcal{D}_u$

**Definition 8.6 (Frame).** Let a frame be a quadruple consisting of the first four components of a VDQML model. Say that a QML-sentence is valid on a frame  $\langle \mathcal{W}_0, \mathcal{R}_0, \mathcal{D}_0, \mathcal{Q}_0 \rangle$  iff it is valid in every model  $\langle \mathcal{W}_0, \mathcal{R}_0, \mathcal{D}_0, \mathcal{Q}_0, \mathcal{I} \rangle$  whose first four components are those of the frame.

Now, we consider two propositions:

**Proposition 8.3.** Show that all instances of (CBF)  $\Box\forall\alpha\phi \rightarrow \forall\alpha\Box\phi$  are valid on a frame  $F = \langle \mathcal{W}, \mathcal{R}, \mathcal{D}, \mathcal{Q} \rangle$  iff  $F$  is increasing i.e.  $\mathcal{R}uw$  implies  $\mathcal{D}_u \subseteq \mathcal{D}_w$ .

*Proof.* ( $\Leftarrow$ ) Suppose  $F$  is increasing i.e.  $\mathcal{R}uw$ , then  $\mathcal{D}_u \subseteq \mathcal{D}_w$ . For reductio,  $V_{\mathcal{M},g}(\Box\forall\alpha\phi \rightarrow \forall\alpha\Box\phi, u) = 0$ . Then

- $V_{\mathcal{M},g}(\Box\forall\alpha\phi, u) = 1$ .  
Hence,  $\forall w \in \mathcal{W}$  s.t.  $\mathcal{R}uw, V_{\mathcal{M},g}(\forall\alpha\phi, w) = 1$ . So,  $\forall d \in \mathcal{D}_w, V_{\mathcal{M},g_d^\alpha}(\phi, w) = 1$
- $V_{\mathcal{M},g}(\forall\alpha\Box\phi, u) = 0$   
Hence,  $\exists d' \in \mathcal{D}_u$  s.t.  $V_{\mathcal{M},g_{d'}^\alpha}(\Box\phi, u) = 0$ . So, for some world  $w' \in \mathcal{W}$  s.t.  $\mathcal{R}uw', V_{\mathcal{M},g_{d'}^\alpha}(\phi, w') = 0$

But this is a contradiction since  $\mathcal{D}_u \subseteq \mathcal{D}_{w'}$ . So, for that  $w', \forall d \in \mathcal{D}_{w'}, V_{g_{d'}^\alpha}(\phi, w) = 1$ , contradicting  $V_{\mathcal{M},g_{d'}^\alpha}(\phi, w') = 0$ .

( $\Rightarrow$ ) Suppose that  $\Box\forall\alpha\phi \rightarrow \forall\alpha\Box\phi$  is valid on a frame  $F$ . Then suppose for reductio that  $F$  is not increasing i.e. there is a pair  $u, w$  such that  $\mathcal{R}uw$  but  $\mathcal{D}_u \not\subseteq \mathcal{D}_w$ . Then, we create a countermodel i.e.  $\mathcal{R}uw$  implies  $\mathcal{D}_u = \mathcal{D}_w$  which satisfies

- $V_{\mathcal{M},g}(\Box\forall\alpha\phi, u) = 1$ .  
Hence,  $\forall w \in \mathcal{W}$  s.t.  $\mathcal{R}uw, V_{\mathcal{M},g}(\forall\alpha\phi, w) = 1$ . So,  $\forall d \in \mathcal{D}_w, V_{\mathcal{M},g_d^\alpha}(\phi, w) = 1$
- $V_{\mathcal{M},g}(\forall\alpha\Box\phi, u) = 0$   
Hence,  $\exists d' \in \mathcal{D}_u$  s.t.  $V_{\mathcal{M},g_{d'}^\alpha}(\Box\phi, u) = 0$ . So, for some world  $w' \in \mathcal{W}$  s.t.  $\mathcal{R}uw', V_{\mathcal{M},g_{d'}^\alpha}(\phi, w') = 0$

i.e. when we let  $\phi = F\alpha$

$$\begin{aligned}\mathcal{W} &= \{w_1, w_2\} \\ \mathcal{R} &= \{\langle w_1, w_2 \rangle\} \\ \mathcal{D}_{w_1} &= \{u, v\} \\ \mathcal{D}_{w_2} &= \{u\} \\ \mathcal{D} &= \{u, v\} \\ \mathcal{I}(F) &= \{\langle u, w_1 \rangle, \langle u, w_2 \rangle\}\end{aligned}$$

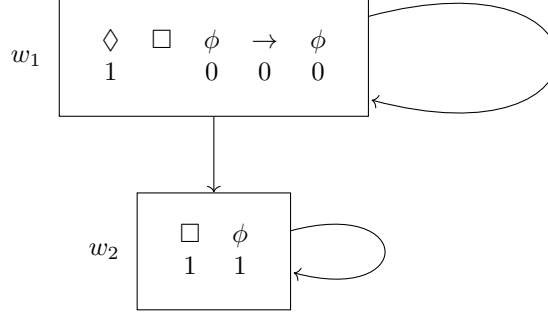
Once we let  $[\alpha]_{g_{d'}^\alpha} = v$ , we find the countermodel ■

**Proposition 8.4.** Show that all instances of (CBF)  $\Box\forall\alpha\phi \rightarrow \forall\alpha\Box\phi$  and (B)  $\Diamond\Box\phi \rightarrow \phi$  are valid on a frame  $F = \langle \mathcal{W}, \mathcal{R}, \mathcal{D}, \mathcal{I} \rangle$  only if  $F$  is locally constant

We first prove the lemma:

**Lemma 8.1.** If  $\Diamond\Box\phi \rightarrow \phi$  is valid on a frame  $F$ ,  $\mathcal{R}uw \implies \mathcal{R}wu$  i.e.  $\mathcal{R}$  is symmetric on  $\mathcal{W}$ .

*Proof.* By contraposition, we assume that  $\exists u, w$  s.t.  $\mathcal{R}wu$  but  $\sim \mathcal{R}uw$ . Then, we have



So, we have such a countermodel. Hence, if there is no such countermodel. then  $\mathcal{R}$  is symmetric on  $\mathcal{W}$  ■

We proceed onto the main proof, which becomes much simpler

*Proof.* We reduce the claim to, using the lemma: if all instances of  $\Box\forall\alpha\phi \rightarrow \forall\alpha\Box\phi$  is valid on  $F$  and  $\mathcal{R}$  is symmetric on  $\mathcal{W}$ , then  $F$  is locally constant.

From **Proposition 8.4**, if all instances of  $\Box\forall\alpha\phi \rightarrow \forall\alpha\Box\phi$  is valid on  $F$ , then  $F$  is increasing i.e.  $\mathcal{R}uw \implies \mathcal{D}_u \subseteq \mathcal{D}_w$ .

By symmetry, this means that  $\mathcal{R}wu$  as well which implies  $\mathcal{D}_w \subseteq \mathcal{D}_u$ . Hence,  $\mathcal{D}_w = \mathcal{D}_u$  i.e.  $F$  is locally constant. ■

## 8.2 2-dimensional quantified modal logic

**Motivation:** QML suffers from expressive weakness. For example, not all readings of ‘All the rich people could’ve been poor’ can be formalised in QML. In particular, if we mean *the* rich people, as in the current rich people, then we cannot formalise this in QML i.e. the rich people all collectively poor in one world. This is because we lack a reference world in our semantics.

We will use SQML as our basis for the formulation of 2DQML.

### 8.2.1 Designated World SQML

The most basic form of 2DQML is in fixing the actual world to  $w_{@}$  i.e. it cannot change.

**Definition 8.7 (Designated-world SQML).** A designated world SQML-model is a quadruple  $\langle \mathcal{W}, w_{@}, \mathcal{D}, \mathcal{I} \rangle$ :

- $\mathcal{W}$  is a non-empty set i.e. the set of worlds
- $w_{@}$  is a member of  $\mathcal{W}$  i.e. the designated world
- $\mathcal{D}$  is a non-empty set i.e. the domain
- $\mathcal{I}$  is a function such that:
  - $\mathcal{I}(\alpha) \in \mathcal{D}$  for each constant  $\alpha$
  - $\mathcal{I}(\Pi^n)$  is a set of  $n + 1$  tuples of the form  $\langle u_1, \dots, u_n, w \rangle$ , where  $u_1, \dots, u_n$ , are members of  $\mathcal{D}$  and  $w \in \mathcal{W}$ , for each  $n$ -place predicate  $\Pi^n$

**Definition 8.8 (dSQML Valuation).** We add the following clause to SQML valuations to achieve dSQML valuations:

- $V_{\mathcal{M},g}(@\phi, w) = 1$  iff  $V_{\mathcal{M},g}(\phi, w_{@}) = 1$

**Definition 8.9 (Truth in dSQML model).** A wff  $\phi$  is true in a dSQML-model  $\mathcal{M} = \langle \mathcal{W}, w_{@}, \mathcal{D}, \mathcal{I} \rangle$  iff  $\phi$  is true at  $w_{@}$  (i.e.  $V_{\mathcal{M},g}(\phi, w_{@}) = 1$  for every assignment  $g$  for  $\mathcal{M}$ )

**Definition 8.10 (dSQML validity).** a wff  $\phi$  is dSQML valid iff  $\phi$  is true in all dSQML-models

### 8.2.2 2DQML

**Motivation:** It is clear that having just one designated world that is fixed is not sufficient. For instance, dSQML cannot formalise all readings of ‘Necessarily, all the rich people could have been poor’ since this means that we go to every world and find a scenario where all the actual rich people *in that world* are all not rich.

*Moral:*  $\Box$  wants us to shift the actual world i.e. permitting the designated world to change.

**Definition 8.11 (2DQML Valuation).** The 2D-valuation,  $V_{\mathcal{M},g}^2$ , for a SQML-model  $\mathcal{M} = \langle \mathcal{W}, \mathcal{D}, \mathcal{I} \rangle$  and variable assignment  $g$  is the unique function that assigns 0 or 1 to each wff relative to each pair of worlds and satisfies the following conditions:

*Atomic Formulas:* for terms  $\alpha, \beta, \dots$ , and  $n$ -ary predicate,  $\Pi^n$

- $V_{\mathcal{M},g}^2(\alpha = \beta, v, w) = 1$  iff  $[\alpha]_{\mathcal{M},g} = [\beta]_{\mathcal{M},g}$
- $V_{\mathcal{M},g}^2(\Pi^n \alpha_1, \dots, \alpha_n, v, w) = 1$  iff  $\langle [\alpha_1]_{\mathcal{M},g}, \dots, [\alpha_n]_{\mathcal{M},g} \rangle \in \mathcal{I}(\Pi^n)$

*Connectives:* for formulas  $\phi$  and  $\psi$ :

- $V_{\mathcal{M},g}^2(\phi \rightarrow \psi, v, w) = 1$  iff  $V_{\mathcal{M},g}^2(\sim \phi, v, w) = 1$  iff  $V_{\mathcal{M},g}^2(\phi, v, w) = 0$

*Quantifiers:* for formula  $\phi$  and variable  $\alpha$ :

- $V_{\mathcal{M},g}^2(\forall \alpha \phi, v, w) = 1$  iff, for every  $d \in \mathcal{D}$ ,  $V_{\mathcal{M},g_d^a}^2(\phi, v, w) = 1$

*Modal Operators:* for formula  $\phi$ :

- $V_{\mathcal{M},g}^2(\Box \phi, v, w) = 1$  iff, for every  $w' \in \mathcal{W}$ ,  $V_{\mathcal{M},g}^2(\phi, v, w') = 1$
- $V_{\mathcal{M},g}^2(@\phi, v, w) = 1$  iff  $V_{\mathcal{M},g}^2(\phi, v, w') = 1$
- $V_{\mathcal{M},g}^2(\times \phi, v, w) = 1$  iff  $V_{\mathcal{M},g}^2(\phi, w, w) = 1$

**Remark:** We can interpret @ and  $\times$  in the following ways

1. For a valuation function  $V_{\mathcal{M},g}^2(\phi, v, w)$ , we call the first world  $v$  the reference world, and the second world  $w$  an evaluation world
2. To evaluate  $V_{\mathcal{M},g}^2(@\phi, v, w)$ , discard the old evaluation world  $w$  and evaluate  $\phi$  at the reference world  $v$
3. To evaluate  $V_{\mathcal{M},g}^2(\times \phi, v, w)$ , set the reference world to  $w$  (i.e. the evaluation world), and evaluate  $\phi$  at  $w$ .

For validity, we have two ways to define it:

The first validity is just truth evaluated at each world  $w$  of every model from the standpoint of the same world  $w$  as the reference world. This is truth *along the diagonal*

**Definition 8.12 (2D-validity).** A wff  $\phi$  is 2D-valid i.e.  $\models_{2D} \phi$  iff  $V_{\mathcal{M},g}^2(\phi, w, w) = 1$  for each SMPL model  $\mathcal{M} = \langle \mathcal{W}, \mathcal{D}, \mathcal{I} \rangle$ , each  $w \in \mathcal{W}$ , and each  $g$  for  $\mathcal{M}$



General validity is truth under every arbitrary pair of reference and evaluation world  $w, v$  of every model – this is why it is called general.

**Definition 8.13 (General 2D validity).** A wff  $\phi$  is generally 2D-valid –  $\models_{\text{G2D}} \phi$  – iff  $V_{\mathcal{M},g}^2(\phi, v, w) = 1$  for each SMPL model  $\mathcal{M} = \langle \mathcal{W}, \mathcal{D}, \mathcal{I} \rangle$ , each  $v, w \in \mathcal{W}$ , and each  $g$  for  $\mathcal{M}$

### Examples for Validity

**Example 8.6.** Show that  $\Box(\phi \leftrightarrow \Box @\phi)$  is neither 2D nor G2D valid.

*Proof.* It suffices to show that  $\not\models_{\text{2D}} \Box(\phi \leftrightarrow \Box @\phi)$  to establish that  $\not\models_{\text{G2D}} \Box(\phi \leftrightarrow \Box @\phi)$ . Let  $\phi$  be  $Fa$ . Our counter-model is

$$\begin{aligned}\mathcal{W} &= \{w_1, w_2\} \\ \mathcal{D} &= \{u\} \\ \mathcal{I}(a) &= u \\ \mathcal{I}(F) &= \{\langle u, w_1 \rangle\}\end{aligned}$$

Hence,  $V_{\mathcal{M},g}^2(\Box(\phi \leftrightarrow \Box @\phi), w_1, w_1) = 0$  since  $V_{\mathcal{M},g}^2(\phi \leftrightarrow \Box @\phi, w_1, w_2) = 0$  as  $V_{\mathcal{M},g}^2(\phi, w_1, w_2) = 0$  while  $V_{\mathcal{M},g}^2(\Box @\phi, w_1, w_2) = 1$ . This means that  $V_{\mathcal{M},g}^2(@\phi, w_1, w_1) = 1$  and  $V_{\mathcal{M},g}^2(@\phi, w_1, w_2) = 1$  since  $V_{\mathcal{M},g}^2(\phi, w_1, w_1) = 1$  ■

**Example 8.7.** Show that  $\Box \times (\phi \leftrightarrow \Box @\phi)$  is 2D and G2D valid.

*Proof.* It suffices to show that  $\models_{\text{G2D}} \Box \times (\phi \leftrightarrow \Box @\phi)$  to establish the first claim as well.

Assume for reductio that  $V_{\mathcal{M},g}^2(\Box \times (\phi \leftrightarrow \Box @\phi), w, w) = 0$ . Therefore, for some  $v \in \mathcal{W}$ ,  $V_{\mathcal{M},g}^2(\times(\phi \leftrightarrow \Box @\phi), w, v) = 0$ , which means that  $V_{\mathcal{M},g}^2(\phi \leftrightarrow \Box @\phi, v, v) = 0$ . Hence,  $V_{\mathcal{M},g}^2(\phi, v, v) \neq V_{\mathcal{M},g}^2(\Box @\phi, v, v)$ . We have two cases:

- Case 1: Suppose  $V_{\mathcal{M},g}^2(\Box @\phi, v, v) = 1, V_{\mathcal{M},g}^2(\phi, v, v) = 0$ . Then  $V_{\mathcal{M},g}^2(@\phi, v, v') = 1$  for all  $v' \in \mathcal{W}$ . So,  $V_{\mathcal{M},g}^2(\phi, v, v) = 1$ , a contradiction.
- Case 2: Suppose  $V_{\mathcal{M},g}^2(\Box @\phi, v, v) = 0, V_{\mathcal{M},g}^2(\phi, v, v) = 1$ . Then  $V_{\mathcal{M},g}^2(@\phi, v, v') = 0$  for some  $v' \in \mathcal{W}$ . So,  $V_{\mathcal{M},g}^2(\phi, v, v) = 0$ , a contradiction.

Hence, since both cases lead to a contradiction, by dichotomy, this is a contradiction. ■

### Formalisation

I think the hardest part is formalising the sentences, especially when there is no intuition for what  $\times$  is.

**Example 8.8.** Formalise ‘The actual inventor invented the zip (or no one singlehandedly invented it).’

**Answer:**  $\exists x(@Ix \wedge \forall y(@Iy \rightarrow y = x) \wedge Ix) \vee \sim \exists x(Ix \wedge \forall y(Iy \rightarrow y = x))$

**Example 8.9.** Formalise ‘Necessarily, all the rich people could have been poor.’

**Answer:**  $\Box \times \Diamond \forall (@Rx \rightarrow Px)$

**Example 8.10.** Formalise ‘Some non-actual thing could exist.’

**Answer:**

- Under the interpretation that it means some non-actual thing could exist, then  $\Diamond \exists x(@ \sim \exists y(y = x))$
- Under the interpretation that it means some non-actual thing could exist *actually*, then  $\exists x(@ \sim \exists y(y = x) \wedge \Diamond \times @ \exists z(z = x))$

**Remark:** ‘ $\sim @\phi$ ’ and ‘ $@ \sim \phi$ ’ have no semantic difference.

## 9 Counterfactuals

**Motivation:** The object of interest is counterfactual conditions i.e. If it had been that  $\phi$ , then it would have been that  $\psi$ . MPL supplies two candidates:

1. The material conditions:  $\phi \rightarrow \psi$
2. The strict conditional:  $\Box(\phi \rightarrow \psi)$ . We abbreviate this as  $\phi \rightarrow \psi$

But these formalisations license the following inference patterns:

1. False antecedent/True consequent

$$\frac{\sim \phi}{\phi \rightarrow \psi} \quad \frac{\psi}{\phi \rightarrow \psi}$$

But (1) It is not snowing, does not license (2) So, if it had snowed, Johnson would have ordered a nuclear strike on Brussels.

And neither does (1) The lecture is on Tuesday license (2) So, if it had been rescheduled for Thursday, it would have been on Tuesday.

2. Augmentation:

$$\frac{\phi \rightarrow \psi}{\phi \wedge \chi \rightarrow \psi} \quad \frac{\phi \rightarrow \psi}{\phi \wedge \chi \rightarrow \psi}$$

But (1) If you had flicked the light switch, the light would be on, does not license (2) So, if you had flicked the switch, having removed the bulb, it would be on.

3. Contraposition:

$$\frac{\phi \rightarrow \psi}{\sim \psi \rightarrow \sim \phi} \quad \frac{\phi \rightarrow \psi}{\sim \psi \rightarrow \sim \phi}$$

But (1) [Speaker looks at the light clouds above] If it rained, it wouldn't rain heavily, does not license (2) So, if it rained heavily, it wouldn't rain.

4. Transitivity:

$$\frac{\phi \rightarrow \psi, \psi \rightarrow \chi}{\phi \rightarrow \chi}$$

### 9.1 Stalnaker's Conditional

We enrich the MPL language with  $\Box \rightarrow$

**Definition 9.1 (SC wff).** We add the following clauses to MPL

- Every sentence letter  $\alpha$  is a wff
- If  $\phi$  and  $\psi$  are wffs, then  $(\phi \rightarrow \psi)$ ,  $\sim \phi$  and  $\Box \phi$  are wffs, and so is  $(\phi \Box \rightarrow \psi)$

**Intuition of SC:** Stalnaker proposes the following truth condition for  $A \Box \rightarrow B$  – Consider a possible world in which  $A$  is true, and which otherwise differs minimally from the actual world. ‘If  $A$ , then  $B$ ’ is true (false) just in case  $B$  is true (false) in that possible world.

Because we want to impose a distance metric onto the space of possible worlds, we have to define what an ordering of worlds is.

#### Order Relations

**Definition 9.2 (Preorder).** A binary relation  $\preceq$  is a (non-strict) preorder of a set  $A$  iff:

- $\preceq$  is reflexive on  $A$ :  $a \preceq a$  for each  $a \in A$
- $\preceq$  is transitive:  $a \preceq c$  for any  $a, b, c$  such that  $a \preceq b$  and  $b \preceq c$

We write  $a \preceq b$  to abbreviate  $\langle a, b \rangle \in \preceq$ .

**Definition 9.3 (Partial Order).** A (non-strict) preorder  $\preceq$  of  $A$  is said to be a (non-strict) partial order of a set  $A$  iff, moreover:

- $\preceq$  is anti-symmetric:  $a = b$  for any  $a$  and  $b$  such that  $a \preceq b$  and  $b \preceq a$

**Definition 9.4 (Linear Order).** A (non-strict) pre- or partial order  $\preceq$  on  $A$  is said to be a linear order iff, moreover:

- $\preceq$  is (strongly) connected on  $A$ :  $a \preceq b$  or  $b \preceq a$  or  $a = b$ , for any  $a, b \in A$

**Definition 9.5 (Least Element).** When  $\preceq$  is a partial order on  $A$ , and  $B \subseteq A$ ,  $b_0$  is said to be the least element of  $B$  wrt  $\preceq$  iff:

- $b_0 \in B$  and every  $b \in B$  is such that  $b_0 \preceq b$

**Remark:** If the least element exists, it is unique.

**Definition 9.6 (SC-model).** An SC-model  $\mathcal{M}$  is an ordered triple  $\langle \mathcal{W}, \preceq, \mathcal{I} \rangle$  where:

- $\mathcal{W}$  is a nonempty set i.e. the set of worlds
- $\mathcal{I}$  is a two place function that assigns either 0 or 1 to each sentence letter relative to each  $w \in \mathcal{W}$  i.e. the interpretation function
- $\preceq$  is a three-place relation over  $\mathcal{W}$  i.e. the nearness relation  
We abbreviate  $\langle u, v, w \rangle$  as  $u \preceq_w v$  i.e.  $u$  is nearer to  $w$  than  $v$
- The valuation function for  $V_{\mathcal{M}}$  for  $\mathcal{M}$  and  $\preceq$  satisfy the following conditions:
  - for any  $w \in \mathcal{W}$ :  $\preceq_w$  is strongly connected in  $\mathcal{W}$
  - for any  $w \in \mathcal{W}$ :  $\preceq_w$  is transitive
  - for any  $w \in \mathcal{W}$ :  $\preceq$  is anti-symmetric
  - for any  $u, v \in \mathcal{W}$ :  $u \preceq_u v$  i.e. the base assumption
  - for any SC-wff,  $\phi$ , provided  $V_{\mathcal{M}}(\phi, v) = 1$  for at least one  $v \in \mathcal{W}$ , then for every  $z \in \mathcal{W}$ , there's some  $w \in \mathcal{W}$  such that  $V_{\mathcal{M}}(\phi, w) = 1$  and such that for any  $u \in \mathcal{W}$ , if  $V_{\mathcal{M}}(\phi, u) = 1$ , then  $w \preceq_z u$

**Definition 9.7 (SC Valuation).** Given an SC-model  $\mathcal{M} = \langle \mathcal{W}, \preceq, \mathcal{I} \rangle$ , the valuation for  $\mathcal{M}$ ,  $V_{\mathcal{M}}$ , is the unique two place function that assigns 0 or 1 to each wff, for each  $w \in \mathcal{W}$ , that meets the following conditions, for any wffs  $\phi$  and  $\psi$ :

- $V_{\mathcal{M}}(\alpha, w) = \mathcal{I}(\alpha, w)$ , for each sentence letter  $\alpha$
- $V_{\mathcal{M}}(\phi \rightarrow \psi, w) = 1$  iff  $V_{\mathcal{M}}(\phi, w) = 0$  or  $V_{\mathcal{M}}(\psi, w) = 1$
- $V_{\mathcal{M}}(\sim \phi, w) = 1$  iff  $V_{\mathcal{M}}(\phi, v) = 1$  for all  $v \in \mathcal{W}$
- $V_{\mathcal{M}}(\phi \Box \rightarrow \psi, w) = 1$  iff for any  $v \in \mathcal{W}$ , IF  $[V_{\mathcal{M}}(\phi, v) = 1$  and for any  $u \in \mathcal{W}$  such that  $V_{\mathcal{M}}(\phi, u) = 1, v \preceq_w u]$  THEN:  $V_{\mathcal{M}}(\psi, v) = 1$

**Remark:** The last clause is how Sider defines a maximally-close  $\phi$ -world. Embedded in the definition is basically: a  $\phi$ -world  $v$  is maximally close to  $w$  if it is in fact a  $\phi$ -world, and that for any other  $\phi$ -worlds  $u$ ,  $v$  is closer to  $w$  than  $u$ . And if it is the case that  $\psi$  is also true in that maximally close world, then  $\phi \Box \rightarrow \psi$  is true at world  $w$ .

We can also define the closest world clause as a selection function i.e.  $V_{\mathcal{M}}(\psi, v) = 1$  where  $v$  is the closest  $\phi$ -world to  $w$ .

**Definition 9.8 (Closest-world selection function).** We can define  $V_{\mathcal{M}}(\psi, v) = 1$  where  $v$  is the closest  $\phi$ -world to  $w$  iff  $V_{\mathcal{M}}(\psi, f(\phi, w)) = 1$ , where

$$f(\phi, w) = \begin{cases} v & \text{if } V_{\mathcal{M}}(\phi, v) = 1 \text{ and } \forall v \in \mathcal{W} \text{ s.t. } V_{\mathcal{M}}(\phi, u) = 1, u \preceq_w v \\ \text{undefined} & \text{if } V_{\mathcal{M}}(\phi, w') = 0, \forall w' \in \mathcal{W} \end{cases}$$

**Remark:** For proof, see Q1 of Week 8 PSet.

**Definition 9.9 (SC-validity).** a wff  $\phi$  is SC-valid –  $\models_{\text{SC}} \phi$  – iff  $V_{\mathcal{M}}(\phi, w) = 1$  for every SC-model  $\mathcal{M} = \langle \mathcal{W}, \preceq, \mathcal{J} \rangle$  and every world  $w \in \mathcal{W}$

**Definition 9.10 (SC semantic consequence).** a wff  $\phi$  is an SC-semantic consequence of a set of wffs  $\Gamma$  –  $\Gamma \models_{\text{SC}} \phi$  – iff  $V_{\mathcal{M}}(\phi, w) = 1$  for every SC-model  $\mathcal{M} = \langle \mathcal{W}, \preceq, \mathcal{J} \rangle$  and every world  $w \in \mathcal{W}$  such that  $V_{\mathcal{M}}(\gamma, w) = 1$  for each  $\gamma \in \Gamma$

As a result, the following inferences are false in SC.

**Example 9.1 (False Inferences).** Show that the following inferences are invalid in SC

1. False antecedent:  $\sim \phi \not\models_{\text{SC}} \phi \Box \rightarrow \psi$
2. True consequent:  $\psi \not\models_{\text{SC}} \phi \Box \rightarrow \psi$
3. Contraposition:  $\phi \Box \rightarrow \psi \not\models_{\text{SC}} \sim \psi \Box \rightarrow \sim \phi$
4. Augmentation:  $\phi \Box \rightarrow \psi \not\models_{\text{SC}} (\phi \wedge \chi) \Box \rightarrow \psi$

## 9.2 Lewis' Conditional

### Motivation

Lewis argues that Stalnaker's assumptions (a) Anti-symmetry and (b) limit are flawed.

1. Against anti-symmetry: why can't there be ties between how similar worlds  $u, v$  are to  $w$ ?
2. Against limit: limit requires that there be a unique  $\phi$ -world closest to  $w$ . But why can't there be two or more equally close  $\phi$ -worlds and why couldn't there be a sequence of  $\phi$ -worlds ever closest to  $w$  with no closest (see density of the rationals in the reals)?

In particular, Lewis proposes a natural language counter-example to SC: 'If the line had been longer than one inch, it would have been one hundred miles.' This is true in SC since there is no closest  $\phi$ -world (as a property of  $\mathbb{R}$ ).

Lewis proposes getting rid of these two assumptions for his LC-models.

**Definition 9.11 (LC-model).** An LC-model is a triple  $\langle \mathcal{W}, \preceq, \mathcal{J} \rangle$  where:

- $\mathcal{W}$  is a non-empty set i.e. the set of possible worlds
- $\preceq$  is a three-place relation over  $\mathcal{W}$ , such that, for each  $w \in \mathcal{W}$ :
  - The binary relation  $\preceq_w$  is a linear preorder on  $\mathcal{W}$
  - if  $u \preceq_w w$ , then  $u = w$ , for any  $u \in \mathcal{W}$  i.e. the modified base assumption
- i.e.  $\preceq$  is the nearness relation
- $\mathcal{J}$  is a two-place function that assigns 0 or 1 to each sentence letter, for each  $w \in \mathcal{W}$

**Remark:** The modified base assumption tells us that any world distinct from  $w$  is not as similar to  $w$  as  $w$  is (this follows from SC anti-symmetry assumption)

**Definition 9.12 (LC Valuation).** Given an LC-model  $\mathcal{M} = \langle \mathcal{W}, \preceq, \mathcal{J} \rangle$ , the valuation for  $\mathcal{M}$ ,  $V_{\mathcal{M}}$ , is the unique two place function that assigns 0 or 1 to each wff, for each  $w \in \mathcal{W}$ , that meets the following conditions, for any wffs  $\phi$  and  $\psi$ :

- $V_{\mathcal{M}}(\alpha, w) = \mathcal{I}(\alpha, w)$ , for each sentence letter  $\alpha$
- $V_{\mathcal{M}}(\phi \rightarrow \psi, w) = 1$  iff  $V_{\mathcal{M}}(\phi, w) = 0$  or  $V_{\mathcal{M}}(\psi, w) = 1$
- $V_{\mathcal{M}}(\sim \phi, w) = 1$  iff  $V_{\mathcal{M}}(\phi, v) = 1$  for all  $v \in \mathcal{W}$
- $V_{\mathcal{M}}(\phi \Box \rightarrow \psi, w) = 1$  iff EITHER  $\phi$  is true in no worlds, OR: there is some world  $v$ , such that  $LV_{\mathcal{M}}(\phi, v) = 1$  and for all  $u$ , if  $u \preceq_w v$ , then  $LV_{\mathcal{M}}(\phi \rightarrow \psi, u) = 1$

**Remark:** the intuition behind Lewis's truth condition for  $\Box \rightarrow$  is this – if  $\phi$  is necessarily false, then  $\phi \Box \rightarrow \psi$  will come out true; but if  $\phi$  is possibly true, then  $\phi \Box \rightarrow \psi$  is true in  $w$  iff there exists some  $\phi$ -world with the feature that no matter how much closer to  $w$  you go, you will never find a  $\phi$ -world where  $\psi$  is false. (If there is a nearest-to- $w$   $\phi$ -world, then  $\phi \Box \rightarrow \psi$  is true in  $w$  iff  $\psi$  is true in all the nearest-to- $w$   $\phi$ -worlds) This avoids vacuity that Stalnaker has because we require that

- for some world  $x$ , the antecedent and consequent are both true in  $x$  and in all worlds that are at least as similar to the actual world as is  $x$ , the antecedent is never true while the consequent is false

But there is no such world  $x$ . For let  $x$  be any world in which the antecedent and consequent are both true. Since the line is one hundred miles long in  $x$ , we can find a world that is more similar to the actual world than  $x$  in which the antecedent is true but the consequent is false. (for more see the essay notes)

Most importantly, LC and SC differs in whether they accept (i) distribution and (ii) Conditional excluded middle i.e.

1. Distribution:  $[\phi \Box \rightarrow (\psi \vee \chi)] \rightarrow [(\phi \Box \rightarrow \psi) \vee (\phi \Box \rightarrow \chi)]$
2. Conditional Excluded Middle:  $(\phi \Box \rightarrow \psi) \vee (\phi \Box \rightarrow \sim \psi)$

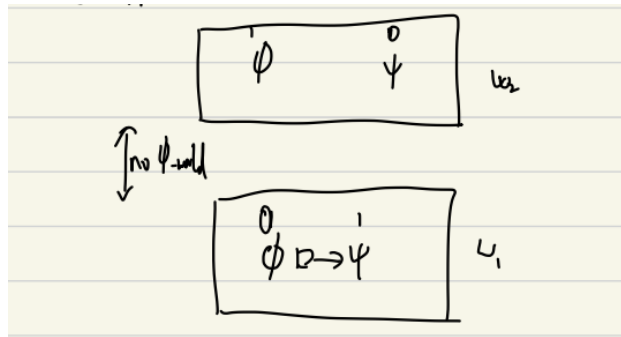
SC accepts these as tautologies, while LC rejects these.

### Illustrations of the different validities in LC and SC

**Example 9.2.** Show that the following hold for Stalnaker's conditional:

- (i)  $\psi \not\models_{SC} \phi \Box \rightarrow \psi$

*Proof.* Our illustration is



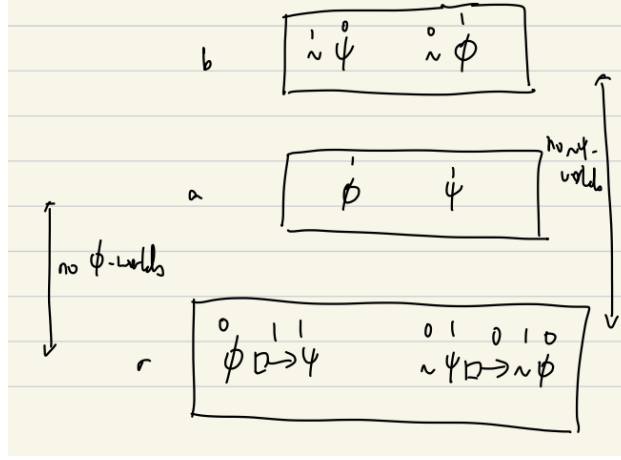
Let  $\phi = P, \psi = R$ . Then,

$$\begin{aligned}
 \mathcal{W} &= \{w_1, w_2\} \\
 \mathcal{I}(P, w_1) &= 0, \mathcal{I}(P, w_2) = 1 \\
 \mathcal{I}(R, w_1) &= 1, \mathcal{I}(R, w_2) = 0 \\
 \preceq_{w_1} &= \{\langle w_1, w_1 \rangle, \langle w_1, w_2 \rangle\}
 \end{aligned}$$

■

(ii)  $\phi \not\models_{SC} \sim \psi \Box \rightarrow \sim \phi$

*Proof.* Our illustration is

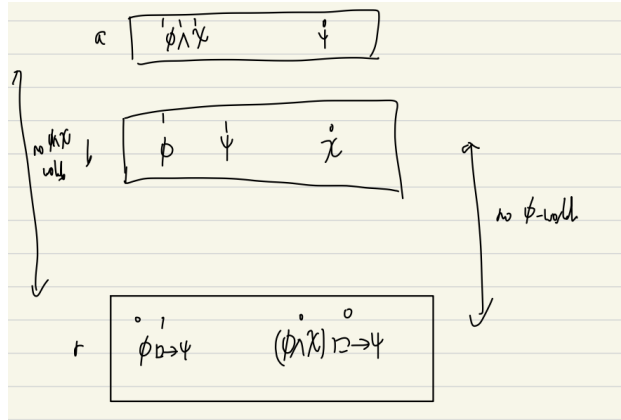


Let  $\phi = P, \psi = R$ . Then, our official countermodel is

$$\begin{aligned} \mathcal{W} &= \{r, a, b\} \\ \preceq_r &= \{\langle a, b \rangle, \dots\} \\ \mathcal{I}(P, a) &= 1, \mathcal{I}(P, b) = 1, \mathcal{I}(P, r) = 0 \\ \mathcal{I}(R, a) &= 1, \mathcal{I}(R, b) = 0, \mathcal{I}(R, r) = 1 \end{aligned}$$

(iii)  $\phi \Box \rightarrow \psi \not\models_{SC} (\phi \wedge \chi) \Box \rightarrow \psi$

*Proof.* Our illustration is

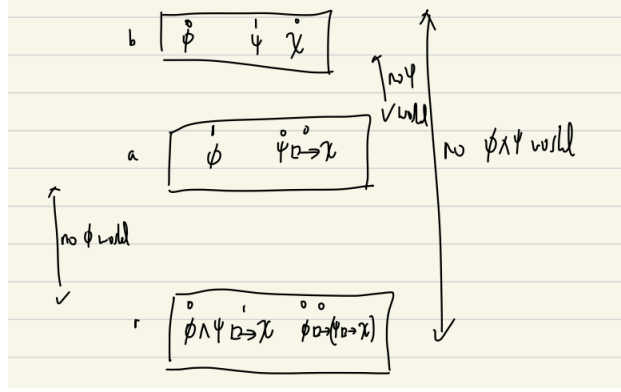


Let  $\phi = P, \chi = Q, \psi = R$ . Our countermodel is officially

$$\begin{aligned} \mathcal{W} &= \{r, a, b\} \\ \preceq_r &= \{\langle b, a \rangle, \dots\} \\ \mathcal{I}(P, a) &= \mathcal{I}(Q, a) = \mathcal{I}(P, b) = \mathcal{I}(R, b) = 1 \\ \text{All else } &0 \end{aligned}$$

$$(iv) (\phi \wedge \psi) \Box \rightarrow \chi \not\models_{SC} (\phi \Box \rightarrow (\psi \Box \rightarrow \chi))$$

*Proof.* Our illustration is



Let  $\phi = P, \psi = R, \chi = Q$ . Our official countermodel is

$$\begin{aligned} \mathcal{W} &= \{r, a, b\} \\ \leq_r &= \{\langle a, b \rangle\} \\ \mathcal{I}(P, r) &= \mathcal{I}(R, r) = \mathcal{I}(P, b) = \mathcal{I}(R, a) = \mathcal{I}(Q, a) = \mathcal{I}(Q, r) = \mathcal{I}(Q, b) = 0 \\ \mathcal{I}(P, a) &= \mathcal{I}(R, b) = 1 \end{aligned}$$

■

**Example 9.3.** Formalise the following argument in the language of SC and demonstrate that the conclusion is a semantic consequence of its premises:

Had I flipped the coin, it would have either landed heads or tails. But it's not the case that it would have definitely landed heads if flipped. So, if I'd flipped it, the coin would have landed tails.

**Formalism:** Our formalism is

$$P \Box \rightarrow R \vee Q, \sim (P \Box \rightarrow R) \models_{SC} P \Box \rightarrow Q$$

The dictionary is

$P$  : I flipped the coin  
 $R$  : The coin lands on head  
 $Q$  : The coin lands on tails

**Semantic Proof for SC:** Assume for reductio that

1.  $V_{\mathcal{M}}(P \Box \rightarrow R \vee Q, w) = 1$
2.  $V_{\mathcal{M}}(\sim (P \Box \rightarrow R), w) = 1$
3.  $V_{\mathcal{M}}(P \Box \rightarrow Q, w) = 0$

From (1), for a closest  $P$ -world,  $v$ ,

- $V_{\mathcal{M}}(P, v) = 1, V_{\mathcal{M}}(R \vee Q, w) = 1$ , or
- $\forall v \in \mathcal{W}, V_{\mathcal{M}}(P, v) = 0$

From (2),  $V_{\mathcal{M}}(P \Box \rightarrow R, w) = 0$ , so for a closest  $P$ -world  $u$ ,  $V_{\mathcal{M}}(P, u) = 1$  and  $V_{\mathcal{M}}(R, u) = 0$ .

From (3), for some closest  $P$ -world  $u'$ ,  $V_{\mathcal{M}}(P, u') = 1$  and  $V_{\mathcal{M}}(Q, u') = 0$ .

Given that  $\exists u \in \mathcal{W}$  s.t.  $V_{\mathcal{M}}(P, u) = 1$  from (2), it cannot be that  $\forall v' \in \mathcal{W}, V_{\mathcal{M}}(P, v') = 0$ .

Hence, for a closest  $P$ -world  $v$ ,  $V_{\mathcal{M}}(R \vee Q, v) = 1$ . But from (2), (3):  $V_{\mathcal{M}}(R, u) = 0, V_{\mathcal{M}}(Q, u') = 0$ .

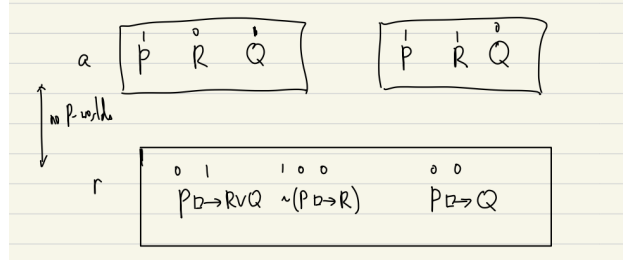
We know that for an SC selection function,  $f(\phi, w)$ , if  $f(\phi, w) = u', f(\phi, w) = v$  and  $f(\phi, w) = u$ , then  $u = u' = v$  due to limit and anti-symmetry.

Hence  $V_{\mathcal{M}}(R, v) = 0$  and  $V_{\mathcal{M}}(Q, v) = 0$ . Therefore  $V_{\mathcal{M}}(R \vee Q, v) = 0$ .

**Counterexample for LC:** We want to show that

$$P \Box \rightarrow R \vee Q, \sim (P \Box \rightarrow R) \not\models_{\text{LC}} P \Box \rightarrow Q$$

Our illustration of the counterexample is



Our official counterexample is

$$\begin{aligned} \mathcal{W} &= \{r, a, b\} \\ \preceq_r &= \{\langle a, b \rangle, \langle b, a \rangle, \dots\} \\ \mathcal{J}(P, a) &= \mathcal{J}(P, b) = \mathcal{J}(R, b) = \mathcal{J}(Q, a) = 1 \\ &\text{Everything else } 0 \end{aligned}$$

**Example 9.4.** Show that

$$1. \models_{\text{SC}} [\phi \Box \rightarrow (\psi \vee \chi)] \rightarrow [(\phi \Box \rightarrow \psi) \vee (\phi \Box \rightarrow \chi)]$$

*Proof.* Assume for reductio that  $V_{\mathcal{M}}((\phi \Box \rightarrow (\psi \vee \chi)) \rightarrow (\phi \Box \rightarrow \psi \vee \phi \Box \rightarrow \chi), w) = 0$ . Then,

$$(a) \ V_{\mathcal{M}}(\phi \Box \rightarrow (\psi \vee \chi), w) = 1$$

$$(b) \ V_{\mathcal{M}}(\phi \Box \rightarrow \psi \vee \phi \Box \rightarrow \chi, w) = 0$$

From (1): either  $\forall u \in \mathcal{W}, V_{\mathcal{M}}(\phi, u) = 0$  or  $V_{\mathcal{M}}(\psi \vee \chi, f(\phi, w)) = 1$ .

- Case 1:  $\forall u \in \mathcal{W}, V_{\mathcal{M}}(\phi, u) = 0$ .

From (2):  $V_{\mathcal{M}}(\phi \Box \rightarrow \psi, w) = 0$  and  $V_{\mathcal{M}}(\phi \Box \rightarrow \chi, w) = 0$ . Hence,  $V_{\mathcal{M}}(\phi, w) = 1$ , thus  $V_{\mathcal{M}}(\phi, w) = 0$

- Case 2:  $V_{\mathcal{M}}(\psi \vee \chi, f(\phi, w)) = 1$ . Suppose  $f(\phi, w) = v$ . By anti-symmetry of SC, we know that  $v$  is unique. Hence,  $V_{\mathcal{M}}(\phi, v) = 1$ , so  $\forall u \in \mathcal{W}$  s.t.  $V_{\mathcal{M}}(\phi, u) = 1, v \preceq_w u$ .

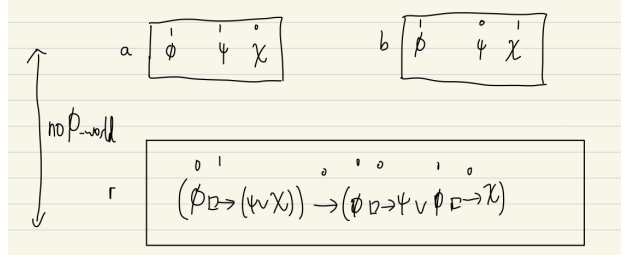
From (2):  $V_{\mathcal{M}}(\phi \Box \rightarrow \psi, w) = 0$  and  $V_{\mathcal{M}}(\phi \Box \rightarrow \chi, w) = 0$ , thus  $V_{\mathcal{M}}(\chi, f(\phi, w)) = 0$ .

Since if  $f(\phi, w) = u, f(\phi, w) = v$ , then  $u = v$ . We know from  $V_{\mathcal{M}}(\psi \vee \chi, f(\phi, w)) = 1$ . Thus,  $V_{\mathcal{M}}(\psi, f(\phi, w)) = 1$  or  $V_{\mathcal{M}}(\chi, f(\phi, w)) = 1$ , a contradiction. ■

$$2. \not\models_{\text{LC}} [\phi \Box \rightarrow (\psi \vee \chi)] \rightarrow [(\phi \Box \rightarrow \psi) \vee (\phi \Box \rightarrow \chi)]$$

*Proof.* Our illustration of the countermodel is





Let  $\phi = P, \psi = R, \chi = Q$ . Our official countermodel is

$$\begin{aligned}\mathcal{W} &= \{r, a, b\} \\ \preceq_r &= \{\langle r, r \rangle, \langle b, b \rangle, \langle a, a \rangle, \langle r, a \rangle, \langle r, b \rangle, \langle a, b \rangle, \langle b, a \rangle\} \\ \mathcal{I}(P, a) &= 1, \mathcal{I}(P, b) = 1, \mathcal{I}(R, a) = 1, \mathcal{I}(Q, a) = 1 \\ &\text{All else } 0\end{aligned}$$

■

$$3. \models_{\text{SC}} (\phi \Box \rightarrow \psi) \vee (\phi \Box \rightarrow \sim \psi)$$

*Proof.* AFC that  $V_{\mathcal{M}}((\phi \Box \rightarrow \psi) \vee (\phi \Box \rightarrow \sim \psi), w) = 0$ . Then

- $V_{\mathcal{M}}(\phi \Box \rightarrow \psi, w) = 0$ , implying that for  $f(\phi, w) = v$ ,  $V_{\mathcal{M}}(\phi, v) = 1$  and  $V_{\mathcal{M}}(\psi, v) = 0$
- $V_{\mathcal{M}}(\phi \Box \rightarrow \sim \psi, w) = 0$ , implying that for  $f(\phi, w) = u$ ,  $V_{\mathcal{M}}(\phi, u) = 1$  and  $V_{\mathcal{M}}(\sim \psi, u) = 0$

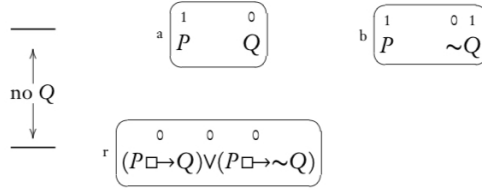
Since we know that for any  $u, v \in \mathcal{W}$ , if  $f(\phi, w) = u$  and  $f(\phi, w) = v$ , then  $u = v$ , it follows that

$$V_{\mathcal{M}}(\psi, u) = 1, V_{\mathcal{M}}(\psi, u) = 0$$

■

$$4. \not\models_{\text{LC}} (\phi \Box \rightarrow \psi) \vee (\phi \Box \rightarrow \sim \psi)$$

*Proof.* Our illustraton of the countermodel is



Let  $\phi = P, \psi = Q$ . Our official countermodel is

$$\begin{aligned}\mathcal{W} &= \{r, a, b\} \\ \preceq_r &= \{\langle a, b \rangle, \langle b, a \rangle, \dots\} \\ \mathcal{I}(P, a) &= \mathcal{I}(P, b) = 1 \\ &\text{All else } 0\end{aligned}$$

■

## 10 Appendix

### 10.1 Mathematical Tools (from Elements of Deductive Logic)

Some simple proof methods that need no exposition

1. Conditional Proof
2. Biconditional Proof
3. Proof by Cases
4. Proof by Contradiction
5. Proof by Generalisation
6. Proof by Contraposition

Of greater importance are the following two theorems:

**Theorem 10.1 (Principle of Mathematical Induction (Ind)).** If (i)  $F(0)$  and (ii) for each  $n \in \mathbb{N}$ :  $F(n)$  implies  $F(n+1)$ , then  $F(n)$  for each  $n \in \mathbb{N}$ . Formally

$$(F(0) \wedge \forall n(F(n) \rightarrow F(n+1))) \rightarrow \forall n F(n)$$

**Theorem 10.2 (Strong Principle of Mathematical Induction (S-Ind)).** If, for each  $n \in \mathbb{N}$ ,  $F(m)$  for all  $m < n$  implies  $F(n)$ , then  $F(n)$  for all  $n \in \mathbb{N}$ . Formally,

$$\forall n((\forall m < n)F(m) \rightarrow F(n)) \rightarrow \forall n F(n)$$

**Theorem 10.3 (Equivalence of Strong and Weak Induction).** S-Ind and Ind are equivalent

**Remark:** The proof of equivalence uses the following property

**Theorem 10.4 (Least Number Principle (LNP)).** If  $M$  is a non-empty set of natural numbers,  $M$  has a least element

We usually use induction on the complexity of formulas

**Definition 10.1 (Complexity).** Let complexity of a wff  $\phi$  be the number of connectives in  $\phi$  – we denote this  $C(\phi)$ .

**Remarks:**

- Sider only includes  $\sim, \rightarrow$  so that when you do induction, you only need to consider those two connectives
- The above complexity measure is for PL. In Predicate Logic, we include quantifiers as well

**Induction Procedure:**

1. *Base Case.* Show that  $G(\alpha)$  holds for any atomic formula  $\alpha$
2. Let  $\phi$  be an arbitrary formula
3. *Induction hypothesis.* Assume  $G(\psi)$  holds for all  $\psi$  with complexity less than  $\phi$
4. *Induction Step.* Prove  $G(\phi)$  using the Induction hypothesis

## 10.2 Soundness and Completeness of Predicate Logic

We want to prove the following adequacy theorem in Predicate Logic:

**Theorem 10.5 (Adequacy Theorem).**  $\models \phi$  iff  $\vdash \phi$

This follows immediately from the following two theorems:

**Theorem 10.6 (Soundness of PC).** If  $\vdash \phi$  then  $\models \phi$

and

**Theorem 10.7 (Completeness of PC).** If  $\models \phi$  then  $\vdash \phi$

## 10.3 Set Theory

**Definition 10.2 (Injective Function).** A function  $f : X \rightarrow Y$  is said to be injective iff for all  $a, b \in X$ , if  $f(a) = f(b)$ , then  $a = b$

**Interpretation:** an injective function  $f$  maps distinct elements of its domain to distinct elements of its codomain. In other words, every element of the function's codomain is the image of at most one element of its domain.

**Remark:** The term one-to-one function must not be confused with one-to-one correspondence that refers to bijective functions, which are functions such that each element in the codomain is an image of exactly one element in the domain.

**Definition 10.3 (Surjective Function).** A function  $f : X \rightarrow Y$  is said to be surjective iff  $\forall y \in Y, \exists x \in X$  such that  $f(x) = y$

**Interpretation:** a function  $f$  is surjective iff for every element in its codomain  $Y$ , there is an element in its domain  $X$  such that  $f(x) = y$  i.e. every element in its codomain is mapped onto by some element in the domain.

**Remark:** A surjection makes no restriction on whether every element in the domain must map onto some element in the codomain, and neither does it make a restriction on how many how many elements in the domain can map onto an element in the codomain.

**Definition 10.4 (Bijective Function).** A function  $f : X \rightarrow Y$  is said to be bijective iff there is a function  $g : Y \rightarrow X$  i.e. the inverse of  $f$ , such that  $g(f(x)) = x$  and  $f(g(y)) = y$ , or equivalently, iff  $f$  is both surjective and injective

**Interpretation:** A bijective function is commonly said to represent a one-to-one correspondence – namely that every element in  $X$  is mapped to exactly one element in  $Y$ , and every element of  $Y$  is mapped to exactly one element in  $X$ .